



# Open Chiplet Atlas Harness (OCAH)

## Integrator Guide

Version 0.0.3, 2026-08-21

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This guide provides instructions and best practices for integrating OCAH subsystems into your chiplet design. Each section covers integration considerations, interfaces, and configuration requirements for the respective subsystem.

# Revision History

This document tracks major revisions and updates to the OCAH Integrator Guide.

*Table 1. Revision History*

| Version | Date       | Author               | Description                                                                                                                                                                                                                                         |
|---------|------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.0.1   | 2026-06-17 | Alessandro Ottaviano | Initial integrator guide draft                                                                                                                                                                                                                      |
| 0.0.2   | 2026-08-13 | Alessandro Ottaviano | Completed the documentation migration: added primitive-cell replacement and CDC constraint guidance, corrected DTP/JTAG integration, and aligned SMC/SEP paths, GPIO and DFX status descriptions, and cross-guide references with the current tree. |
| 0.0.3   | 2026-08-21 | Ben Melton           | Added a Project Defines section covering compile-time defines that apply across OCAH, starting with <b>SIMULATION</b> : how to set it, the testbench hooks and models it enables, and the flows it must not reach.                                  |

# Chapter 1. SMU Top-Level Integration

The System Management Unit (SMU) comprises three peer subsystems: the System Management Controller (SMC), the Debug & Test Ports (DTP), and the Security Processor (SEP). The subsections below describe each subsystem’s interfaces independently, followed by the [SMU Composition](#) section that explains how they are wired together inside the SMU and how to select between the `smu.sv` logic-level core and the `smu_wrapper.sv` reference integration.

## 1.1. Address Map Integration

The OCAH address map has three levels, and the assignment contract splits across the integrator, SEP firmware, and SMC firmware. Assign and program them from this section; the deeper subsystem and crossbar sections referenced below give the per-block detail.

- **Chiplet envelope (integrator).** The SoC/chiplet address map assigns each subsystem a global base and reserved size, and the adopter’s fabric decoder routes those ranges to the SMU. The SMU crossbar preserves offsets and does not constrain envelope size, so the integrator owns non-overlap and containment at the SoC level.
- **SEP and SMC chiplet apertures (firmware).** Each subsystem exposes a chiplet-visible aperture whose base and size are programmable at runtime. Inbound transactions to an aperture are remapped global-to-local before they reach the subsystem.
- **Local-alias views (fixed in RTL).** Internally each subsystem addresses its own resources through a fixed local map: the SMC at `0xC000_0000` (see [Address Map and Aperture](#)), the SEP through its own local map. These are RTL-fixed and not integrator-assigned.

Table 2. Address-map assignment contract

| Level                | Owner        | Assigned / programmed by                                                                       |
|----------------------|--------------|------------------------------------------------------------------------------------------------|
| Chiplet envelope     | Integrator   | SoC/chiplet address map + adopter fabric decode                                                |
| SEP chiplet aperture | SEP firmware | <code>SEP_GLOBAL_BASE_ADDR / SEP_REGION_SIZE</code> (the <code>sep_cpu_ctrl</code> CSRs)       |
| SMC chiplet aperture | SMC firmware | <code>GLOBAL_BASE / REGION_SIZE</code> (the SMC CPU Control CSRs at <code>0xC001_0000</code> ) |
| SEP / SMC local view | Fixed (RTL)  | Offsets within each subsystem’s local map (SMC local base <code>0xC000_0000</code> )           |

Programming order and constraints: firmware programs the SEP and SMC aperture CSRs during early boot, **before any cross-subsystem AXI traffic is issued**, and must keep the two apertures non-overlapping and inside the chiplet envelope the integrator assigned. The full runtime crossbar behaviour and the exact CSR plumbing are in [Firmware programming order](#) under [SMU Composition and Internal Wiring](#).

### 1.1.1. Register collateral

The machine-readable address map ships in the repository tree; integrators consume it from the release checkout or tarball rather than a separate package:

- System tops: `hw/sys/<sys>/regs/<sys>.rdl` (e.g. `hw/sys/sep/regs/sep.rdl`, `hw/sys/smc/regs/smc.rdl`).
- IP blocks: `hw/ip/<ip>/regs/<ip>.rdl`.
- Vended blocks: `vendor/.../overlay/regs/<ip>/regs/<ip>.rdl` or `vendor/.../overlay/rdl/<ip>.rdl`.
- Generated outputs per block: `<block>/regs/gen/{c,sv,svh,py,ral,adoc,html,ipxact}/` — C headers, SystemVerilog packages, Python models, UVM RAL, documentation, and IEEE 1685-2014 IP-XACT component descriptions.

For the architecture-level responsibility split between the integrator, firmware, and the internal SEP/SMC maps, see

## 1.2. System Management Controller (SMC) Integration

The System Management Controller (SMC) is a subsystem responsible for power management, clock control, reset sequencing, and peripheral interface coordination. The `smc.sv` module exposes logic-level interfaces; all technology-specific components (PLLs, PVT sensors, eFuses, memory macros, GPIO padings) are instantiated externally by the adopter. See the [SMU Module Variants](#) section for how the reference wrappers demonstrate this integration.

For detailed architectural documentation about the SMC internal design, refer to [hw/sys/smc/doc/](#). The public integration examples are [hw/top/smc\\_wrapper.sv](#) and [hw/top/smc\\_ip\\_integration.sv](#).

### 1.2.1. Clock and Reset

The SMC requires three external clock inputs: `clk_smc_i` (primary system clock driving the CPU cluster, fabric, and peripherals), `clk_periph_i` (peripheral clock driving the peripherals), and `clk_ref_i` (reference clock for timing-sensitive operations). All must be generated externally by the adopter's PLL infrastructure; the SMC provides AXI-Lite control outputs for PLL configuration (see [Peripheral Interface Connections](#)). The peripheral clock is used to drive the I3C interface, I2C interface, UART interface, and AVSBus interface.

Assert `powergood_i` after power-up sequence completion. Deassert `rst_cold_ni` after the power-up sequence completion.

The SMC generates AASD (async-assert / sync-deassert) reset outputs: `rst_primary_smc_clk_no` for the main clock domain, `rst_primary_periph_clk_no` for the peripheral clock domain, and `rst_primary_ref_clk_no` for the reference clock domain. These ensure clean reset release without metastability across the system hierarchy.

Optional reset override inputs (`rst_cold_override_ni`, `rst_cool_override_ni`, `rst_core_override_ni`) allow external control of specific reset domains for test and debug scenarios. Most integrations leave these override enables deasserted; the DTP drives them through JTAG IC\_RESET controls when enabled (see [SMC-DTP Internal Interconnection](#)).

### 1.2.2. Address Map and Aperture

The SMC has a chiplet-visible aperture and an internal local address view. Internally, the SMC uses a fixed local base address of `0xC000_0000` and a fixed local window size of 16 MiB. SMC-local addresses (registers, memories, remapped regions) are defined as offsets from that base. The chiplet-visible aperture is adopter-defined: the chiplet or SoC address map assigns the global base and size that external masters can use to reach SMC through the SMU crossbar.

When integrating the SMC into a chiplet or SoC, the adopter assigns a system base address and reserved size for the SMC window. The system base (e.g. `0x0800_0000` in a chiplet address map) may be different from the SMC-local base `0xC000_0000`. Firmware programs the SMC `GLOBAL_BASE / REGION_SIZE` CSRs during early boot to match that assignment. The adopter's fabric or address decoder must route the reserved system range to the SMC block so that offsets are preserved: system base + offset maps to SMC-local `0xC000_0000` + offset.

*Table 3. Address Base Mapping*

| View                 | Base Address             | Description                                                         |
|----------------------|--------------------------|---------------------------------------------------------------------|
| SMC-local (internal) | <code>0xC000_0000</code> | Local alias address (fixed). Used by SMC CPU, firmware, and fabric. |
| System (adopter)     | Adopter-defined          | Base of the chiplet/SoC aperture assigned to SMC.                   |

The SMC fabric uses three registers in the CPU Control block (`0xC001_0000`) to distinguish local from remote addresses:

- **LOCAL\_BASE** (read-only, fixed at `0xC000_0000`): The local alias base. Together with the fixed 16 MiB **LOCAL\_ALIAS\_REGION\_SIZE**, this is the address space the SMC CPU and internal fabric use to access SMC resources.
- **GLOBAL\_BASE** (read-write, defaults to `0x4000_0000`): The system-level base address. Firmware must program this during early boot to match the adopter's system address assignment, enabling the SMC to issue outbound transactions using global system addresses.
- **REGION\_SIZE** (read-write, currently resets to `0x0100_0000` / 16 MiB): The size of the SMC global aperture used by SMU-level routing and SEP-to-SMC decode. The SMC input fabric's local/global split uses the fixed **LOCAL\_ALIAS\_REGION\_SIZE** parameter in RTL.

For full details on alias remapping and outbound routing, see the SMC Fabric documentation ([hw/sys/smc/doc/fabric.adoc](#)).

For a detailed breakdown of all components within the SMC local map, refer to the SMC Memory Map documentation ([hw/sys/smc/doc/memmap.adoc](#)).

### 1.2.3. AXI Interface Configuration

The SMC provides three distinct AXI interface groups serving different access paths. The **sys\_axi\_in** interface accepts requests from external system masters requiring access to SMC-managed resources. This interface uses a 56-bit address space with 64-bit data width and 6-bit transaction IDs. Connect this interface to your system fabric for normal operational access from application processors or DMA controllers.

The **sep\_axi\_in** interface provides dedicated access from the Security Processor (SEP) subsystem. This interface shares the same sizing parameters as the system input but operates independently to maintain security domain separation. Use this interface only when integrating alongside the SEP subsystem. The interface may remain unconnected if your integration does not include SEP.

The **output\_axi** interface carries filtered and remapped requests from the SMC to external system resources. This interface uses an 8-bit transaction ID field to accommodate merged traffic from multiple input paths. Connect this interface to your system interconnect to allow SMC components access to external memory and peripherals. The testbench shows example connections using AXI VIP components that can be replaced with your system fabric.

The **jtag\_axi\_in** interface provides debug access through JTAG mechanisms. This interface uses reduced 2-bit transaction IDs and connects to your JTAG debug infrastructure. The testbench demonstrates typical debug access patterns through this interface.

### 1.2.4. GPIO Pad Integration

GPIO pad integration requires careful attention to function assignment and pull requirements. The GPIO table below lists each signal, bump location, function, recommended pull, and integration notes. Default input/output direction is defined by **smc\_padrng\_pkg::DefaultDirectionMap**, not by the table.

The public [hw/top/smc\\_wrapper.sv](#) and [hw/top/smc\\_ip\\_integration.sv](#) examples show the pad connections. For bidirectional pins, implement proper tri-state buffers controlled by the **gpio\_out\_en\_o** signals. Connect multiplexed UART, I2C, I3C, and SPI pins to external devices according to the function assignments in the GPIO table.

Configuration straps are captured separately from the GPIO table. Establish them before cold reset deassertion and keep them stable during capture. The generated reset-unit **STRAPS\_LO** and **STRAPS\_HI** register descriptions define the decoded bit assignments, including **BYPASS\_SRAM\_REPAIR** at **STRAPS\_LO[13]** and **TEST\_EN** at **STRAPS\_LO[14]**.

## 1.2.5. Peripheral Interface Connections

The SMC exposes logic-level interfaces to several technology-specific subsystem components that the adopter instantiates externally. The subsections below describe each connection in turn.

### Adopter Peripheral Window

The `smc_external_req_o` and `smc_external_resp_i` signals form a 32-bit AXI-Lite interface to the adopter's own peripheral register maps. Any technology- or product-specific block that SMC firmware needs to reach sits behind this window rather than on a dedicated per-peripheral port.

The window is the recommended home for the control/status registers of logic that the SMC must configure or monitor but that the SMC itself does not implement—padding configuration logic, clock generation, on-die sensors, memory-technology SHIMs, and similar adopter-owned blocks are typical examples. For example, runtime reconfiguration of pin directions, pull settings, and peripheral multiplexing is expected to be reachable here. The SMC decodes the aperture and forwards the transaction; what lives behind it is entirely adopter-defined.

The window is split into a mandatory region, whose contents and placement the SMC memory map fixes and every integration must implement, and a supplementary region for anything else the adopter chooses to expose. For placement and the full register layout, refer to the SMC Memory Map documentation ([hw/smc/doc/memmap.adoc](http://hw/smc/doc/memmap.adoc)).

### eFuse Interface

The eFuse interface consists of two parts:

- `efuse_bank_ctrl`—a 32-bit AXI-Lite interface used to access the SHIM control CSRs (for example, foundry-specific bank timing such as the initialization delay before sensing can begin).
- `efuse_shim_command`—the custom fuse command interface used to issue read, program, and program/read-back operations to the eFuse bank.

**Adopter responsibility.** The `efuse_shim_command` interface is intentionally generic and technology-neutral. The adopter must implement an Interface SHIM that converts these generic commands into the hardware signals required by their OTP macro. The conversion is driven by the adopter's own read and program state machines inside the SHIM; OCAH only defines the command/response contract, not the macro-side signaling.

The command and response payloads are exchanged as packed structs:

```
// Controller -> SHIM
typedef struct packed {
    addr_t          address;           // Target address (in bits)
    logic           program_data;      // Data to program (when programming)
    access_length_words_t access_length_words; // Number of WORD_WIDTH words to access
    command_t       command;          // READ / PROGRAM / PROGRAM_READ_BACK
    logic           valid;             // Request valid
} fuse_command_req_t;

// SHIM -> Controller
typedef struct packed {
    data_t data; // Read data (one WORD_WIDTH word)
    logic status; // Operation status (0 = OK, 1 = error)
```

```
logic valid; // Response valid (qualifies `data` for one cycle)
} fuse_command_resp_t;
```

**Fuse sensing protocol.** During automatic sensing the controller issues a single read with `access_length_words` set to the entire bank (all `NUM_WORDS` words). The SHIM is expected to stream the bank back one `WORD_WIDTH` word at a time: for each OTP access it drives the read `data` onto `fuse_command_resp_o.data` and asserts `valid` for **exactly one cycle**. The controller consumes one word per cycle that `valid` is high, advancing to the next eFuse word on each assertion.

#### WARNING

`valid` must be asserted for only a single cycle per returned word. If the SHIM holds `valid` high for two cycles, the controller will sample the same `data` value a second time and count it as the data for the **next** eFuse word, shifting all subsequent words and corrupting the sensed shadow contents. The adopter's SHIM FSM is responsible for producing a clean one-cycle `valid` strobe per `WORD_WIDTH` access.

The `shadow_regs_o` output provides read-only access to the sensed fuse values. During bring-up the SMC automatically senses the fuse contents and propagates them through the shadow registers; the `fuse_sense_done_o` signal indicates completion of this sensing operation.

## 1.2.6. Memory Subsystem Integration

The SMC includes an embedded RISC-V CPU cluster with associated memory interfaces. The `rom_intf_req_o` and `rom_intf_rsp_i` signals connect to boot ROM containing the SMC firmware. The ROM must be populated with valid firmware images before initiating the boot sequence.

Scratch RAM interfaces (`scratch_ram_intf_req_o` and `scratch_ram_intf_rsp_i`) connect to 32 independent RAM banks totaling the scratchpad memory space. Cache memory interfaces for L1 instruction and data caches (`l1_icache_*` and `l1_dcache_*`) connect to tag and data RAM banks as specified in the port table.

The `disable_sram_auto_init_i` input controls automatic memory initialization behavior. When deasserted, the SMC automatically zeroes all SRAM contents during bring-up. The `init_mem_done_o` output indicates completion of this initialization sequence. The `SRAM_AUTO_ZERO_DISABLE` strap (`STRAPS_LO[26]`) can disable this automatic zeroing behavior if your integration requires preserved SRAM contents across resets.

## 1.2.7. Memory Repair Integration

The SMC implements Built-In Redundancy Analysis (BIRA) memory repair to address manufacturing defects in SRAM arrays. Repair uses redundant memory elements configured with data stored in eFuse memory, replacing defective cells detected during manufacturing test.

Memory repair executes automatically during boot after eFuse sensing completes and before MBIST operations. The process must complete successfully before the CPU cluster can safely access SRAM arrays. Repair is required on every cold reset but can be bypassed during Function Level Reset (FLR) operations for faster recovery.

#### Control Signals:

- `fuse_sense_done_o`: eFuse controller output indicating fuse sensing completion. External DFT logic monitors this signal to trigger repair operations.
- `skip_mem_repair_o`: Reset unit output indicating repair should be bypassed. Asserts during FLR operations or when the `BYPASS_SRAM_REPAIR` strap is set. When asserted, external repair logic should skip operations and immediately assert completion status.

- `ext_boot_seq_done_i`: External signal indicating the external boot sequence (memory repair and shadow register override) is complete. Held low to pause reset release until the external agent gives the go-ahead; required before the system can continue its boot sequence

### Repair Data:

Repair data is stored in the eFuse BIRA shadow register as a 16-kilobit `repair_data` field. The eFuse controller reads this data during fuse sensing and makes it available through shadow registers. The external memory repair controller must read repair data from `shadow_regs_o` and apply it to SRAM arrays according to your memory architecture.

### Status Monitoring:

Repair completion and success are reported through `DFX_CTRL_STATUS.STATUS_SMU` at `0xC000_B800`. The `mem_repair_done` bit indicates completion regardless of success, while `mem_repair_success` indicates successful repair. Both bits are set by the external repair controller. The SMC ROM firmware monitors these bits during initialization to verify repair completion.

### Bypass Configuration:

The `BYPASS_SRAM_REPAIR` configuration strap (`STRAPS_LO[13]`) bypasses repair logic when asserted high, causing `skip_mem_repair_o` to assert. Use bypass mode only when SRAM arrays are guaranteed defect-free, such as during development or with known-good components.

### Integration Requirements:

Provide an external memory repair controller that monitors `fuse_sense_done_o` to initiate repair, respects `skip_mem_repair_o` for bypass control, reads repair data from eFuse shadow registers, applies repair configuration to SRAM arrays, asserts `ext_boot_seq_done_i` and updates DFT Control Status registers with completion status. The controller operates independently of the SMC core and must complete before the CPU cluster accesses SRAM. The testbench demonstrates example integration patterns adaptable to your DFT infrastructure.

For direct access to the eFuse besides from the shim, i.e. from a DFT fuse controller, adopters must ensure the integrated logic will not interfere with accesses from the OCAH's efuse interface.

During FLR operations, repair is automatically bypassed via `skip_mem_repair_o` to enable rapid reset sequences. Normal cold reset sequences always execute repair unless explicitly bypassed via strap configuration.

## 1.2.8. Interrupt Integration

The SMC aggregates interrupts from multiple sources into `cpu_interrupts_o`, which feeds the PLIC of the embedded RISC-V CPU cluster. The interrupt vector is assembled in `hw/sys/smc/rtl/smc_base.sv` and `hw/sys/smc/rtl/smc_peripherals/rtl/smc_peripherals.sv`. The authoritative bit-level map (1-core and 4-core columns, RTL trace, and PLIC source ID note) is maintained in a single place:

See the *SMC CPU interrupt vector map* in the Technical Reference Manual.

Configure your interrupt controller to monitor the `sync_irq_o` signal and read SMC interrupt status registers to determine active sources. The testbench demonstrates interrupt injection and handling patterns through the various interrupt input ports.

## 1.2.9. Telemetry and Debug Integration

For system-level debug and telemetry, the SMC implements ATB (Advanced Trace Bus) receiver interfaces. The telemetry interface group (`telemetry_atdata_i`, `telemetry_atid_i`, `telemetry_atvalid_i`, `telemetry_atready_o`) operates in a dedicated

`clk_telemetry_i` clock domain with separate `rst_telemetry_ni` reset. Connect these signals to ATB trace sources for debug data collection.

The `ext_debug_bus_i` provides a 512-bit wide debug observation bus for internal signal monitoring. The cross-trigger interface (`xtrigger_pads_i`, `xtrigger_pads_o`, `xtrigger_ss_i`, `xtrigger_ss_o`) enables coordinated debug triggering across multiple subsystems. The `halt_clock_global_i` input array accepts clock halt requests from external debug agents.

Trace memory interfaces (`trace_mem_req_o` and `trace_mem_resp_i`) connect to trace RAM instances for storing captured debug data. The testbench shows example debug infrastructure connections that can be adapted to your debug requirements.

## 1.2.10. Bring-Up Sequence

The recommended bring-up sequence begins with asserting `rst_cold_ni` while stabilizing power supplies and clocks. Once all supplies reach nominal voltages, deassert `rst_cold_ni` to initiate the SMC boot sequence. The SMC immediately begins fuse sensing to read configuration values and security credentials.

Monitor the `fuse_sense_done_o` signal to detect completion of fuse sensing. Configuration straps sampled during this phase determine boot modes and operational settings. After fuse sensing completes, the SMC releases internal subsystem resets based on the programmed reset sequencing configuration.

The CPU cluster begins firmware execution from the boot ROM after reset release. The firmware performs additional initialization including peripheral configuration, memory testing (unless disabled), and system health checks. The `init_mem_done_o` signal indicates completion of memory initialization if automatic SRAM zeroing is enabled.

For Function Level Reset (FLR) support, assert `cfg_flr_pf_active_i` to trigger a warm reset sequence. This isolates subsystems through `isolate_req_o` signals and performs controlled reset propagation. Monitor `ss_reset_complete_i` signals to verify subsystem reset completion before releasing isolation.

The testbench file demonstrates the complete power-on sequence including clock initialization, reset timing, strap configuration, and peripheral bring-up. Reference this testbench when implementing your integration-specific bring-up procedures.

## 1.2.11. SMC Port Interface

The SMC exposes a comprehensive set of interfaces for system integration. The following table defines all SMC top-level ports including clocks, resets, AXI interfaces, peripheral connections, and control signals. Refer to the detailed SMC documentation for internal signal descriptions and timing requirements.

*Table 4. SMC Port Declaration*

| Interface Name                  | Type  | Direction | Width | Description                                                                             | Connection                    | Default value (if applicable) |
|---------------------------------|-------|-----------|-------|-----------------------------------------------------------------------------------------|-------------------------------|-------------------------------|
| <code>clk_smc_i</code>          | logic | Input     | 1     | Primary SMC clock domain. All SMC internal logic operates synchronously to this clock.  | PLL output                    | N/A                           |
| <code>clk_ref_i</code>          | logic | Input     | 1     | Reference clock domain. Used for timing-sensitive operations and peripheral interfaces. | PLL reference clock           | N/A                           |
| <code>clk_periph_i</code>       | logic | Input     | 1     | Peripheral clock domain.                                                                | PLL peripheral clock          | N/A                           |
| <code>powergood_i</code>        | logic | Input     | 1     | Power-good indication. Assert after supply voltages stabilize.                          | Power management controller   | N/A                           |
| <code>powergood_stable_o</code> | logic | Output    | 1     | Stabilized power-good output. Debounced version of <code>powergood_i</code> .           | Reset unit → External systems | N/A                           |

| Interface Name            | Type                               | Direction | Width | Description                                                                                 | Connection                                | Default value (if applicable) |
|---------------------------|------------------------------------|-----------|-------|---------------------------------------------------------------------------------------------|-------------------------------------------|-------------------------------|
| rst_cold_ni               | logic                              | Input     | 1     | Cold reset input (active low). Asynchronous assertion, synchronous deassertion.             | System cold reset                         | N/A                           |
| rst_primary_ref_clk_no    | logic                              | Output    | 1     | Primary reset synchronized to reference clock domain (active low).                          | To subsystems using ref clock             | N/A                           |
| rst_primary_smc_clk_no    | logic                              | Output    | 1     | Primary reset synchronized to SMC clock domain (active low).                                | To subsystems using SMC clock             | N/A                           |
| sys_axi_in_req_i          | smc_sys_in_56_64_12_axi_req_t      | Input     | -     | System AXI input request (56-bit addr, 64-bit data, 6-bit ID, 12-bit user).                 | External system masters → Input Fabric    | N/A, tie to '0 if unused      |
| sys_axi_in_resp_o         | smc_sys_in_56_64_12_axi_resp_t     | Output    | -     | System AXI input response.                                                                  | Input Fabric → External system masters    | N/A, leave floating if unused |
| jtag_axi_in_req_i         | smc_jtag_56_64_12_axi_req_t        | Input     | -     | JTAG debug AXI request (56-bit addr, 64-bit data, 2-bit ID, 12-bit user).                   | JTAG debug port → Input Fabric            | N/A, tie to '0 if unused      |
| jtag_axi_in_resp_o        | smc_jtag_56_64_12_axi_resp_t       | Output    | -     | JTAG debug AXI response.                                                                    | Input Fabric → JTAG debug port            | N/A, leave floating if unused |
| axil_smc_otp_jtag_req_i   | smc_pkg::smc_axil_32_32_req_t      | Input     | -     | eFuse OTP/JTAG AXI-Lite request.                                                            | JTAG → eFuse controller                   | N/A, tie to '0 if unused      |
| axil_smc_otp_jtag_resp_o  | smc_pkg::smc_axil_32_32_resp_t     | Output    | -     | eFuse OTP/JTAG AXI-Lite response.                                                           | eFuse controller → JTAG                   | N/A, leave floating if unused |
| sep_axi_in_req_i          | smc_sep_in_56_64_12_axi_req_t      | Input     | -     | SEP AXI request (56-bit addr, 64-bit data, 6-bit ID, 12-bit user).                          | SEP → Input Fabric                        | N/A, tie to '0 if unused      |
| sep_axi_in_resp_o         | smc_sep_in_56_64_12_axi_resp_t     | Output    | -     | SEP AXI response.                                                                           | Input Fabric → SEP                        | N/A, leave floating if unused |
| output_axi_req_o          | smc_sys_out_56_64_8_12_axi_req_t   | Output    | -     | Filtered and remapped AXI output request (56-bit addr, 64-bit data, 8-bit ID, 12-bit user). | Output Fabric → External system resources | N/A, leave floating if unused |
| output_axi_resp_i         | smc_sys_out_56_64_8_12_axi_resp_t  | Input     | -     | Filtered and remapped AXI output response.                                                  | External system resources → Output Fabric | N/A, tie to '0 if unused      |
| efuse_bank_ctrl_req_o     | smc_pkg::smc_axil_32_32_req_t      | Output    | -     | eFuse bank control AXI-Lite request.                                                        | eFuse controller → eFuse Shim             | N/A                           |
| efuse_bank_ctrl_resp_i    | smc_pkg::smc_axil_32_32_resp_t     | Input     | -     | eFuse bank control AXI-Lite response.                                                       | eFuse Shim → eFuse controller             | N/A                           |
| efuse_shim_command_req_o  | smc_efuse_pkg::fuse_command_req_t  | Output    | -     | eFuse command interface request.                                                            | eFuse controller → eFuse Shim             | N/A                           |
| efuse_shim_command_resp_i | smc_efuse_pkg::fuse_command_resp_t | Input     | -     | eFuse command interface response.                                                           | eFuse Shim → eFuse controller             | N/A                           |
| shadow_regs_o             | smc_efuse_pkg::efuse_map_t         | Output    | -     | eFuse shadow register output.                                                               | eFuse controller → External systems       | N/A                           |
| pll_clk_obs_i             | logic                              | Input     | 1     | PLL clock observation input.                                                                | PLL → GPIO padding (muxed)                | N/A                           |
| pll_clk_obs_en_i          | logic                              | Input     | 1     | PLL clock observation enable.                                                               | Control logic → GPIO padding              | N/A                           |
| pvt_process_clk_obs_i     | logic                              | Input     | 1     | PVT process monitor clock observation input.                                                | PVT → GPIO padding (muxed)                | N/A                           |
| pvt_process_clk_obs_en_i  | logic                              | Input     | 1     | PVT process monitor clock observation enable.                                               | Control logic → GPIO padding              | N/A                           |
| cat_therm_i               | logic                              | Input     | 1     | CAT thermal sensor input.                                                                   | CAT thermal sensor → PVT wrap             | N/A, tie to '0 if unused      |

| Interface Name           | Type                                     | Direction | Width                        | Description                                                                                                                                                                                                       | Connection                                | Default value (if applicable) |
|--------------------------|------------------------------------------|-----------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-------------------------------|
| lsio_interface_select_o  | logic                                    | Output    | [NUM_GPIO_WRAPS-1:0]         | LSIO interface selection for each GPIO wrap.                                                                                                                                                                      | GPIO padding → GPIO wraps                 | N/A                           |
| gpio_in_i                | logic                                    | Input     | [NUM_GPIO_WRAPS-1:0]         | GPIO input data from external pins.                                                                                                                                                                               | External GPIO pins → GPIO padding         | N/A                           |
| gpio_out_o               | logic                                    | Output    | [NUM_GPIO_WRAPS-1:0]         | GPIO output data to external pins.                                                                                                                                                                                | GPIO padding → External GPIO pins         | N/A                           |
| gpio_in_en_o             | logic                                    | Output    | [NUM_GPIO_WRAPS-1:0]         | GPIO input enable control.                                                                                                                                                                                        | GPIO padding → GPIO wraps                 | N/A                           |
| gpio_out_en_o            | logic                                    | Output    | [NUM_GPIO_WRAPS-1:0]         | GPIO output enable control.                                                                                                                                                                                       | GPIO padding → GPIO wraps                 | N/A                           |
| rst_cool_n_from_pin_i    | logic                                    | Input     | 1                            | Cool reset input from GPIO pin 61.                                                                                                                                                                                | GPIO pin 61 → Reset Unit                  | N/A                           |
| clk_telemetry_i          | logic                                    | Input     | 1                            | Telemetry clock domain. Separate clock for ATB telemetry data collection.                                                                                                                                         | Telemetry clock source                    | N/A                           |
| rst_telemetry_n_i        | logic                                    | Input     | 1                            | Telemetry reset (active low). Resets telemetry receiver logic.                                                                                                                                                    | Telemetry reset source                    | N/A                           |
| telemetry_atdata_i       | telemetry_receiver_pkg::telemetry_data_t | Input     | [NUM_TELMETRY_RECEIVERS-1:0] | ATB telemetry data input array.                                                                                                                                                                                   | ATB sources → Telemetry receivers         | N/A                           |
| telemetry_atid_i         | telemetry_receiver_pkg::atb_id_t         | Input     | [NUM_TELMETRY_RECEIVERS-1:0] | ATB telemetry ID input array.                                                                                                                                                                                     | ATB sources → Telemetry receivers         | N/A                           |
| telemetry_atready_o      | logic                                    | Output    | [NUM_TELMETRY_RECEIVERS-1:0] | ATB telemetry ready output.                                                                                                                                                                                       | Telemetry receivers → ATB sources         | N/A                           |
| telemetry_atvalid_i      | logic                                    | Input     | [NUM_TELMETRY_RECEIVERS-1:0] | ATB telemetry valid input.                                                                                                                                                                                        | ATB sources → Telemetry receivers         | N/A                           |
| telemetry_atflushvalid_o | logic                                    | Output    | [NUM_TELMETRY_RECEIVERS-1:0] | ATB flush valid output.                                                                                                                                                                                           | Telemetry receivers → ATB sources         | N/A                           |
| telemetry_atflushready_i | logic                                    | Input     | [NUM_TELMETRY_RECEIVERS-1:0] | ATB flush ready input.                                                                                                                                                                                            | ATB sources → Telemetry receivers         | N/A                           |
| cluster_ded_o            | logic                                    | Output    | 1                            | Cluster double error detection output.                                                                                                                                                                            | CPU cluster → External systems            | N/A                           |
| wdt_first_timeout_o      | logic                                    | Output    | 1                            | First watchdog timeout signal.                                                                                                                                                                                    | CPU cluster WDT → Reset Unit              | N/A                           |
| wdt_second_timeout_o     | logic                                    | Output    | 1                            | Second watchdog timeout signal.                                                                                                                                                                                   | Reset Unit → External systems             | N/A                           |
| smc_region_size_o        | logic                                    | Output    | [31:0]                       | SMC address region size output.                                                                                                                                                                                   | SMC Internal Registers → External systems | N/A                           |
| ext_interrupts_i         | logic                                    | Input     | [NUM_EXT_INTERRUPTS-1:0]     | External interrupt inputs. Width is config-dependent: 32 (1-core) or 256 (4-core). Maps directly to <code>cpu_interrupts_o[NUM_EXT_INTERRUPTS-1:0]</code> .                                                       | External sources → Interrupt aggregator   | N/A, tie to '0 if unused      |
| sep_mailbox_interrupts_i | logic                                    | Input     | [7:0]                        | SEP → SMC mailbox interrupt inputs (8 channels). Routed through <code>peripheral_interrupts[7:0] → cpu_interrupts_o[NUM_EXT_INTERRUPTS+7 : NUM_EXT_INTERRUPTS]</code> (1-core: bits 39–32, 4-core: bits 263–256). | SEP → Interrupt aggregator                | N/A, tie to '0 if unused      |

| Interface Name           | Type                             | Direction | Width                           | Description                                                                                                                                                                                                                                   | Connection                                         | Default value (if applicable)                       |
|--------------------------|----------------------------------|-----------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|-----------------------------------------------------|
| sep_wdt_reset_n_i        | logic                            | Input     | 1                               | SEP watchdog reset (active low). Active-low polarity is inverted before routing to <code>peripheral_interrupts[26]</code> → <code>cpu_interrupts_o[NUM_EXT_INTERRUPTS+26]</code> (1-core: bit 58, 4-core: bit 282).                           | SEP → Interrupt aggregator                         | N/A                                                 |
| fuse_sense_done_o        | logic                            | Output    | 1                               | Fuse sense completion output.                                                                                                                                                                                                                 | eFuse controller → Memory repair logic             | N/A                                                 |
| fuse_reset_n_delayed_o   | logic                            | Output    | 1                               | Delayed fuse reset output (after 16 pipeline stages).                                                                                                                                                                                         | eFuse controller → External systems                | N/A                                                 |
| skip_mem_repair_o        | logic                            | Output    | 1                               | Skip memory repair output.                                                                                                                                                                                                                    | Reset Unit → Memory repair logic                   | N/A                                                 |
| ext_boot_seq_done_i      | logic                            | Input     | 1                               | External boot sequence completion input (memory repair and shadow register override done); gates reset release.                                                                                                                               | External boot/repair controller → eFuse controller | N/A                                                 |
| temp_interrupt_i         | logic                            | Input     | 1                               | PVT temperature sensor interrupt. Routes to <code>peripheral_interrupts[27]</code> → <code>cpu_interrupts_o[NUM_EXT_INTERRUPTS+27]</code> (1-core: bit 59, 4-core: bit 283).                                                                  | PVT wrap → Interrupt aggregator                    | N/A, tie to '0 if unused                            |
| lc_state_i               | logic                            | Input     | [2*LC_STATE_WIDTH-1:0] (8 bits) | Lifecycle state input, differentially encoded.                                                                                                                                                                                                | Lifecycle controller → SMC Internal Registers      | From SEP; tie to 8'hf0 if unused (encoded TEST_DEV) |
| ras_bank_chip_o          | logic                            | Output    | [3:0]                           | RAS bank chip selection output.                                                                                                                                                                                                               | SMC Internal Registers → RAS system                | N/A                                                 |
| ras_bank_instance_o      | logic                            | Output    | [3:0]                           | RAS bank instance selection output.                                                                                                                                                                                                           | SMC Internal Registers → RAS system                | N/A                                                 |
| ext_mailbox_interrupts_o | logic                            | Output    | [NUM_MAILBOXES-1:0]             | External mailbox interrupt outputs (32 interrupts, mapped to IDs 156-187).                                                                                                                                                                    | Mailbox controller → External targets              | N/A                                                 |
| cfg_flr_pf_active_i      | logic                            | Input     | 1                               | Function Level Reset (FLR) PF active signal.                                                                                                                                                                                                  | PCIe/System → Reset Unit                           | N/A, tie to '0 if unused                            |
| isolate_req_o            | logic                            | Output    | [31:0]                          | Isolation request output for 32 subsystems.                                                                                                                                                                                                   | Reset Unit → 32 subsystems                         | N/A                                                 |
| ss_reset_complete_i      | logic                            | Input     | [31:0]                          | Subsystem reset completion status.                                                                                                                                                                                                            | 32 subsystems → Reset Unit                         | 32'hFFFFFFFF                                        |
| ss_config_o              | logic                            | Output    | [31:0]                          | Subsystem configuration output.                                                                                                                                                                                                               | Reset Unit → 32 subsystems                         | N/A                                                 |
| ss_reset_ctrl_o          | smc_reset_unit_pkg::reset_ctrl_t | Output    | [31:0]                          | Subsystem reset control output array.                                                                                                                                                                                                         | Reset Unit → 32 subsystems                         | N/A                                                 |
| sync_irq_o               | logic                            | Output    | 1                               | Synchronized interrupt output.                                                                                                                                                                                                                | Interrupt aggregator → External systems            | N/A                                                 |
| gpio_interrupt_o         | logic                            | Output    | [NUM_GPIO_WRAPS-1:0]            | Raw GPIO interrupt vector from the padding, one bit per GPIO wrap (bonded + unbonded). Software reads GPIO status registers to identify the interrupt source. Already OR-reduced into <code>peripheral_interrupts_o[30:29]</code> internally. | smc_padding → external consumers                   | N/A, leave floating if unused                       |
| uart_interrupt_o         | logic                            | Output    | [NUM_UART-1:0]                  | Raw UART interrupt outputs, one bit per UART instance, in the peripheral clock domain. Already aggregated into <code>peripheral_interrupts_o[21:18]</code> internally via CDC.                                                                | uart_wrap → external consumers                     | N/A, leave floating if unused                       |
| rom_intf_req_o           | rom_req_t                        | Output    | -                               | ROM memory interface request. Width depends on CPU cluster configuration                                                                                                                                                                      | CPU cluster → Boot ROM                             | N/A                                                 |

| Interface Name               | Type                 | Direction | Width                       | Description                                                                     | Connection                           | Default value (if applicable) |
|------------------------------|----------------------|-----------|-----------------------------|---------------------------------------------------------------------------------|--------------------------------------|-------------------------------|
| rom_intf_rsp_i               | rom_rsp_t            | Input     | -                           | ROM memory interface response. Width depends on CPU cluster configuration       | Boot ROM → CPU cluster               | N/A                           |
| scratch_ram_intf_req_o       | scratch_ram_req_t    | Output    | [NUM_SRAM_BANKS2-1:0]       | Scratch RAM interface requests. Width depends on CPU cluster configuration      | CPU cluster → Scratch RAM banks      | N/A                           |
| scratch_ram_intf_rsp_i       | scratch_ram_rsp_t    | Input     | [NUM_SRAM_BANKS2-1:0]       | Scratch RAM interface responses. Width depends on CPU cluster configuration     | Scratch RAM banks → CPU cluster      | N/A                           |
| l1_icache_tag_intf_req_o     | l1_icache_tag_req_t  | Output    | [NUM_ICACHE_TAG_BANKS-1:0]  | L1 I-cache tag interface requests. Width depends on CPU cluster configuration   | CPU cluster → L1 I-cache tag banks   | N/A                           |
| l1_icache_tag_intf_rsp_i     | l1_icache_tag_rsp_t  | Input     | [NUM_ICACHE_TAG_BANKS-1:0]  | L1 I-cache tag interface responses. Width depends on CPU cluster configuration  | L1 I-cache tag banks → CPU cluster   | N/A                           |
| l1_icache_data_intf_req_o    | l1_icache_data_req_t | Output    | [NUM_ICACHE_DATA_BANKS-1:0] | L1 I-cache data interface requests. Width depends on CPU cluster configuration  | CPU cluster → L1 I-cache data banks  | N/A                           |
| l1_icache_data_intf_rsp_i    | l1_icache_data_rsp_t | Input     | [NUM_ICACHE_DATA_BANKS-1:0] | L1 I-cache data interface responses. Width depends on CPU cluster configuration | L1 I-cache data banks → CPU cluster  | N/A                           |
| l1_dcache_tag_intf_req_o     | l1_dcache_tag_req_t  | Output    | [NUM_DCACHE_TAG_BANKS-1:0]  | L1 D-cache tag interface requests. Width depends on CPU cluster configuration   | CPU cluster → L1 D-cache tag banks   | N/A                           |
| l1_dcache_tag_intf_rsp_i     | l1_dcache_tag_rsp_t  | Input     | [NUM_DCACHE_TAG_BANKS-1:0]  | L1 D-cache tag interface responses. Width depends on CPU cluster configuration  | L1 D-cache tag banks → CPU cluster   | N/A                           |
| l1_dcache_data_intf_req_o    | l1_dcache_data_req_t | Output    | [NUM_DCACHE_DATA_BANKS-1:0] | L1 D-cache data interface requests. Width depends on CPU cluster configuration  | CPU cluster → L1 D-cache data banks  | N/A                           |
| l1_dcache_data_intf_rsp_i    | l1_dcache_data_rsp_t | Input     | [NUM_DCACHE_DATA_BANKS-1:0] | L1 D-cache data interface responses. Width depends on CPU cluster configuration | L1 D-cache data banks → CPU cluster  | N/A                           |
| disable_sram_auto_init_i     | logic                | Input     | 1                           | Disable SRAM automatic initialization.                                          | Control logic → CPU cluster          | 1'b0                          |
| init_mem_done_o              | logic                | Output    | 1                           | Memory initialization completion signal.                                        | CPU cluster → Control logic          | N/A                           |
| chiplet_is_primary_i         | logic                | Input     | 1                           | Chiplet is primary signal. Used for System Timer OCTS                           | GPIO Pading Straps (strap 25) → SMC  | N/A                           |
| timer_count_o                | logic                | Output    | [63:0]                      | 64-bit system timer count output for Off-Chip Timer Synchronization (OCTS).     | System Timer OCTS → External systems | N/A                           |
| rst_cold_override_i          | logic                | Input     | 1                           | Override cold reset value (active low).                                         | Reset override control               | 1'b1                          |
| rst_cool_override_i          | logic                | Input     | 1                           | Override cool reset value (active low).                                         | Reset override control               | 1'b1                          |
| rst_core_override_i          | logic                | Input     | 1                           | Override core reset value (active low).                                         | Reset override control               | 1'b1                          |
| rst_cold_override_en_i       | logic                | Input     | 1                           | Enable cold reset override.                                                     | Reset override enable                | 1'b0                          |
| rst_cool_override_en_i       | logic                | Input     | 1                           | Enable cool reset override.                                                     | Reset override enable                | 1'b0                          |
| rst_core_override_en_i       | logic                | Input     | 1                           | Enable core reset override.                                                     | Reset override enable                | 1'b0                          |
| cla_gpio_enable_o            | logic                | Output    | 1                           | CLA GPIO enable output.                                                         | DFD controller → CLA                 | N/A                           |
| tdr_dbg_ctrl_clock_stop_en_i | logic                | Input     | 1                           | DEBUG_CONTROL CLA clock-stop enable.                                            | DTP → DFD                            | 1'b0                          |

| Interface Name                                    | Type                                            | Direction | Width                                   | Description                                                                                                                                                                    | Connection                                    | Default value (if applicable) |
|---------------------------------------------------|-------------------------------------------------|-----------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|-------------------------------|
| <code>tdr_dbg_ctrl_clocks_stopped_by_cla_o</code> | logic                                           | Output    | 1                                       | CLA halt-clock status under enable; also drives <code>cpu_interrupts_o[<a href="#">NUM_EXT_INTERRUPTS+64</a>]</code> (internal interrupt 0) — 1-core: bit 96, 4-core: bit 320. | DFD → DTP / interrupt                         | N/A                           |
| <code>halt_clock_global_i</code>                  | logic                                           | Input     | <a href="#">[5:0]</a>                   | Halt clock global input array.                                                                                                                                                 | External XTrigger agents → DFD controller     | 1'b0                          |
| <code>halt_clock_global_o</code>                  | logic                                           | Output    | 1                                       | Halt clock global output array.                                                                                                                                                | External XTrigger agents → DFD controller     | N/A, leave floating if unused |
| <code>xtrigger_pads_i</code>                      | <code>smc_pkg::xtrigger_t</code>                | Input     | <a href="#">[1:0]</a>                   | XTrigger pad interface input.                                                                                                                                                  | External pads → DFD controller                | N/A, tie to '0 if unused      |
| <code>xtrigger_pads_o</code>                      | <code>smc_pkg::xtrigger_t</code>                | Output    | <a href="#">[1:0]</a>                   | XTrigger pad interface output.                                                                                                                                                 | DFD controller → External pads                | N/A, leave floating if unused |
| <code>xtrigger_ss_i</code>                        | <code>smc_pkg::xtrigger_t</code>                | Input     | <a href="#">[5:0]</a>                   | XTrigger subsystem interface inputs.                                                                                                                                           | External subsystems → DFD controller          | N/A, tie to '0 if unused      |
| <code>xtrigger_ss_o</code>                        | <code>smc_pkg::xtrigger_t</code>                | Output    | -                                       | XTrigger subsystem interface output.                                                                                                                                           | DFD controller → External subsystems          | N/A, leave floating if unused |
| <code>trace_mem_req_o</code>                      | <code>dfd_trace_mem_pkg::SinkMemPktIn_s</code>  | Output    | <a href="#">[TRC_RAM_INSTANCES-1:0]</a> | Trace memory interface requests.                                                                                                                                               | DFD controller → Trace memory                 | N/A, leave floating if unused |
| <code>trace_mem_resp_i</code>                     | <code>dfd_trace_mem_pkg::SinkMemPktOut_s</code> | Input     | <a href="#">[TRC_RAM_INSTANCES-1:0]</a> | Trace memory interface responses.                                                                                                                                              | Trace memory → DFD controller                 | N/A, tie to '0 if unused      |
| <code>ext_debug_bus_i</code>                      | logic                                           | Input     | <a href="#">[511:0]</a>                 | External debug bus input (512 bits, 16-bit aligned).                                                                                                                           | External debug sources → Debug bus aggregator | N/A, tie to '0 if unused      |
| <code>test_en_i</code>                            | logic                                           | Input     | 1                                       | DFT/scan test-enable. Drives clock-gater test ports and AXI cell test inputs.                                                                                                  | Test controller → All modules                 | 1'b0                          |
| <code>scan_rst_ni</code>                          | logic                                           | Input     | 1                                       | Scan reset (active-low). Bypasses reset synchronizers during DFT.                                                                                                              | Test controller → All modules                 | 1'b1                          |

## 1.2.12. GPIO Pin Requirements

The SMC provides GPIO functionality with multiplexed peripheral functions and test interfaces. The following table lists the signal and bump location, assigned function, recommended pull, and GPIO-specific notes. It does not define default direction or decode configuration straps; use `smc_pading_pkg::DefaultDirectionMap` and the generated reset-unit `STRAPS_LO` and `STRAPS_HI` register descriptions for those.

Table 5. OCAH GPIO Requirements

| Index | Signal                               | Bump Loc | Function    | Recommended Pull | GPIO Notes |
|-------|--------------------------------------|----------|-------------|------------------|------------|
| 0     | <code>smc_wrapper.GPIO_PAD[0]</code> | SMC      | SPI.DATA[0] | PD               |            |
| 1     | <code>smc_wrapper.GPIO_PAD[1]</code> | SMC      | SPI.DATA[1] | PD               |            |
| 2     | <code>smc_wrapper.GPIO_PAD[2]</code> | SMC      | SPI.DATA[2] | PD               |            |
| 3     | <code>smc_wrapper.GPIO_PAD[3]</code> | SMC      | SPI.DATA[3] | PD               |            |
| 4     | <code>smc_wrapper.GPIO_PAD[4]</code> | SMC      | SPI.DATA[4] | PD               |            |

| Index | Signal                   | Bump Loc | Function    | Recommended Pull | GPIO Notes |
|-------|--------------------------|----------|-------------|------------------|------------|
| 5     | smc_wrapper.GPIO_PAD[5]  | SMC      | SPI.DATA[5] | PD               |            |
| 6     | smc_wrapper.GPIO_PAD[6]  | SMC      | SPI.DATA[6] | PD               |            |
| 7     | smc_wrapper.GPIO_PAD[7]  | SMC      | SPI.DATA[7] | PD               |            |
| 8     | smc_wrapper.GPIO_PAD[8]  | SMC      | SPI.CS      | PU               |            |
| 9     | smc_wrapper.GPIO_PAD[9]  | SMC      | SPI.CLK     | PD               |            |
| 10    | smc_wrapper.GPIO_PAD[10] | SMC      | SPI.DQS     | PD               |            |
| 11    | smc_wrapper.GPIO_PAD[11] | SMC      | UART[0].RX  | PU               |            |
| 12    | smc_wrapper.GPIO_PAD[12] | SMC      | UART[0].TX  | PU               |            |
| 13    | smc_wrapper.GPIO_PAD[13] | SMC      | UART[0].RTS | PD               |            |
| 14    | smc_wrapper.GPIO_PAD[14] | SMC      | UART[0].CTS | PD               |            |
| 15    | smc_wrapper.GPIO_PAD[15] | SMC      | UART[1].RX  | PU               |            |
| 16    | smc_wrapper.GPIO_PAD[16] | SMC      | UART[1].TX  | PU               |            |
| 17    | smc_wrapper.GPIO_PAD[17] | SMC      | UART[1].RTS | PD               |            |
| 18    | smc_wrapper.GPIO_PAD[18] | SMC      | UART[1].CTS | PD               |            |
| 19    | smc_wrapper.GPIO_PAD[19] | SMC      | UART[2].RX  | PU               |            |
| 20    | smc_wrapper.GPIO_PAD[20] | SMC      | UART[2].TX  | PU               |            |
| 21    | smc_wrapper.GPIO_PAD[21] | SMC      | UART[2].RTS | PD               |            |
| 22    | smc_wrapper.GPIO_PAD[22] | SMC      | UART[2].CTS | PD               |            |
| 23    | smc_wrapper.GPIO_PAD[23] | SMC      | UART[3].RX  | PU               |            |
| 24    | smc_wrapper.GPIO_PAD[24] | SMC      | UART[3].TX  | PU               |            |
| 25    | smc_wrapper.GPIO_PAD[25] | SMC      | UART[3].RTS | PD               |            |
| 26    | smc_wrapper.GPIO_PAD[26] | SMC      | UART[3].CTS | PD               |            |
| 27    | smc_wrapper.GPIO_PAD[27] | SMC      | I3C[0] SCL  | PU               |            |
| 28    | smc_wrapper.GPIO_PAD[28] | SMC      | I3C[0] SDA  | PU               |            |
| 29    | smc_wrapper.GPIO_PAD[29] | SMC      | I3C[2] SCL  | PU               |            |

| Index | Signal                   | Bump Loc | Function            | Recommended Pull | GPIO Notes                                                |
|-------|--------------------------|----------|---------------------|------------------|-----------------------------------------------------------|
| 30    | smc_wrapper.GPIO_PAD[30] | SMC      | I3C[2] SDA          | PU               |                                                           |
| 31    | smc_wrapper.GPIO_PAD[31] | SMC      | I3C[3] SCL          | PU               |                                                           |
| 32    | smc_wrapper.GPIO_PAD[32] | SMC      | I3C[3] SDA          | PU               |                                                           |
| 33    | smc_wrapper.GPIO_PAD[33] | SMC      | I3C[4] SCL          | PU               |                                                           |
| 34    | smc_wrapper.GPIO_PAD[34] | SMC      | I3C[4] SDA          | PU               |                                                           |
| 35    | smc_wrapper.GPIO_PAD[35] | SMC      | I3C[5] SCL          | PU               |                                                           |
| 36    | smc_wrapper.GPIO_PAD[36] | SMC      | I3C[5] SDA          | PU               |                                                           |
| 37    | smc_wrapper.GPIO_PAD[37] | SMC      | I2C[0] SCL          | PU               | Every I2C can be programmed to act as I2C, SMBus or PMBus |
| 38    | smc_wrapper.GPIO_PAD[38] | SMC      | I2C[0] SDA          | PU               |                                                           |
| 39    | smc_wrapper.GPIO_PAD[39] | SMC      | I2C[0] SMB_A        | PU               | SMBus Alert                                               |
| 40    | smc_wrapper.GPIO_PAD[40] | SMC      | I2C[0] SMB_D        | PU               | SMBus Suspend                                             |
| 41    | smc_wrapper.GPIO_PAD[41] | SMC      | I2C[1] SCL          | PU               |                                                           |
| 42    | smc_wrapper.GPIO_PAD[42] | SMC      | I2C[1] SDA          | PU               |                                                           |
| 43    | smc_wrapper.GPIO_PAD[43] | SMC      | I2C[1] SMB_A        | PU               | SMBus Alert                                               |
| 44    | smc_wrapper.GPIO_PAD[44] | SMC      | I2C[1] SMB_D        | PU               | SMBus Suspend                                             |
| 45    | smc_wrapper.GPIO_PAD[45] | SMC      | I2C[2] SCL          | PU               |                                                           |
| 46    | smc_wrapper.GPIO_PAD[46] | SMC      | I2C[2] SDA          | PU               |                                                           |
| 47    | smc_wrapper.GPIO_PAD[47] | SMC      | I2C[2] SMB_A        | PU               | SMBus Alert                                               |
| 48    | smc_wrapper.GPIO_PAD[48] | SMC      | I2C[2] SMB_D        | PU               | SMBus Suspend                                             |
| 49    | smc_wrapper.GPIO_PAD[49] | SMC      | AVS.CLOCK           | PU               |                                                           |
| 50    | smc_wrapper.GPIO_PAD[50] | SMC      | AVS.MDATA           | PU               |                                                           |
| 51    | smc_wrapper.GPIO_PAD[51] | SMC      | AVS.SDATA           | PU               |                                                           |
| 52    | smc_wrapper.GPIO_PAD[52] | SMC      | Thermal Emergency N | PU               |                                                           |
| 53    | smc_wrapper.GPIO_PAD[53] | SMC      | Isolate Request     | PD               |                                                           |
| 54    | smc_wrapper.GPIO_PAD[54] | SMC      | SPI DQS Loopback    | PD               |                                                           |

| Index | Signal                         | Bump Loc | Function                  | Recommended Pull | GPIO Notes                                                                                                                                                                                                                                       |
|-------|--------------------------------|----------|---------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 55    | smc_wrapper.GPIO_PAD[55]       | SMC      | System Timer Sync Load    | PD               |                                                                                                                                                                                                                                                  |
| 56    | smc_wrapper.GPIO_PAD[56]       | SMC      | System Timer Credit Count | PD               |                                                                                                                                                                                                                                                  |
| 57    | smc_wrapper.GPIO_PAD[57]       | SMC      | Boot Stall                | PD               | Stall after fuse sensing to override shadow register values. Boot will resume when this GPIO is released. This value can be overridden by the value specified over JTAG using the boot_stall_jtag_ovrd and boot_stall_jtag_val signals from DTP. |
| 58    | smc_wrapper.GPIO_PAD[58]       | SMC      | OCCP Interface            | PD               |                                                                                                                                                                                                                                                  |
| 59    | smc_wrapper.GPIO_PAD[59]       | SMC      | Reserved                  | PD               |                                                                                                                                                                                                                                                  |
| 60    | smc_wrapper.GPIO_PAD[60]       | SMC      | Reserved                  | PD               |                                                                                                                                                                                                                                                  |
| 61    | smc_wrapper.UNBONDED_GPIO[61]  | SMC      | Cool Reset In             | PU               | Not packaged out                                                                                                                                                                                                                                 |
| 62    | smc_wrapper.UNBONDED_GPIO[62]  | SMC      | Cool Reset Out            | PU               | Not packaged out                                                                                                                                                                                                                                 |
| 63    | smc_wrapper.UNBONDED_GPIO[63]  | SMC      | I3C[1] SCL                | PU               | Not packaged out                                                                                                                                                                                                                                 |
| 64    | smc_wrapper.UNBONDED_GPIO[64]  | SMC      | I3C[1] SDA                | PU               | Not packaged out                                                                                                                                                                                                                                 |
| 65    | smc_wrapper.PAD_JTAG_IN_TCK    | ?        | PTAP TCK                  |                  |                                                                                                                                                                                                                                                  |
| 66    | smc_wrapper.PAD_JTAG_IN_TRSTN  | ?        | PTAP TRST                 |                  |                                                                                                                                                                                                                                                  |
| 67    | smc_wrapper.PAD_JTAG_IN_TDI    | ?        | PTAP TDI                  |                  |                                                                                                                                                                                                                                                  |
| 68    | smc_wrapper.PAD_JTAG_IN_TMS    | ?        | PTAP TMS                  |                  |                                                                                                                                                                                                                                                  |
| 69    | smc_wrapper.PAD_JTAG_IN_TDO    | ?        | PTAP TDO                  |                  |                                                                                                                                                                                                                                                  |
| 70    | smc_wrapper.PAD_JTAG_OUT_TCK   | ?        | STAP TCK                  |                  |                                                                                                                                                                                                                                                  |
| 71    | smc_wrapper.PAD_JTAG_OUT_TRSTN | ?        | STAP TRST                 |                  |                                                                                                                                                                                                                                                  |
| 72    | smc_wrapper.PAD_JTAG_OUT_TDI   | ?        | STAP TDI                  |                  |                                                                                                                                                                                                                                                  |
| 73    | smc_wrapper.PAD_JTAG_OUT_TMS   | ?        | STAP TMS                  |                  |                                                                                                                                                                                                                                                  |
| 74    | smc_wrapper.PAD_JTAG_OUT_TDO   | ?        | STAP TDO                  |                  |                                                                                                                                                                                                                                                  |
| 75    | smc_wrapper.PAD_REFCLK         | SMC      | BP_REFCLK_VDD_SYS         |                  |                                                                                                                                                                                                                                                  |
| 76    | smc_wrapper.PAD_PRSTN          | SMC      | BP_RESETN                 |                  |                                                                                                                                                                                                                                                  |
| 77    | smc_wrapper.PAD_POWERGOOD      | SMC      | BP_POWERGOOD              |                  |                                                                                                                                                                                                                                                  |
| 78    |                                | SEP      | OTP[0]                    |                  |                                                                                                                                                                                                                                                  |
| 79    |                                | SEP      | OTP[1]                    |                  |                                                                                                                                                                                                                                                  |

| Index | Signal                      | Bump Loc | Function      | Recommended Pull | GPIO Notes                |
|-------|-----------------------------|----------|---------------|------------------|---------------------------|
| 80    |                             | SEP      | OTP[2]        |                  |                           |
| 81    |                             | SEP      | OTP[3]        |                  |                           |
| 82    |                             | SEP      | OTP[4]        |                  |                           |
| 83    |                             | SEP      | OTP[5]        |                  |                           |
| 84    |                             | SMC      | PVT[0]        |                  |                           |
| 85    |                             | SMC      | PVT[1]        |                  |                           |
| 86    |                             |          | VDDIO         |                  |                           |
| 87    | smc_wrapper.PLL_CLK_OBS_PAD | SMC      | PLL clock OBS | PD               | PLL clock observation pin |
| 88    | smc_wrapper.PVT_CLK_OBS_PAD | SMC      | PVT RO        | PD               | Process sensor RO output  |
| 89    |                             |          |               |                  |                           |

For multi-chiplet systems, the **PRIMARY\_CHIPLET** strap (**STRAPS\_LO[25]**) determines whether the chiplet operates as primary or secondary. The primary chiplet's System Timer OCTS generates synchronization pulses (**timer\_sync\_load\_o**, **timer\_cnt\_credit\_o**) that secondary chiplets receive to maintain coherent timestamps across the system.

## 1.3. Debug & Test Ports (DTP) Integration

### 1.3.1. Configuration

The cross-trigger topology is fixed at elaboration by **ntp\_pkg**, a committed source file at **hw/sys/ntp/rtl/ntp\_pkg.sv**. **DEFAULT\_NUM\_CTP**, **DEFAULT\_NUM\_INT\_CT**, and **DEFAULT\_NUM\_CLK\_STOP\_REQ** there set the **XTRIG\_NUM\_CTP**, **XTRIG\_NUM\_INT\_CT**, and **XTRIG\_NUM\_CLK\_STOP\_REQ** localparams of **ntp**, so a different port count is applied by editing that package rather than by overriding parameters at the instantiation. The remaining DTP parameters, including **XTRIG\_INT\_CT\_MODE** and every **JTAG\_\*** enable, are set at the instantiation as shown below.

Include the DTP in your build using Bender target **ntp**, which pulls in the DTP RTL together with the JTAG and cross-trigger IPs it instantiates. The legacy filelist **hw/sys/ntp/rtl/config/ntp.f** covers only the JTAG portion of the hierarchy and is not a complete DTP recipe.

### 1.3.2. Instantiation

```
ntp #(
    // JTAG Configuration
    .JTAG_BSR_ENABLE      (1'b1),
    .JTAG_TMP_ENABLE      (1'b1),
    .JTAG_SMC_DBG_ENABLE  (1'b1),
    .JTAG_SEP_DBG_ENABLE  (1'b1),
    .JTAG_IDCODE_MFR_ID   (11'h15A),
    .JTAG_IDCODE_PART_NUM (16'h1234),
    .JTAG_IDCODE_SI_REV   (4'h1),
    // Cross Trigger Configuration
    .XTRIG_INT_CT_MODE    (10'b0000000000)
) u_ntp (
```

```
.clk_i          (clk),
.rst_n_i       (rst_n),
.pwr_on_rst_ni (pwr_on_rst_n),
// JTAG, Boundary Scan, Cross Trigger interfaces...
);
```

### 1.3.3. Clock and Reset

- **TCK Domain:** JTAG logic operates on external Test Clock
- **System Clock** (`clk_i`): Cross trigger logic and JTAG2AXI bridges
- **System Reset** (`rst_n_i`): Resets cross trigger and AXI; does not affect TAP controller
- **Power-On Reset** (`pwr_on_rst_ni`): Asynchronous reset for JTAG state machine

### 1.3.4. Constraints

- JTAG interface: 10-50 MHz TCK frequency
- Cross trigger signals synchronized to system clock
- DTP remains active during clock stop for debug access
- External CTP signals require pad ring integration with ESD protection

### 1.3.5. STAP Scan Chain Topology

The DTP secondary scan path chains multiple STAPs on a single PTAP-managed scan string. The fixed order on this string is: PTAP, I/O STAP (if `JTAG_STAP_IO_ENABLE=1`), SMC debug STAP (if `JTAG_SMC_DBG_ENABLE=1`), SEP debug STAP (if `SEP=1` in the SMU), extra STAP[N-1:0], and return to PTAP.

Both the SMC and SEP debug STAP host interfaces are internal to the SMU and are not exported on the SMU boundary. The SMU wires the SMC debug STAP host outputs to the SMC CPU/system JTAG ports (`smc_cpu_jtag_*`) when `JTAG_SMC_DBG_ENABLE=1`; when disabled, the DTP bypasses this STAP in the secondary scan chain and drives the internal host outputs low. The SEP debug STAP connects to the SEP JTAG debug port when `SEP=1`; when `SEP=0`, the SMU ties `sep_stap_tdo_to_dtp = 1'b0` to bypass the SEP STAP return path.

Each STAP host interface is gated by lifecycle feature controls, synchronized to TCK inside the DTP. `feat_ctrl_i` is enable-polarity (1 = feature enabled), so a STAP host interface is enabled only when all of its required enables are asserted and is disabled when any one of them is deasserted.

| STAP           | Enabled When                                                                              | Host Gating Disable Condition                                                          |
|----------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| SMC debug STAP | <code>feat_ctrl_i.soc_debug &amp; feat_ctrl_i.ap_debug</code>                             | <code>!feat_ctrl_i.soc_debug    !feat_ctrl_i.ap_debug</code>                           |
| SEP debug STAP | <code>feat_ctrl_i.sep_debug &amp; feat_ctrl_i.soc_debug &amp; feat_ctrl_i.ap_debug</code> | <code>!feat_ctrl_i.sep_debug    !feat_ctrl_i.soc_debug    !feat_ctrl_i.ap_debug</code> |

The full set of lifecycle-gated debug paths, including the iJTAG chains and the JTAG2AXI bridges, is tabulated in the DTP JTAG integration section of the TRM.

### 1.3.6. DTP Port Interface

The DTP exposes interfaces for JTAG TAP control, boundary scan, multiple STAP host ports, iJTAG scan chains, AXI debug bridges, cross trigger networks, and CTP GPIO pads. The following table defines all DTP top-level ports. Refer to the dedicated DTP documentation in <hw/sys/dtp/doc/> for internal design details.

Table 6. DTP Port Declaration

| Interface Name                | Type                                          | Direction | Width | Description                                                                                        | Connection           | Default value (if applicable)      |
|-------------------------------|-----------------------------------------------|-----------|-------|----------------------------------------------------------------------------------------------------|----------------------|------------------------------------|
| clk_i                         | logic                                         | Input     | 1     | System clock for cross trigger logic and JTAG2AXI bridges.                                         | System clock         | N/A                                |
| rst_n_i                       | logic                                         | Input     | 1     | System reset (active low). Resets cross trigger and AXI logic; does not affect the TAP controller. | System reset         | N/A                                |
| pwr_on_rst_ni                 | logic                                         | Input     | 1     | Power-on reset for JTAG state machine (active low, asynchronous).                                  | Power-on reset       | N/A                                |
| feat_ctrl_i                   | sep_efuse_pkg::sep_efuse_map_lc_disable_reg_t | Input     | -     | Lifecycle feature control vector gating debug interfaces. Enable-polarity (1 = feature enabled).   | Lifecycle controller | N/A                                |
| jtag_ptap_client_tap_ctrl_i   | jtag_tap_ctrl_t                               | Input     | -     | Primary JTAG TAP control (TCK, TMS, TRST_N).                                                       | External JTAG pads   | N/A                                |
| jtag_ptap_client_tdi_i        | logic                                         | Input     | 1     | Primary JTAG TDI.                                                                                  | External JTAG pads   | N/A                                |
| jtag_ptap_client_tdo_o        | logic                                         | Output    | 1     | Primary JTAG TDO.                                                                                  | External JTAG pads   | N/A                                |
| jtag_ptap_client_tdo_oen_o    | logic                                         | Output    | 1     | Primary JTAG TDO output enable.                                                                    | External JTAG pads   | N/A                                |
| jtag_bsr_host_scan_ctrl_o     | jtag_scan_ctrl_t                              | Output    | -     | Boundary scan control. Tie scan_in to scan_out if BSR is disabled.                                 | Boundary scan chain  | N/A                                |
| jtag_bsr_host_scan_in_i       | logic                                         | Input     | 1     | Boundary scan data in.                                                                             | Boundary scan chain  | N/A                                |
| jtag_bsr_host_scan_out_o      | logic                                         | Output    | 1     | Boundary scan data out.                                                                            | Boundary scan chain  | N/A                                |
| jtag_stap_io_host_tap_ctrl_o  | jtag_tap_ctrl_t                               | Output    | -     | I/O STAP TAP control for chiplet-to-chiplet connectivity.                                          | I/O STAP host        | Tie to '0 if JTAG_STAP_IO_ENABLE=0 |
| jtag_stap_io_host_tdi_i       | logic                                         | Input     | 1     | I/O STAP TDI.                                                                                      | I/O STAP host        | Tie to '0 if unused                |
| jtag_stap_io_host_tdo_o       | logic                                         | Output    | 1     | I/O STAP TDO.                                                                                      | I/O STAP host        | N/A                                |
| jtag_stap_io_host_tdo_oen_o   | logic                                         | Output    | 1     | I/O STAP TDO output enable.                                                                        | I/O STAP host        | N/A                                |
| jtag_stap_smc_host_tap_ctrl_o | jtag_tap_ctrl_t                               | Output    | -     | SMC debug STAP TAP control.                                                                        | SMC JTAG port        | Tie to '0 if JTAG_SMC_DBG_ENABLE=0 |
| jtag_stap_smc_host_tdi_i      | logic                                         | Input     | 1     | SMC debug STAP TDI (from SMC TDO).                                                                 | SMC JTAG port        | Tie to '0 if unused                |
| jtag_stap_smc_host_tdo_o      | logic                                         | Output    | 1     | SMC debug STAP TDO (to SMC TDI).                                                                   | SMC JTAG port        | N/A                                |
| jtag_stap_smc_host_tdo_oen_o  | logic                                         | Output    | 1     | SMC debug STAP TDO output enable.                                                                  | SMC JTAG port        | N/A                                |
| jtag_stap_sep_host_tap_ctrl_o | jtag_tap_ctrl_t                               | Output    | -     | SEP debug STAP TAP control.                                                                        | SEP JTAG port        | Tie to '0 if JTAG_SEP_DBG_ENABLE=0 |
| jtag_stap_sep_host_tdi_i      | logic                                         | Input     | 1     | SEP debug STAP TDI (from SEP TDO).                                                                 | SEP JTAG port        | Tie to '0 if unused                |

| Interface Name                   | Type                | Direction | Width                           | Description                                                           | Connection                   | Default value (if applicable)       |
|----------------------------------|---------------------|-----------|---------------------------------|-----------------------------------------------------------------------|------------------------------|-------------------------------------|
| jtag_stap_sep_host_tdo_o         | logic               | Output    | 1                               | SEP debug STAP TDO (to SEP TDI).                                      | SEP JTAG port                | N/A                                 |
| jtag_stap_sep_host_tdo_oen_o     | logic               | Output    | 1                               | SEP debug STAP TDO output enable.                                     | SEP JTAG port                | N/A                                 |
| jtag_stap_extra_host_tap_ctrl_o  | jtag_tap_ctrl_t     | Output    | [JTAG_NUM_EXTRA_STAP_PORTS-1:0] | Extra STAP TAP control array.                                         | Extra STAP hosts             | Tie to '0 if JTAG_NUM_EXTRA_STAPS=0 |
| jtag_stap_extra_host_tdi_i       | logic               | Input     | [JTAG_NUM_EXTRA_STAP_PORTS-1:0] | Extra STAP TDI array.                                                 | Extra STAP hosts             | Tie to '0 if unused                 |
| jtag_stap_extra_host_tdo_o       | logic               | Output    | [JTAG_NUM_EXTRA_STAP_PORTS-1:0] | Extra STAP TDO array.                                                 | Extra STAP hosts             | N/A                                 |
| jtag_stap_extra_host_tdo_oen_o   | logic               | Output    | [JTAG_NUM_EXTRA_STAP_PORTS-1:0] | Extra STAP TDO output enable array.                                   | Extra STAP hosts             | N/A                                 |
| jtag_stap_host_scan_ctrl_o       | jtag_scan_ctrl_t    | Output    | -                               | Extended STAP scan control. Tie scan_in to scan_out if unused.        | STAP scan chain              | N/A                                 |
| jtag_stap_host_scan_in_i         | logic               | Input     | 1                               | Extended STAP scan data in.                                           | STAP scan chain              | N/A                                 |
| jtag_stap_host_scan_out_o        | logic               | Output    | 1                               | Extended STAP scan data out.                                          | STAP scan chain              | N/A                                 |
| jtag_dfd_host_scan_ctrl_o        | jtag_scan_ctrl_t    | Output    | -                               | DFD iJTAG scan control. Tie scan_in to scan_out if unused.            | DFD iJTAG chain              | N/A                                 |
| jtag_dfd_host_scan_in_i          | logic               | Input     | 1                               | DFD iJTAG scan data in.                                               | DFD iJTAG chain              | N/A                                 |
| jtag_dfd_host_scan_out_o         | logic               | Output    | 1                               | DFD iJTAG scan data out.                                              | DFD iJTAG chain              | N/A                                 |
| jtag_dft_secure_host_scan_ctrl_o | jtag_scan_ctrl_t    | Output    | -                               | Secure DFT iJTAG scan control. Tie scan_in to scan_out if unused.     | Secure DFT iJTAG chain       | N/A                                 |
| jtag_dft_secure_host_scan_in_i   | logic               | Input     | 1                               | Secure DFT iJTAG scan data in.                                        | Secure DFT iJTAG chain       | N/A                                 |
| jtag_dft_secure_host_scan_out_o  | logic               | Output    | 1                               | Secure DFT iJTAG scan data out.                                       | Secure DFT iJTAG chain       | N/A                                 |
| jtag_dft_host_scan_ctrl_o        | jtag_scan_ctrl_t    | Output    | -                               | Non-secure DFT iJTAG scan control. Tie scan_in to scan_out if unused. | Non-secure DFT iJTAG chain   | N/A                                 |
| jtag_dft_host_scan_in_i          | logic               | Input     | 1                               | Non-secure DFT iJTAG scan data in.                                    | Non-secure DFT iJTAG chain   | N/A                                 |
| jtag_dft_host_scan_out_o         | logic               | Output    | 1                               | Non-secure DFT iJTAG scan data out.                                   | Non-secure DFT iJTAG chain   | N/A                                 |
| axi_smc_dbg_req_o                | smc_jtag_axi_req_t  | Output    | -                               | SMC fabric debug AXI request. Active when JTAG_SMC_DBG_ENABLE=1.      | SMC jtag_axi_in_req_i        | Tie to '0 if disabled               |
| axi_smc_dbg_resp_i               | smc_jtag_axi_resp_t | Input     | -                               | SMC fabric debug AXI response.                                        | SMC jtag_axi_in_resp_o       | Tie to '0 if disabled               |
| axil_smc_otp_jtag_req_o          | smc_otp_axil_req_t  | Output    | -                               | SMC OTP debug AXI-Lite request.                                       | SMC axil_smc_otp_jtag_req_i  | Tie to '0 if disabled               |
| axil_smc_otp_jtag_resp_i         | smc_otp_axil_resp_t | Input     | -                               | SMC OTP debug AXI-Lite response.                                      | SMC axil_smc_otp_jtag_resp_o | Tie to '0 if disabled               |
| axil_sep_otp_jtag_req_o          | sep_otp_axil_req_t  | Output    | -                               | SEP OTP debug AXI-Lite request.                                       | SEP axil_sep_otp_jtag_req_i  | Tie to '0 if disabled               |

| Interface Name              | Type                       | Direction | Width                                        | Description                                                                                                                                                                                                                                                               | Connection                                                   | Default value (if applicable) |
|-----------------------------|----------------------------|-----------|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|-------------------------------|
| axil_sep_otp_jtag_resp_i    | sep_otp_axil_resp_t        | Input     | -                                            | SEP OTP debug AXI-Lite response.                                                                                                                                                                                                                                          | SEP <a href="#">axil_sep_otp_jtag_resp_o</a>                 | Tie to '0 if disabled         |
| stop_clks_o                 | logic                      | Output    | 1                                            | Clock stop output from JTAG DEBUG_CONTROL or CLA clock-stop requests via CTN.                                                                                                                                                                                             | PLL clock gates                                              | N/A                           |
| cla_clock_stop_en_o         | logic                      | Output    | 1                                            | CLA clock stop enable from DEBUG_CONTROL register.                                                                                                                                                                                                                        | Clock control logic                                          | N/A                           |
| jtag_boot_stall_ovrd_o      | logic                      | Output    | 1                                            | JTAG boot stall override enable.                                                                                                                                                                                                                                          | Boot control                                                 | N/A                           |
| jtag_boot_stall_o           | logic                      | Output    | 1                                            | JTAG boot stall value when override is asserted.                                                                                                                                                                                                                          | Boot control                                                 | N/A                           |
| jtag_ic_reset_smc_o         | ic_reset_smc_t             | Output    | <a href="#">\$bits(ic_reset_smc_t)</a>       | SMC IC_RESET override slice. Struct contains <a href="#">.ovrd</a> (override enables) and <a href="#">.val</a> (active-low override values), each sized to match the corresponding SMC reset control struct. Tied to '0 when <a href="#">JTAG_IC_RESET_SMC_ENABLE=0</a> . | SMC <a href="#">jtag_smc_reset_ctrl_i</a>                    | N/A                           |
| jtag_ic_reset_sep_o         | ic_reset_sep_t             | Output    | <a href="#">\$bits(ic_reset_sep_t)</a>       | SEP IC_RESET override slice. Struct contains <a href="#">.ovrd</a> (override enables) and <a href="#">.val</a> (active-low override values), each sized to match the corresponding SEP reset control struct. Tied to '0 when <a href="#">JTAG_IC_RESET_SEP_ENABLE=0</a> . | SEP <a href="#">jtag_sep_reset_ctrl_i</a>                    | N/A                           |
| jtag_ic_reset_ext_o         | ic_reset_ext_t             | Output    | <a href="#">\$bits(ic_reset_ext_t)</a>       | External IC_RESET override slice. Struct contains <a href="#">.ovrd</a> (override enables) and <a href="#">.val</a> (active-low override values), each sized per integrator-defined type. Tied to '0 when <a href="#">JTAG_IC_RESET_EXT_ENABLE=0</a> .                    | Integrator-defined reset overrides for logic outside the SMU | N/A                           |
| jtag_ptap_state_o           | tap_state_e                | Output    | -                                            | Current PTAP TAP state.                                                                                                                                                                                                                                                   | Debug infrastructure                                         | N/A                           |
| jtag_ptap_inst_decoded_o    | jtag_instruction_decoded_e | Output    | -                                            | Current decoded PTAP instruction.                                                                                                                                                                                                                                         | Debug infrastructure                                         | N/A                           |
| axil_xtrig_req_i            | xtrig_axil_req_t           | Input     | -                                            | Cross trigger CSR AXI-Lite request.                                                                                                                                                                                                                                       | SMC <a href="#">axil_dtp_csr_req_o</a>                       | N/A                           |
| axil_xtrig_resp_o           | xtrig_axil_resp_t          | Output    | -                                            | Cross trigger CSR AXI-Lite response.                                                                                                                                                                                                                                      | SMC <a href="#">axil_dtp_csr_resp_i</a>                      | N/A                           |
| xtrig_ctm_src_req_o         | logic                      | Output    | <a href="#">[XTRIG_NUM_IN_T_CT-1:0]</a>      | Cross trigger source requests from CTM to destination sinks.                                                                                                                                                                                                              | Cross trigger network                                        | N/A                           |
| xtrig_ctm_src_ack_i         | logic                      | Input     | <a href="#">[XTRIG_NUM_IN_T_CT-1:0]</a>      | Cross trigger source acknowledges. Unused in pulse-sync mode (mode bit = 0).                                                                                                                                                                                              | Cross trigger network                                        | Tie to '0 in pulse-sync mode  |
| xtrig_ctm_dst_req_i         | logic                      | Input     | <a href="#">[XTRIG_NUM_IN_T_CT-1:0]</a>      | Cross trigger destination requests from source generators.                                                                                                                                                                                                                | Cross trigger network                                        | Tie to '0 if unused           |
| xtrig_ctm_dst_ack_o         | logic                      | Output    | <a href="#">[XTRIG_NUM_IN_T_CT-1:0]</a>      | Cross trigger destination acknowledges. Unused in pulse-sync mode (mode bit = 0).                                                                                                                                                                                         | Cross trigger network                                        | N/A                           |
| xtrig_clk_stop_req_i        | logic                      | Input     | <a href="#">[XTRIG_NUM_CLK_STOP_REQ-1:0]</a> | Clock stop requests from CLAs or other devices in the chiplet.                                                                                                                                                                                                            | Clock stop requestors                                        | Tie to '0 if unused           |
| xtrig_ctp_req_out_dout_o    | logic                      | Output    | <a href="#">[XTRIG_NUM_CTP_P-1:0]</a>        | CTP request-out data output (to GPIO pad ring).                                                                                                                                                                                                                           | GPIO pad ring                                                | N/A                           |
| xtrig_ctp_req_out_dout_en_o | logic                      | Output    | <a href="#">[XTRIG_NUM_CTP_P-1:0]</a>        | CTP request-out data output enable.                                                                                                                                                                                                                                       | GPIO pad ring                                                | N/A                           |
| xtrig_ctp_req_out_din_i     | logic                      | Input     | <a href="#">[XTRIG_NUM_CTP_P-1:0]</a>        | CTP request-out data input (from GPIO pad ring).                                                                                                                                                                                                                          | GPIO pad ring                                                | Tie to '0 if unused           |
| xtrig_ctp_req_out_din_en_o  | logic                      | Output    | <a href="#">[XTRIG_NUM_CTP_P-1:0]</a>        | CTP request-out data input enable.                                                                                                                                                                                                                                        | GPIO pad ring                                                | N/A                           |

| Interface Name              | Type  | Direction | Width               | Description                        | Connection    | Default value (if applicable) |
|-----------------------------|-------|-----------|---------------------|------------------------------------|---------------|-------------------------------|
| xtrig_ctp_req_in_dout_o     | logic | Output    | [XTRIG_NUM_CTP-1:0] | CTP request-in data output.        | GPIO pad ring | N/A                           |
| xtrig_ctp_req_in_dout_en_o  | logic | Output    | [XTRIG_NUM_CTP-1:0] | CTP request-in data output enable. | GPIO pad ring | N/A                           |
| xtrig_ctp_req_in_din_i      | logic | Input     | [XTRIG_NUM_CTP-1:0] | CTP request-in data input.         | GPIO pad ring | Tie to '0 if unused           |
| xtrig_ctp_req_in_din_en_o   | logic | Output    | [XTRIG_NUM_CTP-1:0] | CTP request-in data input enable.  | GPIO pad ring | N/A                           |
| xtrig_ctp_ack_in_dout_o     | logic | Output    | [XTRIG_NUM_CTP-1:0] | CTP ack-in data output.            | GPIO pad ring | N/A                           |
| xtrig_ctp_ack_in_dout_en_o  | logic | Output    | [XTRIG_NUM_CTP-1:0] | CTP ack-in data output enable.     | GPIO pad ring | N/A                           |
| xtrig_ctp_ack_in_din_i      | logic | Input     | [XTRIG_NUM_CTP-1:0] | CTP ack-in data input.             | GPIO pad ring | Tie to '0 if unused           |
| xtrig_ctp_ack_in_din_en_o   | logic | Output    | [XTRIG_NUM_CTP-1:0] | CTP ack-in data input enable.      | GPIO pad ring | N/A                           |
| xtrig_ctp_ack_out_dout_o    | logic | Output    | [XTRIG_NUM_CTP-1:0] | CTP ack-out data output.           | GPIO pad ring | N/A                           |
| xtrig_ctp_ack_out_dout_en_o | logic | Output    | [XTRIG_NUM_CTP-1:0] | CTP ack-out data output enable.    | GPIO pad ring | N/A                           |
| xtrig_ctp_ack_out_din_i     | logic | Input     | [XTRIG_NUM_CTP-1:0] | CTP ack-out data input.            | GPIO pad ring | Tie to '0 if unused           |
| xtrig_ctp_ack_out_din_en_o  | logic | Output    | [XTRIG_NUM_CTP-1:0] | CTP ack-out data input enable.     | GPIO pad ring | N/A                           |

## 1.4. Security Processor (SEP) Integration

The Security Processor (SEP) is the Root of Trust for the chiplet, providing a physically isolated execution environment for security-critical services such as secure boot, lifecycle control, device attestation, cryptographic acceleration, and key management. This section provides practical integration procedures for adopters integrating the SEP as part of the SMU or as a standalone out-of-context IP.

For detailed architectural documentation about the SEP internal design, refer to the dedicated SEP documentation located in [hw/sys/sep/doc/](#). The SEP hardware overview and interconnect fabric are described in [hw/sys/sep/doc/overview.adoc](#) and [hw/sys/sep/doc/fabric.adoc](#) respectively.

### 1.4.1. Module Overview

The `sep.sv` module provides the complete SEP functionality with logic-level interfaces. Technology-specific components — SRAM, ROM, and eFuse shims — are instantiated by the adopter outside SEP, following the same vendor-neutral philosophy used for the SMC.

Separately, SEP exposes interfaces for supplementary peripherals an adopter may add on rather than substitute: an external TRNG and an additional SPI controller. These are not technology-specific — SEP contains its own entropy source and SPI host — so the interfaces exist to supplement what SEP already provides, and may be left unused.

When integrated inside the SMU, the SEP's external interfaces are passed through the SMU boundary with a `sep_` prefix.

## 1.4.2. Clock and Reset Requirements

The SEP requires two clock inputs. Connect `clk_i` to the primary system clock driving all SEP internal logic. Connect `clk_wdt_i` to a 200 kHz clock source for the watchdog timer; this clock must be independent of the system clock so that the WDT can detect system clock failures.

The primary reset `rst_ni` follows the standard active-low, asynchronous assertion / synchronous deassertion convention. The `dbg_rstb_i` input provides a separate reset for the EL2 CPU debug module. The `wdt_rst_ni` input receives the watchdog reset generated by the SMC reset unit. The SEP outputs `wdt_timer_rst_req_o` when the WDT triggers a reset request.

## 1.4.3. AXI Interfaces

The SEP connects to the system through three AXI paths. The `smn_outbound_axi` master interface allows SEP firmware to access system resources outside the SEP domain, such as SMC registers or external memory. The `smn_inbound_axi` slave interface accepts requests from system masters requiring access to SEP-managed resources. The `sep_ext_to_smc_axi` output provides a dedicated AXI path for SEP-to-SMC transactions, connecting to the SMC's `sep_axi_in` port.

An additional AXI extension port (`axi_extension_axi`) enables adopters to attach custom peripherals to the SEP interconnect fabric without modifying the SEP core. When unused, tie the response input to return DECERR.

## 1.4.4. Memory Interfaces

All SEP memory macros are instantiated externally and connected through dedicated request/response interfaces. The `sep_cpu_tcm` interface provides tightly-coupled memory (ICCM and DCCM) for the VeeR EL2 CPU. The `sep_sram` interface connects scratch SRAM used for firmware data. The `sep_boot_rom` interface provides the BL0 ROM containing the first-stage bootloader.

The crypto subsystem requires additional memory interfaces. `sep_crypto_pka_imem_sram` and `sep_crypto_pka_dmem_sram` connect to the OTBN public key accelerator instruction and data memories respectively. `km_rom_mem` and `km_sram_mem` connect to the Key Manager's dedicated ROM and SRAM macros.

## 1.4.5. Cryptographic Subsystem Connections

The SEP includes an internal entropy source that requires a ring-oscillator sample clock connected to `entropy_rosc_sample_clk_i`. This clock must be asynchronous to `clk_i` and typically runs between 100 and 400 MHz.

For adopters providing an external TRNG, the `ext_trng_axil` AXI-Lite interface connects to the external TRNG's control registers, while the `ext_trng_axis` AXI-Stream interfaces carry entropy data. The `ext_trng_irq_i` input is routed to the PIC; `ext_trng_alarm_i` is reserved for RAS.

## 1.4.6. eFuse and Lifecycle

The SEP accesses its own eFuse bank through the `efuse_bank_ctrl` AXI-Lite interface and the `efuse_shim_command` custom interface, mirroring the pattern used by the SMC. The SEP must wait for the SMC fuse sense to complete (`smc_fuse_sense_done_i`) before initiating its own fuse operations, and it signals completion through `sep_fuse_sense_done_o`.

The lifecycle controller outputs `lc_state_o` (differentially encoded), `feat_ctrl_o` (feature control vector), `lc_sigint_err_o` (signal integrity error), and `security_disable_o`. These outputs gate debug access through the DTP and control feature availability across the chiplet. The `lcc_demote_state` outputs report lifecycle demotion events to the SMC.

An OTP debug AXI-Lite port (`axil_sep_otp_jtag`) connects to the DTP for JTAG access to SEP fuses. When SEP is disabled (`SEP=0` at the SMU level), the SMU instantiates an AXI-Lite error slave on this path.

### 1.4.7. SPI Host Interface

The SEP provides a SPI host controller through the `sep_io_spi` interface, supporting single and quad modes. The controller's own interrupt leaves the module as the `irq` field of `sep_io_spi_req_o`, so no separate interrupt output is needed.

The `spi_irq_i` input is a separate SPI interrupt request routed to the SEP PIC. Adopters that replace or supplement the internal SPI host with an external SPI controller connect that controller's interrupt here; adopters that use only the internal controller may loop back the `irq` field of `sep_io_spi_req_o` or, if a mux over multiple SPI sources is needed, drive the selected interrupt from their own integration logic. Tie the input low when unused.

### 1.4.8. Bring-Up Considerations

The SEP boot sequence begins after the SMC completes fuse sensing and releases the SEP reset. The SEP BL0 ROM firmware verifies the BL1 software image using non-volatile public keys before transferring control. Feature controls derived from secure boot authentication and lifecycle state govern what operations are permitted after boot.

The `sep_straps_i` input provides configuration straps for SEP-specific settings. The `rst_vec`, `jtag_id`, and `core_id` inputs must be tied to appropriate constants in the top-level integration.

### 1.4.9. SEP Port Interface

The SEP exposes interfaces for clocks, resets, JTAG debug, CPU control, memory macros, AXI bus connections, cryptographic subsystem, eFuse, lifecycle, SPI host, and mailbox. The following table defines all SEP top-level ports.

Table 7. SEP Port Declaration

| Interface Name                   | Type  | Direction | Width | Description                                                                             | Connection                 | Default value (if applicable) |
|----------------------------------|-------|-----------|-------|-----------------------------------------------------------------------------------------|----------------------------|-------------------------------|
| <code>clk_i</code>               | logic | Input     | 1     | Primary SEP clock. All SEP internal logic operates synchronously to this clock.         | System clock               | N/A                           |
| <code>clk_wdt_i</code>           | logic | Input     | 1     | 200 kHz clock for watchdog timer.                                                       | Low-frequency clock source | N/A                           |
| <code>rst_ni</code>              | logic | Input     | 1     | Primary reset (active low). Asynchronous assertion, synchronous deassertion.            | System reset               | N/A                           |
| <code>dbg_rstb_i</code>          | logic | Input     | 1     | EL2 debugger reset (active low).                                                        | JTAG debug infrastructure  | N/A                           |
| <code>wdt_rst_ni</code>          | logic | Input     | 1     | Watchdog timer reset from SMC (active low).                                             | SMC reset unit             | N/A                           |
| <code>wdt_timer_rst_req_o</code> | logic | Output    | 1     | WDT reset request output. Triggers memory interface reset in <code>sep_wrapper</code> . | SEP wrapper                | N/A                           |
| <code>jtag_tck</code>            | logic | Input     | 1     | JTAG test clock.                                                                        | JTAG pads (via DTP STAP)   | N/A                           |
| <code>jtag_tms</code>            | logic | Input     | 1     | JTAG test mode select.                                                                  | JTAG pads (via DTP STAP)   | N/A                           |
| <code>jtag_tdi</code>            | logic | Input     | 1     | JTAG test data in.                                                                      | JTAG pads (via DTP STAP)   | N/A                           |
| <code>jtag_trst_n</code>         | logic | Input     | 1     | JTAG test reset (active low).                                                           | JTAG pads (via DTP STAP)   | N/A                           |

| Interface Name           | Type                             | Direction | Width  | Description                                                                                               | Connection                      | Default value (if applicable) |
|--------------------------|----------------------------------|-----------|--------|-----------------------------------------------------------------------------------------------------------|---------------------------------|-------------------------------|
| jtag_tdo                 | logic                            | Output    | 1      | JTAG test data out.                                                                                       | JTAG pads (via DTP STAP)        | N/A                           |
| jtag_tdoEn               | logic                            | Output    | 1      | JTAG test data out enable.                                                                                | JTAG pads (via DTP STAP)        | N/A                           |
| axil_sep_otp_jtag_req_i  | sep_efuse_pkg::efuse_axil_req_t  | Input     | -      | OTP debug AXI-Lite request from JTAG.                                                                     | DTP axil_sep_otp_jtag_req_o     | Tie to '0 if unused           |
| axil_sep_otp_jtag_resp_o | sep_efuse_pkg::efuse_axil_resp_t | Output    | -      | OTP debug AXI-Lite response to JTAG.                                                                      | DTP axil_sep_otp_jtag_resp_i    | N/A                           |
| mpc_debug_halt_req       | logic                            | Input     | 1      | Async halt request from MPC.                                                                              | Debug controller                | Tie to '0 if unused           |
| mpc_debug_run_req        | logic                            | Input     | 1      | Async run request from MPC.                                                                               | Debug controller                | Tie to '0 if unused           |
| mpc_reset_run_req        | logic                            | Input     | 1      | Run/halt selection after reset.                                                                           | Debug controller                | 1'b1                          |
| i_cpu_halt_req           | logic                            | Input     | 1      | Async halt request to CPU.                                                                                | Debug controller                | Tie to '0 if unused           |
| i_cpu_run_req            | logic                            | Input     | 1      | Async restart request to CPU.                                                                             | Debug controller                | Tie to '0 if unused           |
| test_en_i                | logic                            | Input     | 1      | DFT/scan test-enable. Drives clock-gater test ports and AXI cell test inputs.                             | Test controller                 | 1'b0                          |
| scan_rst_ni              | logic                            | Input     | 1      | Scan reset (active-low). Bypasses reset synchronizers during DFT.                                         | Test controller                 | 1'b1                          |
| ext_boot_seq_done_i      | logic                            | Input     | 1      | External boot sequence completion (memory repair and shadow register override done); gates reset release. | External boot/repair controller | 1'b1                          |
| dmi_core_enable          | logic                            | Input     | 1      | DMI core enable.                                                                                          | Debug infrastructure            | Tie to '0 if unused           |
| dmi_uncore_enable        | logic                            | Input     | 1      | DMI uncore enable.                                                                                        | Debug infrastructure            | Tie to '0 if unused           |
| dmi_uncore_en            | logic                            | Output    | 1      | DMI uncore access enable.                                                                                 | Debug infrastructure            | N/A                           |
| dmi_uncore_wr_en         | logic                            | Output    | 1      | DMI uncore write enable.                                                                                  | Debug infrastructure            | N/A                           |
| dmi_uncore_addr          | logic                            | Output    | [6:0]  | DMI uncore address.                                                                                       | Debug infrastructure            | N/A                           |
| dmi_uncore_wdata         | logic                            | Output    | [31:0] | DMI uncore write data.                                                                                    | Debug infrastructure            | N/A                           |
| dmi_uncore_rdata         | logic                            | Input     | [31:0] | DMI uncore read data.                                                                                     | Debug infrastructure            | N/A                           |
| dmi_active               | logic                            | Output    | 1      | DMI active indicator.                                                                                     | Debug infrastructure            | N/A                           |
| sep_cpu_trace            | sep_cpu_trace_t                  | Output    | -      | CPU trace output for debug.                                                                               | Trace infrastructure            | N/A                           |
| jtag_id                  | logic                            | Input     | [31:1] | JTAG ID value.                                                                                            | Configuration tie-off           | N/A                           |
| core_id                  | logic                            | Input     | [31:4] | CPU core ID.                                                                                              | Configuration tie-off           | N/A                           |
| timer_int                | logic                            | Input     | 1      | Timer interrupt.                                                                                          | Internal timer                  | Tie to '0 if unused           |

| Interface Name               | Type                                       | Direction | Width                   | Description                                                                             | Connection               | Default value (if applicable) |
|------------------------------|--------------------------------------------|-----------|-------------------------|-----------------------------------------------------------------------------------------|--------------------------|-------------------------------|
| soft_int                     | logic                                      | Input     | 1                       | Software interrupt.                                                                     | Software IRQ source      | Tie to '0 if unused           |
| extintsrc_req                | logic                                      | Input     | [NUM_EXTERNAL_IRQS-1:0] | External interrupt sources for PIC.                                                     | External IRQ sources     | Tie to '0 if unused           |
| sep_cpu_tcm_req_o            | sep_cpu_tcm_req_t                          | Output    | -                       | CPU tightly-coupled memory (TCM) request (ICCM + DCCM).                                 | SRAM macros              | N/A                           |
| sep_cpu_tcm_rsp_i            | sep_cpu_tcm_rsp_t                          | Input     | -                       | CPU TCM response.                                                                       | SRAM macros              | N/A                           |
| sep_sram_req                 | sep_sram_req_t                             | Output    | -                       | Scratch SRAM request.                                                                   | SRAM macros              | N/A                           |
| sep_sram_rsp                 | sep_sram_rsp_t                             | Input     | -                       | Scratch SRAM response.                                                                  | SRAM macros              | N/A                           |
| sep_boot_rom_req             | sep_sram_req_t                             | Output    | -                       | Boot ROM (BL0) request.                                                                 | ROM macro                | N/A                           |
| sep_boot_rom_rsp             | sep_sram_rsp_t                             | Input     | -                       | Boot ROM (BL0) response.                                                                | ROM macro                | N/A                           |
| smn_outbound_axi_req_o       | sep_system_peripherals_outbound_axi_req_t  | Output    | -                       | SMN outbound AXI request (system bus master).                                           | SMN fabric               | N/A                           |
| smn_outbound_axi_resp_i      | sep_system_peripherals_outbound_axi_resp_t | Input     | -                       | SMN outbound AXI response.                                                              | SMN fabric               | N/A                           |
| smn_inbound_axi_req_i        | sep_system_peripherals_internal_axi_req_t  | Input     | -                       | SMN inbound AXI request (system bus slave).                                             | SMN fabric               | Tie to '0 if unused           |
| smn_inbound_axi_resp_o       | sep_system_peripherals_internal_axi_resp_t | Output    | -                       | SMN inbound AXI response.                                                               | SMN fabric               | N/A                           |
| sep_ext_to_smc_axi_req_o     | sep_system_peripherals_internal_axi_req_t  | Output    | -                       | SEP-to-SMC AXI request. Dedicated path for SEP to access SMC resources.                 | SMC<br>sep_axi_in_req_i  | N/A                           |
| sep_ext_to_smc_axi_resp_i    | sep_system_peripherals_internal_axi_resp_t | Input     | -                       | SEP-to-SMC AXI response.                                                                | SMC<br>sep_axi_in_resp_o | N/A                           |
| entropy_rosc_sample_clk_i    | logic                                      | Input     | 1                       | Ring-oscillator sample clock for entropy source (async vs clk_i; ~100-400 MHz typical). | Ring oscillator          | N/A                           |
| sep_crypto_pka_imem_sram_req | sep_crypto_pka_imem_sram_req_t             | Output    | -                       | OTBN PKA instruction memory SRAM request.                                               | SRAM macro               | N/A                           |
| sep_crypto_pka_imem_sram_rsp | sep_crypto_pka_imem_sram_rsp_t             | Input     | -                       | OTBN PKA instruction memory SRAM response.                                              | SRAM macro               | N/A                           |
| sep_crypto_pka_dmem_sram_req | sep_crypto_pka_dmem_sram_req_t             | Output    | -                       | OTBN PKA data memory SRAM request.                                                      | SRAM macro               | N/A                           |
| sep_crypto_pka_dmem_sram_rsp | sep_crypto_pka_dmem_sram_rsp_t             | Input     | -                       | OTBN PKA data memory SRAM response.                                                     | SRAM macro               | N/A                           |
| ext_trng_axil_req_o          | sep_32_32_axil_req_t                       | Output    | -                       | External TRNG AXI-Lite passthrough request (to sep_ip_integration).                     | External TRNG shim       | N/A                           |
| ext_trng_axil_resp_i         | sep_32_32_axil_resp_t                      | Input     | -                       | External TRNG AXI-Lite passthrough response.                                            | External TRNG shim       | N/A                           |
| ext_trng_axis_req_i          | ext_trng_axis_req_t                        | Input     | [EXT_TRNG_NUM_AXIS-1:0] | External TRNG AXI-Stream request array (from sep_ip_integration).                       | External TRNG shim       | N/A                           |
| ext_trng_axis_rsp_o          | ext_trng_axis_rsp_t                        | Output    | [EXT_TRNG_NUM_AXIS-1:0] | External TRNG AXI-Stream response array.                                                | External TRNG shim       | N/A                           |
| ext_trng_irq_i               | logic                                      | Input     | 1                       | External TRNG interrupt (routed to PIC).                                                | External TRNG shim       | Tie to '0 if unused           |
| ext_trng_alarm_i             | logic                                      | Input     | 1                       | External TRNG alarm (reserved for RAS).                                                 | External TRNG shim       | Tie to '0 if unused           |

| Interface Name            | Type                                          | Direction | Width                      | Description                                                                                         | Connection               | Default value (if applicable) |
|---------------------------|-----------------------------------------------|-----------|----------------------------|-----------------------------------------------------------------------------------------------------|--------------------------|-------------------------------|
| km_rom_mem_req_o          | km_intf_pkg::km_rom_mem_req_t                 | Output    | -                          | Key Manager ROM memory request.                                                                     | ROM macro                | N/A                           |
| km_rom_mem_rsp_i          | km_intf_pkg::km_rom_mem_rsp_t                 | Input     | -                          | Key Manager ROM memory response.                                                                    | ROM macro                | N/A                           |
| km_sram_mem_req_o         | km_intf_pkg::km_sram_mem_req_t                | Output    | -                          | Key Manager SRAM memory request.                                                                    | SRAM macro               | N/A                           |
| km_sram_mem_rsp_i         | km_intf_pkg::km_sram_mem_rsp_t                | Input     | -                          | Key Manager SRAM memory response.                                                                   | SRAM macro               | N/A                           |
| efuse_bank_ctrl_req_o     | sep_efuse_pkg::efuse_axi_req_t                | Output    | -                          | eFuse bank control AXI-Lite request.                                                                | eFuse shim               | N/A                           |
| efuse_bank_ctrl_resp_i    | sep_efuse_pkg::efuse_axi_resp_t               | Input     | -                          | eFuse bank control AXI-Lite response.                                                               | eFuse shim               | N/A                           |
| efuse_shim_command_req_o  | sep_efuse_pkg::fuse_command_req_t             | Output    | -                          | eFuse command interface request.                                                                    | eFuse shim               | N/A                           |
| efuse_shim_command_resp_i | sep_efuse_pkg::fuse_command_resp_t            | Input     | -                          | eFuse command interface response.                                                                   | eFuse shim               | N/A                           |
| lcc_demote_state_1_o      | logic                                         | Output    | [1:0]                      | Lifecycle demote state output 1 (to SMC).                                                           | SMC lifecycle logic      | N/A                           |
| lcc_demote_state_2_o      | logic                                         | Output    | [1:0]                      | Lifecycle demote state output 2 (to SMC).                                                           | SMC lifecycle logic      | N/A                           |
| sep_io_spi_req_o          | sep_io_spi_req_t                              | Output    | -                          | SPI host controller request.                                                                        | SPI pad interface        | N/A                           |
| sep_io_spi_rsp_i          | sep_io_spi_rsp_t                              | Input     | -                          | SPI host controller response.                                                                       | SPI pad interface        | N/A                           |
| spi_irq_i                 | logic                                         | Input     | 1                          | SPI interrupt request into the SEP PIC, driven by whichever SPI controller the integration selects. | SPI controller           | Tie to '0 if unused           |
| lc_state_o                | logic                                         | Output    | [2*LC_STATE_BIT_WIDTH-1:0] | Lifecycle state output (differentially encoded).                                                    | SMC, DTP                 | N/A                           |
| feat_ctrl_o               | sep_efuse_pkg::sep_efuse_map_lc_disable_reg_t | Output    | -                          | Lifecycle feature control vector.                                                                   | SMC, DTP                 | N/A                           |
| lc_sigint_err_o           | logic                                         | Output    | 1                          | Lifecycle signal integrity error.                                                                   | SMC                      | N/A                           |
| security_disable_o        | logic                                         | Output    | 1                          | Security disable output.                                                                            | SMC                      | N/A                           |
| smc_mailbox_interrupt_o   | logic                                         | Output    | [NUM_MAILBOXES-1:0]        | Mailbox interrupt outputs to SMC.                                                                   | SMC interrupt aggregator | N/A                           |
| smc_fuse_sense_done_i     | logic                                         | Input     | 1                          | SMC fuse sense done indication.                                                                     | SMC fuse_sense_done_o    | N/A                           |
| sep_fuse_sense_done_o     | logic                                         | Output    | 1                          | SEP fuse sense done output.                                                                         | System integration       | N/A                           |
| sep_straps_i              | sep_pkg::sep_straps_t                         | Input     | -                          | SEP configuration straps.                                                                           | GPIO padding / tie-off   | N/A                           |
| axi_extension_axi_req_o   | sep_32_64_6_12_axi_req_t                      | Output    | -                          | AXI extension port request for adopter peripheral expansion.                                        | Adopter peripherals      | N/A                           |
| axi_extension_axi_resp_i  | sep_32_64_6_12_axi_resp_t                     | Input     | -                          | AXI extension port response.                                                                        | Adopter peripherals      | Tie to DECERR if unused       |
| smc_global_base_addr_i    | logic                                         | Input     | [55:0]                     | SMC global base address for outbound address translation.                                           | System configuration     | N/A                           |
| smc_region_size_i         | logic                                         | Input     | [55:0]                     | SMC region size for outbound address translation.                                                   | SMC smc_region_size_o    | N/A                           |

| Interface Name                      | Type  | Direction | Width   | Description                      | Connection           | Default value (if applicable) |
|-------------------------------------|-------|-----------|---------|----------------------------------|----------------------|-------------------------------|
| <code>km_unrecoverable_err_o</code> | logic | Output    | 1       | Key Manager unrecoverable error. | RAS / error handling | N/A                           |
| <code>km_recoverable_err_o</code>   | logic | Output    | 1       | Key Manager recoverable error.   | RAS / error handling | N/A                           |
| <code>ext_debug_bus_o</code>        | logic | Output    | [511:0] | External debug observation bus.  | Debug infrastructure | N/A                           |

## 1.5. SMU Composition and Internal Wiring

The System Management Unit (SMU) combines the SMC, DTP, and optionally the SEP into a unified subsystem. The interface descriptions in the preceding sections apply directly when integrating the SMU, since the SMU inherits all three subsystems' ports. This section covers the module variants available to adopters, how the SMU wires its subsystems together internally, and what additional configuration it exposes.

### 1.5.1. Module Variants and IP Integration

OCAH core modules (`smc.sv`, `sep.sv`, `dtp.sv`, and the composite `smu.sv`) expose logic-level interfaces only. All technology-specific components — PLLs, PVT sensors, eFuse shims, memory macros, GPIO padrings, SPI controllers, and external TRNG— are instantiated outside the core modules by the adopter. This separation keeps the OCAH deliverable technology-neutral.

The repository provides reference wrappers that demonstrate how to bridge the gap between OCAH core interfaces and adopter-specific IP:

- `smu_wrapper.sv` wraps `smu.sv` with pad-level signal handling for clocks, resets, GPIO, eFuse, and JTAG interfaces. It internally instantiates `smc_ip_integration.sv` for SMC-related vendor IP and (when `SEP=1`) `sep_ip_integration.sv` for SEP-related vendor IP.
- `smc_wrapper.sv` provides an equivalent reference wrapping `smc.sv` alone, useful when integrating the SMC without the DTP/SEP subsystems.

Neither the wrappers nor the IP integration modules are part of the OCAH deliverable; they exist as integration guides and for Tenstorrent internal usage. Adopters must develop their own equivalents following the interface patterns demonstrated in these modules. The interface contracts at the module boundaries ensure technology substitution without modifying the verified OCAH logic.

The figure below illustrates the relationship between the OCAH core modules, the wrapper layer, and the IP integration modules.

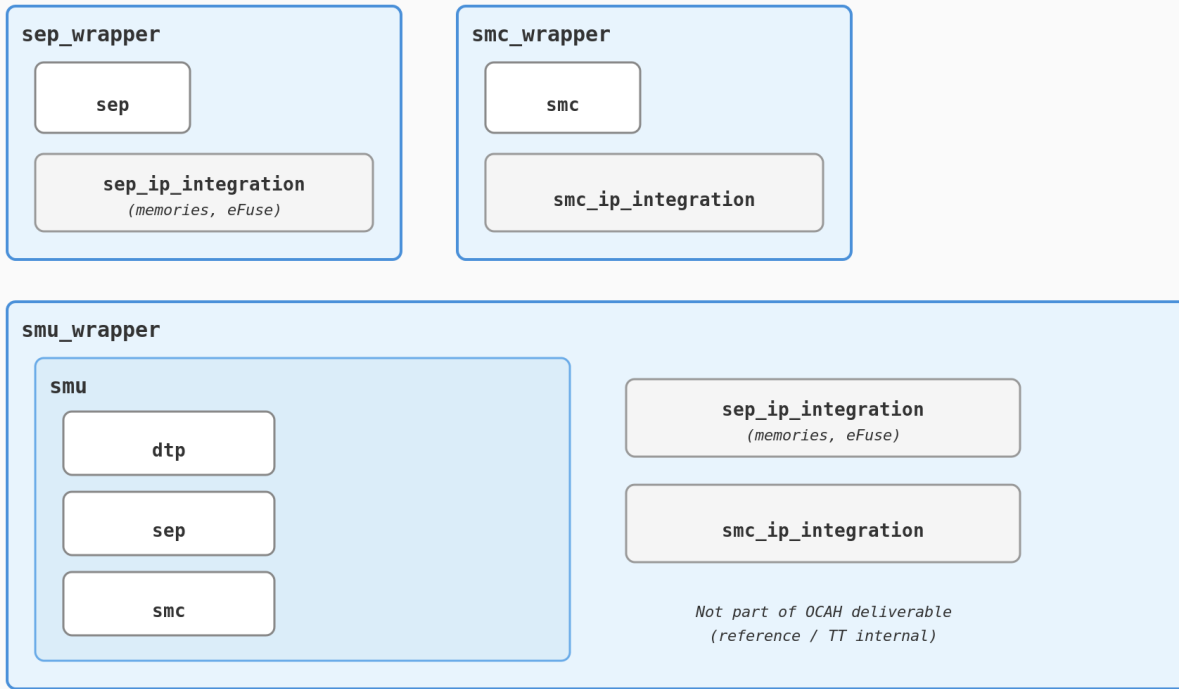


Figure 1. SMU Integration Components

## IP Integration Modules

`smc_ip_integration.sv` demonstrates integration of SMC-side technology components: PLL wrappers (generating `pll_cgm_clk_o` and `pll_awm_clk_o`), PVT sensor interfaces, eFuse shims, GPIO padding logic, memory macros (ROM, scratchpad RAM, cache tag and data arrays), and trace memory instances.

`sep_ip_integration.sv` demonstrates integration of SEP-side technology components: SEP memory macros (TCM, scratch SRAM, boot ROM, OTBN IMEM/DMEM, Key Manager ROM/SRAM), eFuse shim, external TRNG, and SPI controller selection.

## Wrapper Clock and PLL Architecture

The `smu_wrapper.sv` (and `smc_wrapper.sv`) reference accepts `BP_REFCLK` at the pad level and processes it through the integrated padding to produce internal reference timing. A `pll_wrapper` module generates system clocks `pll_cgm_clk_o` and `pll_awm_clk_o` from the padding-processed reference. The SMC controls this PLL through the same AXI-Lite interface exposed by the core module. However, the core receives its clocks through DFX inputs `pll_cgm_clk_dfx_i` and `ref_clk_vdd_sys_dfx_i`, so the adopter must route PLL outputs through the clock distribution network and feed them back through these DFX inputs.

### 1.5.2. Clock and Reset Architecture

Connect `clk_smu_i` to the primary system clock and `clk_ref_i` to a stable reference source. The DTP operates on TCK for JTAG logic and on the system clock for cross-trigger and AXI bridges. Reset sequencing follows the SMC flow; the DTP JTAG state machine uses `pwr_on_rst_ni`, which the SMU connects to `rst_cold_ni`. The SMU internally connects DTP debug paths to the SMC and optionally instantiates the SEP when the `SEP` parameter is set to 1. The full interconnection details are described in [SMC-DTP Internal Interconnection](#) and [SEP-Conditional Wiring](#) below.

### 1.5.3. SMU AXI Crossbar Address Map

Inside the SMU, the `smu_axi_xbar` module routes traffic between three AXI4 manager/subordinate pairs: the SEP outbound master (`sep_out`), the SMC outbound master (`smc_out`), and the external SMU AXI input (`ext_in` / `smu_axi_in`). The corresponding targets are the SEP and SMC inbound subordinates, and the external SMU AXI output (`ext_out` / `smu_axi_out`).

See the *SMU AXI Crossbar* and *OCAH Memory Map* sections of the Technical Reference Manual for connectivity and the responsibility split between hardware-fixed and firmware-programmable apertures.

The crossbar uses a **runtime** address map. Only the SEP and SMC apertures are programmable at runtime; the `ext_out` rule is static:

- `sep_global_base_addr_i` / `sep_region_size_i` come from the `sep_cpu_ctrl` CSRs `SEP_GLOBAL_BASE_ADDR` / `SEP_REGION_SIZE`, exposed at both the `sep.sv` and `smu.sv` boundaries as `sep_global_base_addr_o` / `sep_region_size_o` (both 56-bit). `SEP_REGION_SIZE` is a 32-bit CSR field, zero-extended to 56-bit at the SEP system CSR; the xbar input takes only the lower 32 bits. `SEP_REGION_SIZE` defines the SEP global/chiplet aperture for SMU routing and SEP inbound global-to-local remap. It is distinct from the SEP CPU's architectural local-alias span. `SEP_REGION_SIZE` currently resets to `0x0100_0000` (16 MiB). Tied to '0' at the `smu.sv` boundary when the `SEP` parameter is 0.
- `smc_global_base_addr_i` / `smc_region_size_i` come from the SMC CPU control CSRs `GLOBAL_BASE` / `REGION_SIZE`, exposed at both the `smc.sv` and `smu.sv` boundaries as `smc_global_base_o` (56-bit) / `smc_region_size_o` (32-bit). `REGION_SIZE` resets to `0x0100_0000` (16 MiB).

SEP, SMC, and `smu_axi_xbar` all run on `clk_smu_i`, so these quasi-static CSR fields need no clock-domain crossing. The crossbar's address arithmetic is done on 57-bit widened sums; `ext_out` covers `0x0000_8000_0000` up to the top of the 56-bit address space.

### Firmware programming order

Firmware must program `SEP_GLOBAL_BASE_ADDR` / `SEP_REGION_SIZE` and SMC `GLOBAL_BASE` / `REGION_SIZE` during early boot, **before any cross-subsystem AXI traffic is issued**, and must ensure the two apertures do not overlap and both fall within the chiplet-visible SMU envelope assigned by the integrator. The SMU crossbar does not itself constrain envelope size.

See the *OCAH Memory Map* section of the Technical Reference Manual for that responsibility split.

### 1.5.4. SMU Default Parameters

The SMU exposes a set of configurable parameters that control DTP JTAG features, identification, cross trigger, and pipeline depths. The table below lists each parameter with its default value.

| Parameter                              | Value | Description                       |
|----------------------------------------|-------|-----------------------------------|
| <code>JTAG_BSR_ENABLE</code>           | 1     | Boundary scan enabled             |
| <code>JTAG_EXTTEST_TRAIN_ENABLE</code> | 1     | EXTTEST_TRAIN instruction enabled |
| <code>JTAG_EXTTEST_PULSE_ENABLE</code> | 1     | EXTTEST_PULSE instruction enabled |
| <code>JTAG_INTEST_ENABLE</code>        | 1     | INTEST instruction enabled        |
| <code>JTAG_CLAMP_ENABLE</code>         | 1     | CLAMP instruction enabled         |
| <code>JTAG_HIGHZ_ENABLE</code>         | 1     | HIGHZ instruction enabled         |
| <code>JTAG_RUNBIST_ENABLE</code>       | 1     | RUNBIST instruction enabled       |
| <code>JTAG_TMP_ENABLE</code>           | 1     | TMP controller enabled            |

| Parameter            | Value    | Description                                                                                                                                                   |
|----------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| JTAG_IC_RESET_ENABLE | 1        | Enables the external IC_RESET TDR slice (SMU drives this into the DTP's JTAG_IC_RESET_EXT_ENABLE). SMC and SEP slices are enabled unconditionally by the SMU. |
| JTAG_SMC_DBG_ENABLE  | 1        | SMC debug interface enabled                                                                                                                                   |
| JTAG_STAP_IO_ENABLE  | 1        | I/O STAP enabled                                                                                                                                              |
| JTAG_NUM_EXTRA_STAPS | 1        | Number of additional STAPs                                                                                                                                    |
| JTAG_IDCODE_MFR_ID   | 110h000  | Manufacturer ID                                                                                                                                               |
| JTAG_IDCODE_PART_NUM | 160h0000 | Part number                                                                                                                                                   |
| JTAG_IDCODE_SI_REV   | 40h0     | Silicon revision                                                                                                                                              |
| JTAG_OCH_VER         | 80h00    | OCH IP version                                                                                                                                                |
| XTRIG_INT_CT_MODE    | 80h00    | Internal CT protocol mode bits (0=pulse sync, 1=req/ack)                                                                                                      |
| SMC_OTP_RD_PL_DEPTH  | 20h3     | SMC OTP read pipeline depth                                                                                                                                   |
| SMC_OTP_WR_PL_DEPTH  | 20h3     | SMC OTP write pipeline depth                                                                                                                                  |
| SMC_RD_PL_DEPTH      | 20h3     | SMC fabric read pipeline depth                                                                                                                                |
| SMC_WR_PL_DEPTH      | 20h3     | SMC fabric write pipeline depth                                                                                                                               |

SEP-related DTP configuration is not exposed as an SMU config-struct field. The SMU derives SEP debug/STAP enablement from the module parameter `SEP` and wires the corresponding DTP interfaces internally to the SEP block when `SEP=1`.

The DTP IC\_RESET TDR is composed of three independent struct-typed slices. The SMU passes `smc_pkg::jtag_smc_reset_ctrl_t` into the DTP's `ic_reset_smc_t` parameter (with `JTAG_IC_RESET_SMC_ENABLE=1`), `sep_pkg::jtag_sep_reset_ctrl_t` into `ic_reset_sep_t` (with `JTAG_IC_RESET_SEP_ENABLE=SEP`), and exposes the external slice via the SMU's own `ic_reset_ext_t` type parameter and `jtag_ic_reset_ext_o` port (gated by `JTAG_IC_RESET_ENABLE`). Each slice's `.ovrd/.val` sub-structs must be the same width; `jtag_ptap` interleaves them into the `{enable, control}` bit order required by the JTAG specification.

The SMC and SEP slices are consumed inside the SMU and have no port. The external slice is reserved for the adopter: set `ic_reset_ext_t` to a struct carrying the reset overrides needed by logic instantiated outside the SMU and route `jtag_ic_reset_ext_o` to that logic. The struct is defined by the adopter, not by the OCAH packages; the parameter default (`jtag_tap_pkg::jtag_ic_reset_default_t`) is a single-override stub. Clear `JTAG_IC_RESET_ENABLE` when no external overrides are required.

### 1.5.5. DTP Parameter Pass-Through

The SMU passes through most DTP parameters to allow integrator customization. Type parameters are set internally based on SMC package types.

#### JTAG Configuration Parameters

| Parameter                | SMU Default | Pass-through | Notes                 |
|--------------------------|-------------|--------------|-----------------------|
| JTAG_BSR_ENABLE          | 1           | Yes          | Boundary scan enabled |
| JTAG_EXTEST_TRAIN_ENABLE | 1           | Yes          |                       |
| JTAG_EXTEST_PULSE_ENABLE | 1           | Yes          |                       |
| JTAG_INTTEST_ENABLE      | 1           | Yes          |                       |
| JTAG_CLAMP_ENABLE        | 1           | Yes          |                       |
| JTAG_HIGHZ_ENABLE        | 1           | Yes          |                       |

| Parameter            | SMU Default | Pass-through | Notes                                                                                                                                                    |
|----------------------|-------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| JTAG_RUNBIST_ENABLE  | 1           | Yes          |                                                                                                                                                          |
| JTAG_TMP_ENABLE      | 1           | Yes          |                                                                                                                                                          |
| JTAG_IC_RESET_ENABLE | 1           | No           | SMU routes this to the DTP's JTAG_IC_RESET_EXT_ENABLE; SMC and SEP slices are enabled inside the SMU instantiation (SMC unconditionally, SEP when SEP=1) |
| JTAG_SMC_DBG_ENABLE  | 1           | Yes          |                                                                                                                                                          |
| JTAG_SEP_DBG_ENABLE  | 1 (SEP=1)   | No           | Derived internally from the SMU module parameter SEP; not exposed in smu_cfg_t                                                                           |
| JTAG_STAP_IO_ENABLE  | 1           | Yes          |                                                                                                                                                          |
| JTAG_NUM_EXTRA_STAPS | 1           | Yes          | Port array size equals parameter value; when set to 0, array size is 1 (tie-off required)                                                                |

## JTAG ID Parameters

| Parameter            | SMU Default | Pass-through | Notes            |
|----------------------|-------------|--------------|------------------|
| JTAG_IDCODE_MFR_ID   | 110h000     | Yes          | Manufacturer ID  |
| JTAG_IDCODE_PART_NUM | 160h0000    | Yes          | Part number      |
| JTAG_IDCODE_SI_REV   | 40h0        | Yes          | Silicon revision |
| JTAG_OCH_VER         | 80h00       | Yes          | OCH IP version   |

## Cross Trigger Parameters

| Parameter         | SMU Default | DTP Internal | Pass-through | Notes                                  |
|-------------------|-------------|--------------|--------------|----------------------------------------|
| XTRIG_INT_CT_MODE | [7:0]       | [9:0]        | Yes          | 8-bit on SMU; DTP bits [1:0]=0 for SMC |

The SMU parameter width is two less than DTP's for cross trigger (8 vs 10) because internal cross trigger ports [1:0] and clock stop request #0 are connected to the SMC internally. The SMU remaps DTP ports [9:2] to SMU external ports [7:0].

## Pipeline Depth Parameters

| Parameter           | SMU Default | Pass-through | Notes                                                     |
|---------------------|-------------|--------------|-----------------------------------------------------------|
| SMC_OTP_RD_PL_DEPTH | 20h3        | Yes          | SMC OTP read pipeline depth                               |
| SMC_OTP_WR_PL_DEPTH | 20h3        | Yes          | SMC OTP write pipeline depth                              |
| SEP_OTP_RD_PL_DEPTH | 20h3        | No           | Fixed to 20h3 in SMU; used when SEP=1, ignored when SEP=0 |
| SEP_OTP_WR_PL_DEPTH | 20h3        | No           | Fixed to 20h3 in SMU; used when SEP=1, ignored when SEP=0 |
| SMC_RD_PL_DEPTH     | 20h3        | Yes          | SMC fabric read pipeline depth                            |
| SMC_WR_PL_DEPTH     | 20h3        | Yes          | SMC fabric write pipeline depth                           |

## Type Parameters

The following type parameters are set internally by the SMU and are not exposed for integrator configuration.

| Parameter          | Set To                                 | Notes                    |
|--------------------|----------------------------------------|--------------------------|
| jtag_tap_ctrl_t    | prim_jtag_pkg::jtag_tap_ctrl_t         | JTAG TAP control struct  |
| jtag_scan_ctrl_t   | prim_jtag_pkg::jtag_scan_ctrl_t        | JTAG scan control struct |
| smc_jtag_axi_req_t | smc_pkg::smc_jtag_56_64_2_12_axi_req_t | SMC debug AXI request    |

| Parameter           | Set To                                  | Notes                                              |
|---------------------|-----------------------------------------|----------------------------------------------------|
| smc_jtag_axi_resp_t | smc_pkg::smc_jtag_56_64_2_12_axi_resp_t | SMC debug AXI response                             |
| smc_otp_axil_req_t  | smc_efuse_pkg::efuse_axil_req_t         | SMC OTP AXI-Lite request                           |
| smc_otp_axil_resp_t | smc_efuse_pkg::efuse_axil_resp_t        | SMC OTP AXI-Lite response                          |
| sep_otp_axil_req_t  | smc_efuse_pkg::efuse_axil_req_t         | SEP OTP AXI-Lite request (error slave when SEP=0)  |
| sep_otp_axil_resp_t | smc_efuse_pkg::efuse_axil_resp_t        | SEP OTP AXI-Lite response (error slave when SEP=0) |
| xtrig_axil_req_t    | smc_pkg::smc_axil_32_32_req_t           | Cross trigger CSR request                          |
| xtrig_axil_resp_t   | smc_pkg::smc_axil_32_32_resp_t          | Cross trigger CSR response                         |

## 1.5.6. DTP Port Mapping

The following tables define the complete DTP port list as connected within the SMU. Ports marked "SMU external port" are exposed on the SMU boundary for integrator connection. Ports marked with a subsystem name (e.g. SMC, SEP) are wired internally within the SMU.

### Clock and Reset

| Port          | Direction | Width | SMU Connection                                  |
|---------------|-----------|-------|-------------------------------------------------|
| clk_i         | input     | 1     | SMU clk_smu_i (shared with SMC clk_smc_i)       |
| rst_n_i       | input     | 1     | SMC rst_primary_smc_clk_no                      |
| pwr_on_rst_ni | input     | 1     | SMU rst_cold_ni (power-on reset for JTAG logic) |

### Primary JTAG TAP Interface

| Port                        | Direction | Width           | SMU Connection    |
|-----------------------------|-----------|-----------------|-------------------|
| jtag_ptap_client_tap_ctrl_i | input     | jtag_tap_ctrl_t | SMU external port |
| jtag_ptap_client_tdi_i      | input     | 1               | SMU external port |
| jtag_ptap_client_tdo_o      | output    | 1               | SMU external port |
| jtag_ptap_client_tdo_oen_o  | output    | 1               | SMU external port |

### Boundary Scan Interface

| Port                      | Direction | Width            | SMU Connection    |
|---------------------------|-----------|------------------|-------------------|
| jtag_bsr_host_scan_ctrl_o | output    | jtag_scan_ctrl_t | SMU external port |
| jtag_bsr_host_scan_in_i   | input     | 1                | SMU external port |
| jtag_bsr_host_scan_out_o  | output    | 1                | SMU external port |

### I/O STAP Interface

| Port                         | Direction | Width           | SMU Connection    |
|------------------------------|-----------|-----------------|-------------------|
| jtag_stap_io_host_tap_ctrl_o | output    | jtag_tap_ctrl_t | SMU external port |
| jtag_stap_io_host_tdi_i      | input     | 1               | SMU external port |
| jtag_stap_io_host_tdo_o      | output    | 1               | SMU external port |
| jtag_stap_io_host_tdo_oen_o  | output    | 1               | SMU external port |

## SMC Debug STAP Interface

| Port                          | Direction | Width           | SMU Connection                                                                                                         |
|-------------------------------|-----------|-----------------|------------------------------------------------------------------------------------------------------------------------|
| jtag_stap_smc_host_tap_ctrl_o | output    | jtag_tap_ctrl_t | SMC smc_cpu_jtag_{TCK,TMS}_i; trst_n inverted drives SMC smc_cpu_jtag_reset_i when JTAG_SMC_DBG_ENABLE=1, else tie-off |
| jtag_stap_smc_host_tdi_i      | input     | 1               | SMC smc_cpu_jtag_TDO_data_o when JTAG_SMC_DBG_ENABLE=1, else 1'b0                                                      |
| jtag_stap_smc_host_tdo_o      | output    | 1               | SMC smc_cpu_jtag_TDI_i when JTAG_SMC_DBG_ENABLE=1, else unused                                                         |
| jtag_stap_smc_host_tdo_oen_o  | output    | 1               | Internal only; not consumed by SMC                                                                                     |

## SEP Debug STAP Interface

| Port                          | Direction | Width           | SMU Connection                                    |
|-------------------------------|-----------|-----------------|---------------------------------------------------|
| jtag_stap_sep_host_tap_ctrl_o | output    | jtag_tap_ctrl_t | SEP jtag_{tck,tms,trst_n} when SEP=1, else unused |
| jtag_stap_sep_host_tdi_i      | input     | 1               | SEP jtag_tdo when SEP=1, else 1'b0                |
| jtag_stap_sep_host_tdo_o      | output    | 1               | SEP jtag_tdi when SEP=1, else unused              |
| jtag_stap_sep_host_tdo_oen_o  | output    | 1               | Internal only; not consumed by SEP at SMU level   |

## Extra STAP Interfaces

| Port                            | Direction | Width                | SMU Connection    |
|---------------------------------|-----------|----------------------|-------------------|
| jtag_stap_extra_host_tap_ctrl_o | output    | jtag_tap_ctrl_t[1:0] | SMU external port |
| jtag_stap_extra_host_tdi_i      | input     | [1:0]                | SMU external port |
| jtag_stap_extra_host_tdo_o      | output    | [1:0]                | SMU external port |
| jtag_stap_extra_host_tdo_oen_o  | output    | [1:0]                | SMU external port |

## Extended STAP Scan Interface

| Port                       | Direction | Width            | SMU Connection    |
|----------------------------|-----------|------------------|-------------------|
| jtag_stap_host_scan_ctrl_o | output    | jtag_scan_ctrl_t | SMU external port |
| jtag_stap_host_scan_in_i   | input     | 1                | SMU external port |
| jtag_stap_host_scan_out_o  | output    | 1                | SMU external port |

## DFD iJTAG Interface

| Port                      | Direction | Width            | SMU Connection    |
|---------------------------|-----------|------------------|-------------------|
| jtag_dfd_host_scan_ctrl_o | output    | jtag_scan_ctrl_t | SMU external port |
| jtag_dfd_host_scan_in_i   | input     | 1                | SMU external port |
| jtag_dfd_host_scan_out_o  | output    | 1                | SMU external port |

## DFT iJTAG Interface

| Port                      | Direction | Width            | SMU Connection    |
|---------------------------|-----------|------------------|-------------------|
| jtag_dft_host_scan_ctrl_o | output    | jtag_scan_ctrl_t | SMU external port |
| jtag_dft_host_scan_in_i   | input     | 1                | SMU external port |
| jtag_dft_host_scan_out_o  | output    | 1                | SMU external port |

## SMC Fabric Debug AXI Interface

| Port                            | Direction | Width                            | SMU Connection                      |
|---------------------------------|-----------|----------------------------------|-------------------------------------|
| <code>axi_smc_dbg_req_o</code>  | output    | <code>smc_jtag_axi_req_t</code>  | SMC <code>jtag_axi_in_req_i</code>  |
| <code>axi_smc_dbg_resp_i</code> | input     | <code>smc_jtag_axi_resp_t</code> | SMC <code>jtag_axi_in_resp_o</code> |

## SMC OTP Debug AXI-Lite Interface

| Port                                  | Direction | Width                            | SMU Connection                            |
|---------------------------------------|-----------|----------------------------------|-------------------------------------------|
| <code>axil_smc_otp_jtag_req_o</code>  | output    | <code>smc_otp_axil_req_t</code>  | SMC <code>axil_smc_otp_jtag_req_i</code>  |
| <code>axil_smc_otp_jtag_resp_i</code> | input     | <code>smc_otp_axil_resp_t</code> | SMC <code>axil_smc_otp_jtag_resp_o</code> |

## SEP OTP Debug AXI-Lite Interface

When `SEP=1`, this interface connects to the SEP `axil_sep_otp_jtag_*` ports. When `SEP=0`, the SMU instantiates `u_sep_otp_axil_err_slv`, which responds with DECERR to all transactions.

| Port                                  | Direction | Width                            | SMU Connection                                                                                                                        |
|---------------------------------------|-----------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <code>axil_sep_otp_jtag_req_o</code>  | output    | <code>sep_otp_axil_req_t</code>  | SEP <code>axil_sep_otp_jtag_req_i</code> when <code>SEP=1</code> , else AXI-Lite error slave ( <code>u_sep_otp_axil_err_slv</code> )  |
| <code>axil_sep_otp_jtag_resp_i</code> | input     | <code>sep_otp_axil_resp_t</code> | SEP <code>axil_sep_otp_jtag_resp_o</code> when <code>SEP=1</code> , else AXI-Lite error slave ( <code>u_sep_otp_axil_err_slv</code> ) |

## Clock Control

| Port                     | Direction | Width | SMU Connection                                                       |
|--------------------------|-----------|-------|----------------------------------------------------------------------|
| <code>stop_clks_o</code> | output    | 1     | SMU external port: <code>dtb_stop_clks_o</code> (to PLL clock gates) |

## Boot Stall Control

| Port                                | Direction | Width | SMU Connection |
|-------------------------------------|-----------|-------|----------------|
| <code>jtag_boot_stall_ovrd_o</code> | output    | 1     | Disconnected   |
| <code>jtag_boot_stall_o</code>      | output    | 1     | Disconnected   |

## JTAG Reset Control

| Port                             | Direction | Width                                       | SMU Connection                                                                                                                                                              |
|----------------------------------|-----------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>jtag_ic_reset_smc_o</code> | output    | <code>smc_pkg::jtag_smc_reset_ctrl_t</code> | SMC <code>jtag_smc_reset_ctrl_i</code> (packed struct with <code>.ovrd</code> override enables and <code>.val</code> active-low override values)                            |
| <code>jtag_ic_reset_sep_o</code> | output    | <code>sep_pkg::jtag_sep_reset_ctrl_t</code> | SEP <code>jtag_sep_reset_ctrl_i</code> (packed struct with <code>.ovrd</code> override enables and <code>.val</code> active-low override values); SMU-internal, no SMU port |
| <code>jtag_ic_reset_ext_o</code> | output    | <code>ic_reset_ext_t</code>                 | SMU <code>jtag_ic_reset_ext_o</code> external port (integrator-defined packed struct; carries reset overrides for logic outside the SMU)                                    |

## JTAG State Outputs

| Port                     | Direction | Width                      | SMU Connection    |
|--------------------------|-----------|----------------------------|-------------------|
| jtag_ptap_state_o        | output    | tap_state_e                | SMU external port |
| jtag_ptap_inst_decoded_o | output    | jtag_instruction_decoded_e | SMU external port |

## Cross Trigger CSR AXI-Lite Interface

| Port              | Direction | Width             | SMU Connection          |
|-------------------|-----------|-------------------|-------------------------|
| axil_xtrig_req_i  | input     | xtrig_axil_req_t  | SMC axil_dtp_csr_req_o  |
| axil_xtrig_resp_o | output    | xtrig_axil_resp_t | SMC axil_dtp_csr_resp_i |

## Cross Trigger Matrix Interface

| Port                | Direction | Width                  | SMU Connection                                                                  |
|---------------------|-----------|------------------------|---------------------------------------------------------------------------------|
| xtrig_ctm_src_req_o | output    | [XTRIG_NUM_INT_CT-1:0] | [1:0]: SMU internal (SMC), [9:2]: SMU external<br>xtrig_ctm_src_req_o[7:0]      |
| xtrig_ctm_src_ack_i | input     | [XTRIG_NUM_INT_CT-1:0] | [1:0]: Tie-off (SMC), [9:2]: SMU external<br>xtrig_ctm_src_ack_i[7:0]           |
| xtrig_ctm_dst_req_i | input     | [XTRIG_NUM_INT_CT-1:0] | [1:0]: SMC xtrigger_ss_o[1:0], [9:2]: SMU external<br>xtrig_ctm_dst_req_i[7:0]  |
| xtrig_ctm_dst_ack_o | output    | [XTRIG_NUM_INT_CT-1:0] | [1:0]: Unused (SMC connection), [9:2]: SMU external<br>xtrig_ctm_dst_ack_o[7:0] |

Internal cross trigger ports [1:0] are reserved for SMC (both bits of `xtrigger_ss_o`). The SMU exposes ports [9:2] as external ports [7:0], so the SMU `XTRIG_NUM_INT_CT` parameter is two less than DTP's (8 vs 10). The SMC uses pulse-synchronization mode, so ack signals for ports [1:0] are unused: `xtrig_ctm_src_ack_i[1:0]` should be tied to `2'b00` and `xtrig_ctm_dst_ack_o[1:0]` are left unconnected. The usage of ack signals depends on the `XTRIG_INT_CT_MODE` parameter: when a mode bit is 0 (pulse synchronization), the corresponding ack is unused; when 1 (req/ack handshaking), the ack must be connected to the cross trigger endpoint.

## Clock Stop Request Interface

| Port                 | Direction | Width                        | SMU Connection                                                                                     |
|----------------------|-----------|------------------------------|----------------------------------------------------------------------------------------------------|
| xtrig_clk_stop_req_i | input     | [XTRIG_NUM_CLK_STOP_REQ-1:0] | [0]: SMC<br>tdr_dbg_ctrl_clocks_stopped_by_cla_o, [8:1]:<br>SMU external xtrig_clk_stop_req_i[7:0] |

Clock stop request port #0 is reserved for SMC. The SMU exposes ports [8:1] as external ports [7:0], so the SMU `XTRIG_NUM_CLK_STOP_REQ` parameter should be one less than DTP's (8 vs 9).

## Cross Trigger Port GPIO Interface

The 16 CTP signals from DTP are ported through the SMU to `smu_wrapper` as external ports. In `smu_wrapper`, they connect to internal signals destined for `smc_ip_integration`. Currently, input signals are tied off to '0' and output signals are dangling pending proper CTP-to-GPIO connection implementation.

| Port                        | Direction | Width  | SMU Connection                                    |
|-----------------------------|-----------|--------|---------------------------------------------------|
| xtrig_ctp_req_out_dout_o    | output    | [15:0] | SMU external → smu_wrapper internal (dangling)    |
| xtrig_ctp_req_out_dout_en_o | output    | [15:0] | SMU external → smu_wrapper internal (dangling)    |
| xtrig_ctp_req_out_din_i     | input     | [15:0] | SMU external → smu_wrapper internal (tied to '0') |
| xtrig_ctp_req_out_din_en_o  | output    | [15:0] | SMU external → smu_wrapper internal (dangling)    |

| Port                        | Direction | Width  | SMU Connection                                    |
|-----------------------------|-----------|--------|---------------------------------------------------|
| xtrig_ctp_req_in_dout_o     | output    | [15:0] | SMU external → smu_wrapper internal (dangling)    |
| xtrig_ctp_req_in_dout_en_o  | output    | [15:0] | SMU external → smu_wrapper internal (dangling)    |
| xtrig_ctp_req_in_din_i      | input     | [15:0] | SMU external → smu_wrapper internal (tied to '0') |
| xtrig_ctp_req_in_din_en_o   | output    | [15:0] | SMU external → smu_wrapper internal (dangling)    |
| xtrig_ctp_ack_in_dout_o     | output    | [15:0] | SMU external → smu_wrapper internal (dangling)    |
| xtrig_ctp_ack_in_dout_en_o  | output    | [15:0] | SMU external → smu_wrapper internal (dangling)    |
| xtrig_ctp_ack_in_din_i      | input     | [15:0] | SMU external → smu_wrapper internal (tied to '0') |
| xtrig_ctp_ack_in_din_en_o   | output    | [15:0] | SMU external → smu_wrapper internal (dangling)    |
| xtrig_ctp_ack_out_dout_o    | output    | [15:0] | SMU external → smu_wrapper internal (dangling)    |
| xtrig_ctp_ack_out_dout_en_o | output    | [15:0] | SMU external → smu_wrapper internal (dangling)    |
| xtrig_ctp_ack_out_din_i     | input     | [15:0] | SMU external → smu_wrapper internal (tied to '0') |
| xtrig_ctp_ack_out_din_en_o  | output    | [15:0] | SMU external → smu_wrapper internal (dangling)    |

## 1.5.7. SMC Port Mapping

The following tables define the complete SMC port list as connected within the SMU. Ports marked "SMU external port" are exposed on the SMU boundary. Ports marked with "DTP" or "SEP" are wired internally.

### Clocks

| Port         | Direction | Width | SMU Connection                                                   |
|--------------|-----------|-------|------------------------------------------------------------------|
| clk_smc_i    | input     | 1     | SMU <code>clk_smu_i</code> (shared with DTP <code>clk_i</code> ) |
| clk_periph_i | input     | 1     | SMU external port                                                |
| clk_ref_i    | input     | 1     | SMU external port                                                |

### Resets

| Port                   | Direction | Width | SMU Connection                               |
|------------------------|-----------|-------|----------------------------------------------|
| rst_cold_ni            | input     | 1     | SMU external port                            |
| rst_primary_ref_clk_no | output    | 1     | SMU external port                            |
| rst_primary_smc_clk_no | output    | 1     | SMU external port + DTP <code>rst_n_i</code> |

### System AXI Input

| Port              | Direction | Width                                         | SMU Connection    |
|-------------------|-----------|-----------------------------------------------|-------------------|
| sys_axi_in_req_i  | input     | <code>smc_sys_in_56_64_6_12_axi_req_t</code>  | SMU external port |
| sys_axi_in_resp_o | output    | <code>smc_sys_in_56_64_6_12_axi_resp_t</code> | SMU external port |

### JTAG AXI Input

| Port               | Direction | Width                                       | SMU Connection                      |
|--------------------|-----------|---------------------------------------------|-------------------------------------|
| jtag_axi_in_req_i  | input     | <code>smc_jtag_56_64_2_12_axi_req_t</code>  | DTP <code>axi_smc_dbg_req_o</code>  |
| jtag_axi_in_resp_o | output    | <code>smc_jtag_56_64_2_12_axi_resp_t</code> | DTP <code>axi_smc_dbg_resp_i</code> |

## SMC OTP JTAG AXI-Lite

| Port                     | Direction | Width             | SMU Connection               |
|--------------------------|-----------|-------------------|------------------------------|
| axil_smc_otp_jtag_req_i  | input     | efuse_axil_req_t  | DTP axil_smc_otp_jtag_req_o  |
| axil_smc_otp_jtag_resp_o | output    | efuse_axil_resp_t | DTP axil_smc_otp_jtag_resp_i |

## SEP AXI Input

When **SEP=0**, the SEP AXI subordinate input is tied off to '0' and the response output is left unconnected. When **SEP=1**, the SMU drives this path from the SEP-to-SMC dedicated AXI connection.

| Port              | Direction | Width                            | SMU Connection                              |
|-------------------|-----------|----------------------------------|---------------------------------------------|
| sep_axi_in_req_i  | input     | smc_sep_in_56_64_6_12_axi_req_t  | Tied off to '0' (no incoming transactions)  |
| sep_axi_in_resp_o | output    | smc_sep_in_56_64_6_12_axi_resp_t | Unconnected (no transactions to respond to) |

## AXI Output

| Port              | Direction | Width                             | SMU Connection    |
|-------------------|-----------|-----------------------------------|-------------------|
| output_axi_req_o  | output    | smc_sys_out_56_64_8_12_axi_req_t  | SMU external port |
| output_axi_resp_i | input     | smc_sys_out_56_64_8_12_axi_resp_t | SMU external port |

## DTP CSR Interface

| Port                | Direction | Width                 | SMU Connection        |
|---------------------|-----------|-----------------------|-----------------------|
| axil_dtp_csr_req_o  | output    | smc_axil_32_32_req_t  | DTP axil_xtrig_req_i  |
| axil_dtp_csr_resp_i | input     | smc_axil_32_32_resp_t | DTP axil_xtrig_resp_o |

## Adopter Peripheral Window AXI-Lite

| Port                | Direction | Width                 | SMU Connection                               |
|---------------------|-----------|-----------------------|----------------------------------------------|
| smc_external_req_o  | output    | smc_axil_32_32_req_t  | SMU external port (to adopter peripherals)   |
| smc_external_resp_i | input     | smc_axil_32_32_resp_t | SMU external port (from adopter peripherals) |

## eFuse Interface

| Port                      | Direction | Width               | SMU Connection    |
|---------------------------|-----------|---------------------|-------------------|
| efuse_bank_ctrl_req_o     | output    | efuse_axil_req_t    | SMU external port |
| efuse_bank_ctrl_resp_i    | input     | efuse_axil_resp_t   | SMU external port |
| efuse_shim_command_req_o  | output    | fuse_command_req_t  | SMU external port |
| efuse_shim_command_resp_i | input     | fuse_command_resp_t | SMU external port |
| shadow_regs_o             | output    | efuse_map_t         | SMU external port |

## GPIO Data Signals

| Port                    | Direction | Width                | SMU Connection    |
|-------------------------|-----------|----------------------|-------------------|
| lsio_interface_select_o | output    | [NUM_GPIO_WRAPS-1:0] | SMU external port |

| Port          | Direction | Width                | SMU Connection    |
|---------------|-----------|----------------------|-------------------|
| gpio_in_i     | input     | [NUM_GPIO_WRAPS-1:0] | SMU external port |
| gpio_out_o    | output    | [NUM_GPIO_WRAPS-1:0] | SMU external port |
| gpio_in_en_o  | output    | [NUM_GPIO_WRAPS-1:0] | SMU external port |
| gpio_out_en_o | output    | [NUM_GPIO_WRAPS-1:0] | SMU external port |

## GPIO External Pins

| Port                  | Direction | Width | SMU Connection    |
|-----------------------|-----------|-------|-------------------|
| rst_cool_n_from_pin_i | input     | 1     | SMU external port |

## ATB Telemetry

| Port                | Direction | Width                                         | SMU Connection    |
|---------------------|-----------|-----------------------------------------------|-------------------|
| clk_telemetry_i     | input     | 1                                             | SMU external port |
| rst_telemetry_ni    | input     | 1                                             | SMU external port |
| telemetry_atdata_i  | input     | telemetry_data_t[NUM_TELEMETRY_RECEIVERS-1:0] | SMU external port |
| telemetry_atid_i    | input     | atb_id_t[NUM_TELEMETRY_RECEIVERS-1:0]         | SMU external port |
| telemetry_atready_o | output    | [NUM_TELEMETRY_RECEIVERS-1:0]                 | SMU external port |
| telemetry_atvalid_i | input     | [NUM_TELEMETRY_RECEIVERS-1:0]                 | SMU external port |
| telemetry_afvalid_o | output    | [NUM_TELEMETRY_RECEIVERS-1:0]                 | SMU external port |
| telemetry_afready_i | input     | [NUM_TELEMETRY_RECEIVERS-1:0]                 | SMU external port |

## DED/WDT

| Port                 | Direction | Width | SMU Connection    |
|----------------------|-----------|-------|-------------------|
| cluster_ded_o        | output    | 1     | SMU external port |
| wdt_first_timeout_o  | output    | 1     | SMU external port |
| wdt_second_timeout_o | output    | 1     | SMU external port |

## Region Size

| Port              | Direction | Width  | SMU Connection    |
|-------------------|-----------|--------|-------------------|
| smc_region_size_o | output    | [31:0] | SMU external port |

## External Interrupts

| Port                     | Direction | Width                    | SMU Connection                                                                      |
|--------------------------|-----------|--------------------------|-------------------------------------------------------------------------------------|
| ext_interrupts_i         | input     | [NUM_EXT_INTERRUPTS-1:0] | SMU external port                                                                   |
| sep_mailbox_interrupts_i | input     | [7:0]                    | Internal: from SEP <code>smc_mailbox_interrupt_o</code> when SEP=1, else tied to '0 |
| sep_wdt_reset_n_i        | input     | 1                        | Internal: <code>~sep_wdt_timer_rst_req</code> when SEP=1, else tied to 10b1         |

## Fuse Signals

| Port                                | Direction | Width | SMU Connection    |
|-------------------------------------|-----------|-------|-------------------|
| <code>fuse_sense_done_o</code>      | output    | 1     | SMU external port |
| <code>fuse_reset_n_delayed_o</code> | output    | 1     | SMU external port |

## DFT Signals

| Port                             | Direction | Width | SMU Connection    |
|----------------------------------|-----------|-------|-------------------|
| <code>skip_mem_repair_o</code>   | output    | 1     | SMU external port |
| <code>ext_boot_seq_done_i</code> | input     | 1     | SMU external port |

## PVT

| Port                          | Direction | Width | SMU Connection    |
|-------------------------------|-----------|-------|-------------------|
| <code>temp_interrupt_i</code> | input     | 1     | SMU external port |

## Lifecycle State

| Port                         | Direction | Width                  | SMU Connection                                                                                    |
|------------------------------|-----------|------------------------|---------------------------------------------------------------------------------------------------|
| <code>lc_state_o</code>      | output    | [2*LC_STATE_WIDTH-1:0] | SEP lifecycle state when <code>SEP=1</code> , else <code>80hf0</code> ; ported through SMU        |
| <code>lc_sigint_err_o</code> | output    | 1                      | SEP lifecycle integrity error when <code>SEP=1</code> , else SMC <code>efuse_lc_sigint_err</code> |

## RAS Bank Settings

| Port                             | Direction | Width | SMU Connection    |
|----------------------------------|-----------|-------|-------------------|
| <code>ras_bank_chip_o</code>     | output    | [3:0] | SMU external port |
| <code>ras_bank_instance_o</code> | output    | [3:0] | SMU external port |

## Mailbox Interrupts

| Port                                  | Direction | Width               | SMU Connection    |
|---------------------------------------|-----------|---------------------|-------------------|
| <code>ext_mailbox_interrupts_o</code> | output    | [NUM_MAILBOXES-1:0] | SMU external port |

## Reset Unit Signals

| Port                             | Direction | Width                           | SMU Connection    |
|----------------------------------|-----------|---------------------------------|-------------------|
| <code>cfg_flr_pf_active_i</code> | input     | 1                               | SMU external port |
| <code>isolate_req_o</code>       | output    | [31:0]                          | SMU external port |
| <code>ss_reset_complete_i</code> | input     | [31:0]                          | SMU external port |
| <code>ss_config_o</code>         | output    | [31:0]                          | SMU external port |
| <code>ss_reset_ctrl_o</code>     | output    | <code>reset_ctrl_t[31:0]</code> | SMU external port |
| <code>sync_irq_o</code>          | output    | 1                               | SMU external port |

## CPU Memory Signals

| Port                                   | Direction | Width                                  | SMU Connection    |
|----------------------------------------|-----------|----------------------------------------|-------------------|
| <code>rom_intf_req_o</code>            | output    | <code>rom_req_t</code>                 | SMU external port |
| <code>rom_intf_rsp_i</code>            | input     | <code>rom_rsp_t</code>                 | SMU external port |
| <code>scratch_ram_intf_req_o</code>    | output    | <code>scratch_ram_req_t[31:0]</code>   | SMU external port |
| <code>scratch_ram_intf_rsp_i</code>    | input     | <code>scratch_ram_rsp_t[31:0]</code>   | SMU external port |
| <code>l1_icache_tag_intf_req_o</code>  | output    | <code>l1_icache_tag_req_t[3:0]</code>  | SMU external port |
| <code>l1_icache_tag_intf_rsp_i</code>  | input     | <code>l1_icache_tag_rsp_t[3:0]</code>  | SMU external port |
| <code>l1_icache_data_intf_req_o</code> | output    | <code>l1_icache_data_req_t[7:0]</code> | SMU external port |
| <code>l1_icache_data_intf_rsp_i</code> | input     | <code>l1_icache_data_rsp_t[7:0]</code> | SMU external port |
| <code>l1_dcache_tag_intf_req_o</code>  | output    | <code>l1_dcache_tag_req_t[3:0]</code>  | SMU external port |
| <code>l1_dcache_tag_intf_rsp_i</code>  | input     | <code>l1_dcache_tag_rsp_t[3:0]</code>  | SMU external port |
| <code>l1_dcache_data_intf_req_o</code> | output    | <code>l1_dcache_data_req_t[3:0]</code> | SMU external port |
| <code>l1_dcache_data_intf_rsp_i</code> | input     | <code>l1_dcache_data_rsp_t[3:0]</code> | SMU external port |

## Memory Init

| Port                                  | Direction | Width | SMU Connection    |
|---------------------------------------|-----------|-------|-------------------|
| <code>disable_sram_auto_init_i</code> | input     | 1     | SMU external port |
| <code>init_mem_done_o</code>          | output    | 1     | SMU external port |

## System Timer OCTS Interface

| Port                       | Direction | Width  | SMU Connection    |
|----------------------------|-----------|--------|-------------------|
| <code>timer_count_o</code> | output    | [63:0] | SMU external port |

## Reset Overrides (from DTP)

| Port                           | Direction | Width                                       | SMU Connection                                                                                                                                 |
|--------------------------------|-----------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>jtag_reset_ctrl_i</code> | input     | <code>smc_pkg::jtag_smc_reset_ctrl_t</code> | DTP <code>jtag_ic_reset_smc_o</code> (packed struct with <code>.ovrd</code> override enables and <code>.val</code> active-low override values) |

## DFD Signals

| Port                                              | Direction | Width                               | SMU Connection                                                                  |
|---------------------------------------------------|-----------|-------------------------------------|---------------------------------------------------------------------------------|
| <code>cla_gpio_enable_o</code>                    | output    | 1                                   | SMU external port                                                               |
| <code>tdr_dbg_ctrl_clock_stop_en_i</code>         | input     | 1                                   | DTP <code>cla_clock_stop_en_o</code>                                            |
| <code>tdr_dbg_ctrl_clocks_stopped_by_cla_o</code> | output    | 1                                   | DTP <code>xtrig_clk_stop_req_i[0]</code> (SMU connects SMC output to DTP input) |
| <code>halt_clock_global_i</code>                  | input     | [NUM_MAX_EXTERNAL_XTRIG_AGENTS-1:0] | SMU external port                                                               |

## Cross Trigger Interface

| Port            | Direction | Width                                         | SMU Connection                                         |
|-----------------|-----------|-----------------------------------------------|--------------------------------------------------------|
| xtrigger_pads_i | input     | xtrigger_t                                    | SMU external port                                      |
| xtrigger_pads_o | output    | xtrigger_t                                    | SMU external port                                      |
| xtrigger_ss_i   | input     | xtrigger_t[NUM_MAX_EXTERNAL_XTRIG_AGENTS-1:0] | SMU external port                                      |
| xtrigger_ss_o   | output    | xtrigger_t                                    | [1:0]: DTP xtrig_ctm_dst_req_i[1:0] (both bits to DTP) |

## Trace Memory

| Port             | Direction | Width                                  | SMU Connection    |
|------------------|-----------|----------------------------------------|-------------------|
| trace_mem_req_o  | output    | SinkMemPktIn_s[TRC_RAM_INSTANCES-1:0]  | SMU external port |
| trace_mem_resp_i | input     | SinkMemPktOut_s[TRC_RAM_INSTANCES-1:0] | SMU external port |

## External Debug Bus

| Port            | Direction | Width   | SMU Connection    |
|-----------------|-----------|---------|-------------------|
| ext_debug_bus_i | input     | [511:0] | SMU external port |

## Test Mode

| Port       | Direction | Width | SMU Connection    |
|------------|-----------|-------|-------------------|
| test_en_i  | input     | 1     | SMU external port |
| scan_rst_i | input     | 1     | SMU external port |

## Captured Straps

| Port              | Direction | Width  | SMU Connection    |
|-------------------|-----------|--------|-------------------|
| captured_straps_i | input     | [63:0] | SMU external port |

Please refer to the `reset_unit` STRAPS\_LO and STRAPS\_HI register descriptions for the decoded value of the straps.

## 1.5.8. SMC-DTP Internal Interconnection

The following table summarizes the key internal connections between the DTP and SMC within the SMU. These connections are not visible on the SMU boundary.

| Connection    | DTP Port                 | SMC Port                 | Direction | Notes |
|---------------|--------------------------|--------------------------|-----------|-------|
| AXI Debug Bus | axi_smc_dbg_req_o        | jtag_axi_in_req_i        | DTP → SMC |       |
|               | axi_smc_dbg_resp_i       | jtag_axi_in_resp_o       | SMC → DTP |       |
| AXI OTP Bus   | axil_smc_otp_jtag_req_o  | axil_smc_otp_jtag_req_i  | DTP → SMC |       |
|               | axil_smc_otp_jtag_resp_i | axil_smc_otp_jtag_resp_o | SMC → DTP |       |
| DTP CSR Bus   | axil_xtrig_req_i         | axil_dtp_csr_req_o       | SMC → DTP |       |
|               | axil_xtrig_resp_o        | axil_dtp_csr_resp_i      | DTP → SMC |       |

| Connection               | DTP Port                              | SMC Port                                          | Direction | Notes                                                                                                                   |
|--------------------------|---------------------------------------|---------------------------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------|
| JTAG SMC Reset Overrides | <code>jtag_ic_reset_smc_o</code>      | <code>jtag_reset_ctrl_i</code>                    | DTP → SMC | Packed <code>smc_pkg::jtag_smc_reset_ctrl_t</code> struct carrying all SMC reset override enables and active-low values |
| DTP Reset                | <code>rst_n_i</code>                  | <code>rst_primary_smc_clk_no</code>               | SMC → DTP |                                                                                                                         |
| Cross Trigger [1:0]      | <code>xtrig_ctm_dst_req_i[1:0]</code> | <code>xtrigger_ss_o[1:0]</code>                   | SMC → DTP | Both bits                                                                                                               |
| Clock Stop #0            | <code>xtrig_clk_stop_req_i[0]</code>  | <code>tdr_dbg_ctrl_clocks_stopped_by_cla_o</code> | SMC → DTP |                                                                                                                         |
| Boot Stall               | <code>jtag_boot_stall_ovrd_o</code>   | Disconnected                                      | N/A       |                                                                                                                         |
|                          | <code>jtag_boot_stall_o</code>        | Disconnected                                      | N/A       |                                                                                                                         |

## 1.5.9. SEP-Conditional Wiring

SEP-related interfaces are handled internally within the SMU. Their behavior depends on the module parameter `SEP`.

| Module/Signal       | Implementation                                                                                           | Description                                                              |
|---------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| SMC SEP AXI Input   | <code>sep_axi_in_req_i</code> = SEP traffic when <code>SEP=1</code> , else '0                            | Dedicated SEP-to-SMC subordinate path                                    |
| DTP SEP OTP JTAG    | SEP <code>axil_sep_otp_jtag_*</code> when <code>SEP=1</code> , else <code>u_sep_otp_axil_err_slv</code>  | SEP OTP debug AXI-Lite path                                              |
| DTP SEP STAP return | SEP <code>jtag_tdo</code> when <code>SEP=1</code> , else <code>sep_stap_tdo_to_dtp = 1'b0</code>         | SEP JTAG STAP return path                                                |
| SMU                 | <code>sep_mailbox_interrupts</code>                                                                      | Internal SEP-to-SMC mailbox interrupts when <code>SEP=1</code> , else '0 |
| SMC                 | <code>sep_wdt_reset_n_i</code> = <code>~sep_wdt_timer_rst_req</code> when <code>SEP=1</code> , else 1'b1 | SEP watchdog reset input to SMC                                          |

## Other Internal Tie-offs

| Module | Port                                  | Tie-off Value | Reason                   |
|--------|---------------------------------------|---------------|--------------------------|
| DTP    | <code>xtrig_ctm_src_ack_i[1:0]</code> | 2'b00         | SMC uses pulse-sync mode |

`xtrig_ctm_src_ack_i[9:2]` (exposed as SMU `xtrig_ctm_src_ack_i[7:0]`) is ported out through the SMU. The integrator should tie off to '0 when using pulse-sync mode (`XTRIG_INT_CT_MODE` bit = 0), or connect to actual ack signals when using req/ack mode (bit = 1).

## 1.5.10. SMU Port Interface

The SMU exposes a single consolidated boundary that an adopter connects to at the chip or chiplet level. The following table defines all SMU top-level ports including clock, reset, JTAG, cross trigger, SMC-originated interfaces (AXI, GPIO, eFuse, PLL, PVT, memory), and SEP passthrough interfaces (memory macros, eFuse, crypto SRAM, SPI, AXI extension). Ports prefixed with `sep_` are only active when the module parameter `SEP=1`.

Table 8. SMU Port Declaration

| Interface Name         | Type  | Direction | Width | Description                                                   | Connection | Default value (if applicable) |
|------------------------|-------|-----------|-------|---------------------------------------------------------------|------------|-------------------------------|
| <code>clk_smu_i</code> | logic | Input     | 1     | Primary SMU clock domain. Shared by SMC and DTP system logic. | PLL output | N/A                           |

| Interface Name                  | Type                            | Direction | Width                           | Description                                                                                                  | Connection              | Default value (if applicable) |
|---------------------------------|---------------------------------|-----------|---------------------------------|--------------------------------------------------------------------------------------------------------------|-------------------------|-------------------------------|
| clk_ref_i                       | logic                           | Input     | 1                               | Reference clock domain for timing-sensitive operations.                                                      | PLL reference clock     | N/A                           |
| rst_cold_ni                     | logic                           | Input     | 1                               | Cold reset (active low). Asynchronous assertion, synchronous deassertion. Also serves as DTP power-on reset. | System cold reset       | N/A                           |
| rst_cold_stable_ref_clk_no      | logic                           | Output    | 1                               | Stable cold reset synchronized to reference clock domain (active low).                                       | Downstream subsystems   | N/A                           |
| powergood_i                     | logic                           | Input     | 1                               | Power-good indication from supply supervisor.                                                                | Power supply supervisor | N/A                           |
| jtag_ptap_client_tap_ctrl_i     | prim_jtag_pkg::jtag_tap_ctrl_t  | Input     | -                               | Primary JTAG TAP control (TCK, TMS, TRST_N).                                                                 | External JTAG pads      | N/A                           |
| jtag_ptap_client_tdi_i          | logic                           | Input     | 1                               | Primary JTAG TDI.                                                                                            | External JTAG pads      | N/A                           |
| jtag_ptap_client_tdo_o          | logic                           | Output    | 1                               | Primary JTAG TDO.                                                                                            | External JTAG pads      | N/A                           |
| jtag_ptap_client_tdo_oen_o      | logic                           | Output    | 1                               | Primary JTAG TDO output enable.                                                                              | External JTAG pads      | N/A                           |
| jtag_bsr_host_scan_ctrl_o       | prim_jtag_pkg::jtag_scan_ctrl_t | Output    | -                               | Boundary scan control.                                                                                       | Boundary scan chain     | N/A                           |
| jtag_bsr_host_scan_in_i         | logic                           | Input     | 1                               | Boundary scan data in.                                                                                       | Boundary scan chain     | N/A                           |
| jtag_bsr_host_scan_out_o        | logic                           | Output    | 1                               | Boundary scan data out.                                                                                      | Boundary scan chain     | N/A                           |
| jtag_stap_io_host_tap_ctrl_o    | prim_jtag_pkg::jtag_tap_ctrl_t  | Output    | -                               | I/O STAP TAP control.                                                                                        | I/O STAP host           | N/A                           |
| jtag_stap_io_host_tdi_i         | logic                           | Input     | 1                               | I/O STAP TDI.                                                                                                | I/O STAP host           | Tie to '0 if unused           |
| jtag_stap_io_host_tdo_o         | logic                           | Output    | 1                               | I/O STAP TDO.                                                                                                | I/O STAP host           | N/A                           |
| jtag_stap_io_host_tdo_oen_o     | logic                           | Output    | 1                               | I/O STAP TDO output enable.                                                                                  | I/O STAP host           | N/A                           |
| jtag_stap_extra_host_tap_ctrl_o | prim_jtag_pkg::jtag_tap_ctrl_t  | Output    | [JTAG_NUM_EXTRA_STAP_PORTS-1:0] | Extra STAP TAP control array.                                                                                | Extra STAP hosts        | Tie to '0 if unused           |
| jtag_stap_extra_host_tdi_i      | logic                           | Input     | [JTAG_NUM_EXTRA_STAP_PORTS-1:0] | Extra STAP TDI array.                                                                                        | Extra STAP hosts        | Tie to '0 if unused           |
| jtag_stap_extra_host_tdo_o      | logic                           | Output    | [JTAG_NUM_EXTRA_STAP_PORTS-1:0] | Extra STAP TDO array.                                                                                        | Extra STAP hosts        | N/A                           |
| jtag_stap_extra_host_tdo_oen_o  | logic                           | Output    | [JTAG_NUM_EXTRA_STAP_PORTS-1:0] | Extra STAP TDO output enable array.                                                                          | Extra STAP hosts        | N/A                           |
| jtag_stap_host_scan_ctrl_o      | prim_jtag_pkg::jtag_scan_ctrl_t | Output    | -                               | Extended STAP scan control.                                                                                  | STAP scan chain         | N/A                           |
| jtag_stap_host_scan_in_i        | logic                           | Input     | 1                               | Extended STAP scan data in.                                                                                  | STAP scan chain         | N/A                           |
| jtag_stap_host_scan_out_o       | logic                           | Output    | 1                               | Extended STAP scan data out.                                                                                 | STAP scan chain         | N/A                           |
| jtag_dfd_host_scan_ctrl_o       | prim_jtag_pkg::jtag_scan_ctrl_t | Output    | -                               | DFD iJTAG scan control.                                                                                      | DFD iJTAG chain         | N/A                           |

| Interface Name                   | Type                                          | Direction | Width                              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Connection                         | Default value (if applicable) |
|----------------------------------|-----------------------------------------------|-----------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|-------------------------------|
| jtag_dfd_host_scan_in_i          | logic                                         | Input     | 1                                  | DFD iJTAG scan data in.                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | DFD iJTAG chain                    | N/A                           |
| jtag_dfd_host_scan_out_o         | logic                                         | Output    | 1                                  | DFD iJTAG scan data out.                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | DFD iJTAG chain                    | N/A                           |
| jtag_dft_secure_host_scan_ctrl_o | prim_jtag_pkg::jtag_scan_ctrl_t               | Output    | -                                  | Secure DFT iJTAG scan control.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Secure DFT iJTAG chain             | N/A                           |
| jtag_dft_secure_host_scan_in_i   | logic                                         | Input     | 1                                  | Secure DFT iJTAG scan data in.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Secure DFT iJTAG chain             | N/A                           |
| jtag_dft_secure_host_scan_out_o  | logic                                         | Output    | 1                                  | Secure DFT iJTAG scan data out.                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Secure DFT iJTAG chain             | N/A                           |
| jtag_dft_host_scan_ctrl_o        | prim_jtag_pkg::jtag_scan_ctrl_t               | Output    | -                                  | Non-secure DFT iJTAG scan control.                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Non-secure DFT iJTAG chain         | N/A                           |
| jtag_dft_host_scan_in_i          | logic                                         | Input     | 1                                  | Non-secure DFT iJTAG scan data in.                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Non-secure DFT iJTAG chain         | N/A                           |
| jtag_dft_host_scan_out_o         | logic                                         | Output    | 1                                  | Non-secure DFT iJTAG scan data out.                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Non-secure DFT iJTAG chain         | N/A                           |
| dtp_stop_clks_o                  | logic                                         | Output    | 1                                  | Clock stop output from DTP (JTAG_DEBUG_CONTROL or CLA clock-stop).                                                                                                                                                                                                                                                                                                                                                                                                                             | PLL clock gates                    | N/A                           |
| jtag_ptap_state_o                | jtag_tap_pkg::tap_state_e                     | Output    | -                                  | Current PTAP TAP state.                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Debug infrastructure               | N/A                           |
| jtag_ptap_inst_decoded_o         | jtag_inst_reg_pkg::jtag_instruction_decoded_e | Output    | -                                  | Current decoded PTAP instruction.                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Debug infrastructure               | N/A                           |
| jtag_ic_reset_ext_o              | ic_reset_ext_t                                | Output    | $\$bits(ic\_reset\_ext\_t)$        | External JTAG IC_RESET override slice. Integrator-defined packed struct with <code>.ovrd</code> override enables and <code>.val</code> active-low override values. Driven by the DTP's <code>jtag_ic_reset_ext_o</code> when <code>Cfg.JTAG_IC_RESET_ENABLE=1</code> , otherwise tied to '0. An integrator needing JTAG reset overrides for logic outside the SMU sets <code>ic_reset_ext_t</code> to their own override struct and routes this port to it; the SMU does not define that type. | Integrator-defined reset overrides | N/A                           |
| xtrig_ctm_src_req_o              | logic                                         | Output    | $[XTRIG\_NUM\_INT\_CT-1:0]$        | Cross trigger source requests (SMU external ports [7:0]; DTP [1:0] reserved for SMC).                                                                                                                                                                                                                                                                                                                                                                                                          | Cross trigger network              | N/A                           |
| xtrig_ctm_src_ack_i              | logic                                         | Input     | $[XTRIG\_NUM\_INT\_CT-1:0]$        | Cross trigger source acknowledges. Unused in pulse-sync mode (mode bit = 0).                                                                                                                                                                                                                                                                                                                                                                                                                   | Cross trigger network              | Tie to '0 in pulse-sync mode  |
| xtrig_ctm_dst_req_i              | logic                                         | Input     | $[XTRIG\_NUM\_INT\_CT-1:0]$        | Cross trigger destination requests.                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Cross trigger network              | Tie to '0 if unused           |
| xtrig_ctm_dst_ack_o              | logic                                         | Output    | $[XTRIG\_NUM\_INT\_CT-1:0]$        | Cross trigger destination acknowledges. Unused in pulse-sync mode.                                                                                                                                                                                                                                                                                                                                                                                                                             | Cross trigger network              | N/A                           |
| xtrig_clk_stop_req_i             | logic                                         | Input     | $[XTRIG\_NUM\_CLK\_STOP\_REQ-1:0]$ | Clock stop requests (SMU external ports [7:0]; port [0] reserved for SMC internally).                                                                                                                                                                                                                                                                                                                                                                                                          | Clock stop requestors              | Tie to '0 if unused           |
| xtrig_ctp_req_out_dout_o         | logic                                         | Output    | $[XTRIG\_NUM\_CTP-1:0]$            | CTP request-out data output.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | GPIO pad ring                      | N/A                           |
| xtrig_ctp_req_out_dout_en_o      | logic                                         | Output    | $[XTRIG\_NUM\_CTP-1:0]$            | CTP request-out data output enable.                                                                                                                                                                                                                                                                                                                                                                                                                                                            | GPIO pad ring                      | N/A                           |
| xtrig_ctp_req_out_din_i          | logic                                         | Input     | $[XTRIG\_NUM\_CTP-1:0]$            | CTP request-out data input.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | GPIO pad ring                      | Tie to '0 if unused           |
| xtrig_ctp_req_out_din_en_o       | logic                                         | Output    | $[XTRIG\_NUM\_CTP-1:0]$            | CTP request-out data input enable.                                                                                                                                                                                                                                                                                                                                                                                                                                                             | GPIO pad ring                      | N/A                           |

| Interface Name                | Type                               | Direction | Width               | Description                                                          | Connection            | Default value (if applicable) |
|-------------------------------|------------------------------------|-----------|---------------------|----------------------------------------------------------------------|-----------------------|-------------------------------|
| xtrig_ctp_req_in_dout_o       | logic                              | Output    | [XTRIG_NUM_CTP-1:0] | CTP request-in data output.                                          | GPIO pad ring         | N/A                           |
| xtrig_ctp_req_in_dout_en_o    | logic                              | Output    | [XTRIG_NUM_CTP-1:0] | CTP request-in data output enable.                                   | GPIO pad ring         | N/A                           |
| xtrig_ctp_req_in_din_i        | logic                              | Input     | [XTRIG_NUM_CTP-1:0] | CTP request-in data input.                                           | GPIO pad ring         | Tie to '0 if unused           |
| xtrig_ctp_req_in_din_en_o     | logic                              | Output    | [XTRIG_NUM_CTP-1:0] | CTP request-in data input enable.                                    | GPIO pad ring         | N/A                           |
| xtrig_ctp_ack_in_dout_o       | logic                              | Output    | [XTRIG_NUM_CTP-1:0] | CTP ack-in data output.                                              | GPIO pad ring         | N/A                           |
| xtrig_ctp_ack_in_dout_en_o    | logic                              | Output    | [XTRIG_NUM_CTP-1:0] | CTP ack-in data output enable.                                       | GPIO pad ring         | N/A                           |
| xtrig_ctp_ack_in_din_i        | logic                              | Input     | [XTRIG_NUM_CTP-1:0] | CTP ack-in data input.                                               | GPIO pad ring         | Tie to '0 if unused           |
| xtrig_ctp_ack_in_din_en_o     | logic                              | Output    | [XTRIG_NUM_CTP-1:0] | CTP ack-in data input enable.                                        | GPIO pad ring         | N/A                           |
| xtrig_ctp_ack_out_dout_o      | logic                              | Output    | [XTRIG_NUM_CTP-1:0] | CTP ack-out data output.                                             | GPIO pad ring         | N/A                           |
| xtrig_ctp_ack_out_dout_en_o   | logic                              | Output    | [XTRIG_NUM_CTP-1:0] | CTP ack-out data output enable.                                      | GPIO pad ring         | N/A                           |
| xtrig_ctp_ack_out_din_i       | logic                              | Input     | [XTRIG_NUM_CTP-1:0] | CTP ack-out data input.                                              | GPIO pad ring         | Tie to '0 if unused           |
| xtrig_ctp_ack_out_din_en_o    | logic                              | Output    | [XTRIG_NUM_CTP-1:0] | CTP ack-out data input enable.                                       | GPIO pad ring         | N/A                           |
| rst_primary_ref_clk_no        | logic                              | Output    | 1                   | Primary reset synchronized to reference clock domain (active low).   | Downstream subsystems | N/A                           |
| rst_primary_smc_clk_no        | logic                              | Output    | 1                   | Primary reset synchronized to SMC clock domain (active low).         | Downstream subsystems | N/A                           |
| smu_axi_in_req_i              | smu_axi_xbar_pkg::axi_56_64_req_t  | Input     | -                   | SMU AXI crossbar inbound request (from SMN).                         | SMN fabric            | Tie to '0 if unused           |
| smu_axi_in_resp_o             | smu_axi_xbar_pkg::axi_56_64_resp_t | Output    | -                   | SMU AXI crossbar inbound response.                                   | SMN fabric            | N/A                           |
| smu_axi_out_req_o             | smu_axi_xbar_pkg::axi_out_req_t    | Output    | -                   | SMU AXI crossbar outbound request (to SMN).                          | SMN fabric            | N/A                           |
| smu_axi_out_resp_i            | smu_axi_xbar_pkg::axi_out_resp_t   | Input     | -                   | SMU AXI crossbar outbound response.                                  | SMN fabric            | Tie to '0 if unused           |
| smc_external_req_o            | smc_pkg::smc_axil_32_req_t         | Output    | -                   | SMC AXI-Lite external port request for adopter peripheral expansion. | Adopter peripherals   | N/A                           |
| smc_external_resp_i           | smc_pkg::smc_axil_32_resp_t        | Input     | -                   | SMC AXI-Lite external port response.                                 | Adopter peripherals   | Tie to DECERR if unused       |
| smc_efuse_bank_ctrl_req_o     | smc_pkg::smc_axil_32_req_t         | Output    | -                   | SMC eFuse bank control AXI-Lite request.                             | eFuse shim            | N/A                           |
| smc_efuse_bank_ctrl_resp_i    | smc_pkg::smc_axil_32_resp_t        | Input     | -                   | SMC eFuse bank control AXI-Lite response.                            | eFuse shim            | N/A                           |
| smc_efuse_shim_command_req_o  | smc_efuse_pkg::fuse_command_req_t  | Output    | -                   | SMC eFuse command interface request.                                 | eFuse shim            | N/A                           |
| smc_efuse_shim_command_resp_i | smc_efuse_pkg::fuse_command_resp_t | Input     | -                   | SMC eFuse command interface response.                                | eFuse shim            | N/A                           |
| smc_shadow_regs_o             | smc_efuse_pkg::efuse_map_t         | Output    | -                   | SMC eFuse shadow register output.                                    | External systems      | N/A                           |

| Interface Name           | Type                                     | Direction | Width                         | Description                                     | Connection                 | Default value (if applicable) |
|--------------------------|------------------------------------------|-----------|-------------------------------|-------------------------------------------------|----------------------------|-------------------------------|
| pll_clk_obs_i            | logic                                    | Input     | 1                             | PLL clock observation input.                    | PLL shim                   | N/A                           |
| pll_clk_obs_en_i         | logic                                    | Input     | 1                             | PLL clock observation enable.                   | Control logic              | N/A                           |
| pvt_process_clk_obs_i    | logic                                    | Input     | 1                             | PVT process monitor clock observation input.    | PVT shim                   | N/A                           |
| pvt_process_clk_obs_en_i | logic                                    | Input     | 1                             | PVT process monitor clock observation enable.   | Control logic              | N/A                           |
| cat_therm_i              | logic                                    | Input     | 1                             | CAT thermal sensor input.                       | CAT thermal sensor         | Tie to '0 if unused           |
| lsio_interface_select_o  | logic                                    | Output    | [NUM_GPIO_WRAP S-1:0]         | LSIO interface selection.                       | GPIO padring               | N/A                           |
| gpio_in_i                | logic                                    | Input     | [NUM_GPIO_WRAP S-1:0]         | GPIO input data from external pins.             | External GPIO pins         | N/A                           |
| gpio_out_o               | logic                                    | Output    | [NUM_GPIO_WRAP S-1:0]         | GPIO output data to external pins.              | External GPIO pins         | N/A                           |
| gpio_in_en_o             | logic                                    | Output    | [NUM_GPIO_WRAP S-1:0]         | GPIO input enable control.                      | GPIO padring               | N/A                           |
| gpio_out_en_o            | logic                                    | Output    | [NUM_GPIO_WRAP S-1:0]         | GPIO output enable control.                     | GPIO padring               | N/A                           |
| rst_cool_n_from_pin_i    | logic                                    | Input     | 1                             | Cool reset input from GPIO pin.                 | GPIO pin                   | N/A                           |
| clk_telemetry_i          | logic                                    | Input     | 1                             | Telemetry clock domain for ATB data collection. | Telemetry clock source     | N/A                           |
| rst_telemetry_n_i        | logic                                    | Input     | 1                             | Telemetry reset (active low).                   | Telemetry reset source     | N/A                           |
| telemetry_atdata_i       | telemetry_receiver_pkg::telemetry_data_t | Input     | [NUM_TELEMETRY_RECEIVERS-1:0] | ATB telemetry data input array.                 | ATB sources                | N/A                           |
| telemetry_atid_i         | telemetry_receiver_pkg::atb_id_t         | Input     | [NUM_TELEMETRY_RECEIVERS-1:0] | ATB telemetry ID input array.                   | ATB sources                | N/A                           |
| telemetry_atready_o      | logic                                    | Output    | [NUM_TELEMETRY_RECEIVERS-1:0] | ATB telemetry ready output.                     | ATB sources                | N/A                           |
| telemetry_atvalid_i      | logic                                    | Input     | [NUM_TELEMETRY_RECEIVERS-1:0] | ATB telemetry valid input.                      | ATB sources                | N/A                           |
| telemetry_atflushvalid_o | logic                                    | Output    | [NUM_TELEMETRY_RECEIVERS-1:0] | ATB flush valid output.                         | ATB sources                | N/A                           |
| telemetry_atflushready_i | logic                                    | Input     | [NUM_TELEMETRY_RECEIVERS-1:0] | ATB flush ready input.                          | ATB sources                | N/A                           |
| cluster_ded_o            | logic                                    | Output    | 1                             | Cluster double error detection output.          | External systems           | N/A                           |
| wdt_first_timeout_o      | logic                                    | Output    | 1                             | First watchdog timeout signal.                  | Reset unit                 | N/A                           |
| wdt_second_timeout_o     | logic                                    | Output    | 1                             | Second watchdog timeout signal.                 | External systems           | N/A                           |
| smc_region_size_o        | logic                                    | Output    | [31:0]                        | SMC address region size output.                 | External systems           | N/A                           |
| ext_interrupts_i         | logic                                    | Input     | [Cfg.NUM_INT_TIO_SMC-1:0]     | External interrupt inputs to SMC.               | External interrupt sources | Tie to '0 if unused           |

| Interface Name           | Type                             | Direction | Width                   | Description                                                                                                                                                                            | Connection                      | Default value (if applicable) |
|--------------------------|----------------------------------|-----------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-------------------------------|
| fuse_sense_done_o        | logic                            | Output    | 1                       | SMC fuse sense completion output.                                                                                                                                                      | Memory repair logic             | N/A                           |
| fuse_reset_n_delayed_o   | logic                            | Output    | 1                       | Delayed fuse reset output.                                                                                                                                                             | External systems                | N/A                           |
| skip_mem_repair_o        | logic                            | Output    | 1                       | Skip memory repair output.                                                                                                                                                             | Memory repair logic             | N/A                           |
| ext_boot_seq_done_i      | logic                            | Input     | 1                       | External boot sequence completion input (memory repair and shadow register override done); gates reset release.                                                                        | External boot/repair controller | N/A                           |
| temp_interrupt_i         | logic                            | Input     | 1                       | Temperature interrupt input.                                                                                                                                                           | PVT wrap                        | Tie to '0 if unused           |
| lc_state_o               | logic                            | Output    | [2*LC_STATE_WIDTH-1:0]  | Lifecycle state output (driven by SEP when SEP=1, else 0xf0).                                                                                                                          | External systems                | N/A                           |
| lc_sigint_err_o          | logic                            | Output    | 1                       | Lifecycle signal integrity error.                                                                                                                                                      | External systems                | N/A                           |
| ras_bank_chip_o          | logic                            | Output    | [3:0]                   | RAS bank chip selection output.                                                                                                                                                        | RAS system                      | N/A                           |
| ras_bank_instance_o      | logic                            | Output    | [3:0]                   | RAS bank instance selection output.                                                                                                                                                    | RAS system                      | N/A                           |
| ndmreset_request_i       | logic                            | Input     | [CPU_CLUSTER_COUNT-1:0] | Non-debug-module reset requests.                                                                                                                                                       | Debug infrastructure            | Tie to '0 if unused           |
| ndmreset_process_o       | logic                            | Output    | [CPU_CLUSTER_COUNT-1:0] | Non-debug-module reset process indicators.                                                                                                                                             | Debug infrastructure            | N/A                           |
| ext_mailbox_interrupts_o | logic                            | Output    | [NUM_MAILBOXES-1:0]     | External mailbox interrupt outputs.                                                                                                                                                    | External targets                | N/A                           |
| cfg_flr_pf_active_i      | logic                            | Input     | 1                       | Function Level Reset (FLR) PF active signal.                                                                                                                                           | PCIe / system                   | Tie to '0 if unused           |
| isolate_req_o            | logic                            | Output    | [31:0]                  | Isolation request output for 32 subsystems.                                                                                                                                            | Subsystems                      | N/A                           |
| ss_reset_complete_i      | logic                            | Input     | [31:0]                  | Subsystem reset completion status.                                                                                                                                                     | Subsystems                      | 32'hFFFFFFFF                  |
| ss_config_o              | logic                            | Output    | [31:0]                  | Subsystem configuration output.                                                                                                                                                        | Subsystems                      | N/A                           |
| ss_reset_ctrl_o          | smc_reset_unit_pkg::reset_ctrl_t | Output    | [31:0]                  | Subsystem reset control output array.                                                                                                                                                  | Subsystems                      | N/A                           |
| sync_irq_o               | logic                            | Output    | 1                       | Synchronized interrupt output.                                                                                                                                                         | External systems                | N/A                           |
| gpio_interrupt_o         | logic                            | Output    | [NUM_GPIO_WRAPS-1:0]    | Raw GPIO interrupt vector from the padding, one bit per GPIO wrap (bonded + unbonded). Passed through from SMC. Software reads GPIO status registers to identify the interrupt source. | SMC → external consumers        | N/A, leave floating if unused |
| uart_interrupt_o         | logic                            | Output    | [NUM_UART-1:0]          | Raw UART interrupt outputs, one bit per UART instance, in the peripheral clock domain. Passed through from SMC.                                                                        | SMC → external consumers        | N/A, leave floating if unused |
| rom_intf_req_o           | rom_req_t                        | Output    | -                       | SMC ROM memory interface request.                                                                                                                                                      | Boot ROM                        | N/A                           |
| rom_intf_rsp_i           | rom_rsp_t                        | Input     | -                       | SMC ROM memory interface response.                                                                                                                                                     | Boot ROM                        | N/A                           |
| scratch_ram_intf_req_o   | scratch_ram_req_t                | Output    | [31:0]                  | SMC scratch RAM interface requests.                                                                                                                                                    | Scratch RAM banks               | N/A                           |

| Interface Name            | Type                               | Direction | Width                   | Description                                                                                                                                                     | Connection          | Default value (if applicable) |
|---------------------------|------------------------------------|-----------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------|
| scratch_ram_intf_rsp_i    | scratch_ram_rsp_t                  | Input     | [31:0]                  | SMC scratch RAM interface responses.                                                                                                                            | Scratch RAM banks   | N/A                           |
| l1_icache_tag_intf_req_o  | l1_icache_tag_req_t                | Output    | [3:0]                   | SMC L1 I-cache tag interface requests.                                                                                                                          | I-cache tag banks   | N/A                           |
| l1_icache_tag_intf_rsp_i  | l1_icache_tag_rsp_t                | Input     | [3:0]                   | SMC L1 I-cache tag interface responses.                                                                                                                         | I-cache tag banks   | N/A                           |
| l1_icache_data_intf_req_o | l1_icache_data_req_t               | Output    | [7:0]                   | SMC L1 I-cache data interface requests.                                                                                                                         | I-cache data banks  | N/A                           |
| l1_icache_data_intf_rsp_i | l1_icache_data_rsp_t               | Input     | [7:0]                   | SMC L1 I-cache data interface responses.                                                                                                                        | I-cache data banks  | N/A                           |
| l1_dcache_tag_intf_req_o  | l1_dcache_tag_req_t                | Output    | [3:0]                   | SMC L1 D-cache tag interface requests.                                                                                                                          | D-cache tag banks   | N/A                           |
| l1_dcache_tag_intf_rsp_i  | l1_dcache_tag_rsp_t                | Input     | [3:0]                   | SMC L1 D-cache tag interface responses.                                                                                                                         | D-cache tag banks   | N/A                           |
| l1_dcache_data_intf_req_o | l1_dcache_data_req_t               | Output    | [3:0]                   | SMC L1 D-cache data interface requests.                                                                                                                         | D-cache data banks  | N/A                           |
| l1_dcache_data_intf_rsp_i | l1_dcache_data_rsp_t               | Input     | [3:0]                   | SMC L1 D-cache data interface responses.                                                                                                                        | D-cache data banks  | N/A                           |
| disable_sram_auto_init_i  | logic                              | Input     | 1                       | Disable SRAM automatic initialization.                                                                                                                          | Control logic       | 1'b0                          |
| init_mem_done_o           | logic                              | Output    | 1                       | Memory initialization completion signal.                                                                                                                        | Control logic       | N/A                           |
| chiplet_is_primary_i      | logic                              | Input     | 1                       | Chiplet is primary (for System Timer OCTS).                                                                                                                     | GPIO padding straps | N/A                           |
| timer_count_o             | logic                              | Output    | [63:0]                  | 64-bit system timer count output (OCTS).                                                                                                                        | External systems    | N/A                           |
| trace_mem_req_o           | dfd_trace_mem_pkg::SinkMemPktIn_s  | Output    | [TRC_RAM_INSTANCES-1:0] | Trace memory interface requests.                                                                                                                                | Trace memory        | N/A                           |
| trace_mem_resp_i          | dfd_trace_mem_pkg::SinkMemPktOut_s | Input     | [TRC_RAM_INSTANCES-1:0] | Trace memory interface responses.                                                                                                                               | Trace memory        | N/A                           |
| test_en_i                 | logic                              | Input     | 1                       | DFT/scan test-enable. Drives clock-gater test ports and AXI cell test inputs.                                                                                   | Test controller     | 1'b0                          |
| scan_rst_ni               | logic                              | Input     | 1                       | Scan reset (active-low). Bypasses reset synchronizers during DFT.                                                                                               | Test controller     | 1'b1                          |
| captured_straps_i         | logic                              | Input     | [63:0]                  | Captured strap configuration. Please refer to the <a href="#">reset_unit</a> STRAPS_LO and STRAPS_HI register descriptions for the decoded value of the straps. | GPIO padding        | N/A                           |
| mem_repair_done_i         | logic                              | Input     | 1                       | Memory repair done indicator.                                                                                                                                   | DFT controller      | 1'b0                          |
| mem_repair_success_i      | logic                              | Input     | 1                       | Memory repair success indicator.                                                                                                                                | DFT controller      | 1'b0                          |
| mem_repair_abort_i        | logic                              | Input     | 1                       | Memory repair abort indicator.                                                                                                                                  | DFT controller      | 1'b0                          |
| mbist_done_i              | logic                              | Input     | 1                       | MBIST done indicator.                                                                                                                                           | DFT controller      | 1'b0                          |
| mbist_pass_i              | logic                              | Input     | 1                       | MBIST pass indicator.                                                                                                                                           | DFT controller      | 1'b0                          |
| mbist_abort_i             | logic                              | Input     | 1                       | MBIST abort indicator.                                                                                                                                          | DFT controller      | 1'b0                          |
| sep_sram_req_o            | sep_pkg::sep_sram_req_t            | Output    | -                       | SEP scratch SRAM request passthrough.                                                                                                                           | SEP SRAM macros     | N/A                           |
| sep_sram_rsp_i            | sep_pkg::sep_sram_rsp_t            | Input     | -                       | SEP scratch SRAM response passthrough.                                                                                                                          | SEP SRAM macros     | N/A                           |

| Interface Name                 | Type                                           | Direction | Width | Description                                                                                         | Connection                 | Default value (if applicable) |
|--------------------------------|------------------------------------------------|-----------|-------|-----------------------------------------------------------------------------------------------------|----------------------------|-------------------------------|
| sep_boot_rom_req_o             | sep_pkg::sep_sram_req_t                        | Output    | -     | SEP boot ROM request passthrough.                                                                   | SEP ROM macro              | N/A                           |
| sep_boot_rom_rsp_i             | sep_pkg::sep_sram_rsp_t                        | Input     | -     | SEP boot ROM response passthrough.                                                                  | SEP ROM macro              | N/A                           |
| sep_cpu_tcm_req_o              | sep_pkg::sep_cpu_tcm_req_t                     | Output    | -     | SEP CPU TCM request passthrough (ICCM + DCCM).                                                      | SEP SRAM macros            | N/A                           |
| sep_cpu_tcm_rsp_i              | sep_pkg::sep_cpu_tcm_rsp_t                     | Input     | -     | SEP CPU TCM response passthrough.                                                                   | SEP SRAM macros            | N/A                           |
| sep_efuse_bank_ctrl_req_o      | sep_efuse_pkg::efuse_axil_req_t                | Output    | -     | SEP eFuse bank control AXI-Lite request passthrough.                                                | SEP eFuse shim             | N/A                           |
| sep_efuse_bank_ctrl_rsp_i      | sep_efuse_pkg::efuse_axil_rsp_t                | Input     | -     | SEP eFuse bank control AXI-Lite response passthrough.                                               | SEP eFuse shim             | N/A                           |
| sep_efuse_shim_command_req_o   | sep_efuse_pkg::fuse_command_req_t              | Output    | -     | SEP eFuse command interface request passthrough.                                                    | SEP eFuse shim             | N/A                           |
| sep_efuse_shim_command_rsp_i   | sep_efuse_pkg::fuse_command_rsp_t              | Input     | -     | SEP eFuse command interface response passthrough.                                                   | SEP eFuse shim             | N/A                           |
| sep_crypto_pka_imem_sram_req_o | sep_crypto_pkg::sep_crypto_pka_imem_sram_req_t | Output    | -     | SEP OTBN PKA instruction memory SRAM request passthrough.                                           | SRAM macro                 | N/A                           |
| sep_crypto_pka_imem_sram_rsp_i | sep_crypto_pkg::sep_crypto_pka_imem_sram_rsp_t | Input     | -     | SEP OTBN PKA instruction memory SRAM response passthrough.                                          | SRAM macro                 | N/A                           |
| sep_crypto_pka_dmem_sram_req_o | sep_crypto_pkg::sep_crypto_pka_dmem_sram_req_t | Output    | -     | SEP OTBN PKA data memory SRAM request passthrough.                                                  | SRAM macro                 | N/A                           |
| sep_crypto_pka_dmem_sram_rsp_i | sep_crypto_pkg::sep_crypto_pka_dmem_sram_rsp_t | Input     | -     | SEP OTBN PKA data memory SRAM response passthrough.                                                 | SRAM macro                 | N/A                           |
| sep_km_rom_mem_req_o           | km_intf_pkg::km_rom_mem_req_t                  | Output    | -     | SEP Key Manager ROM memory request passthrough.                                                     | ROM macro                  | N/A                           |
| sep_km_rom_mem_rsp_i           | km_intf_pkg::km_rom_mem_rsp_t                  | Input     | -     | SEP Key Manager ROM memory response passthrough.                                                    | ROM macro                  | N/A                           |
| sep_km_sram_mem_req_o          | km_intf_pkg::km_sram_mem_req_t                 | Output    | -     | SEP Key Manager SRAM memory request passthrough.                                                    | SRAM macro                 | N/A                           |
| sep_km_sram_mem_rsp_i          | km_intf_pkg::km_sram_mem_rsp_t                 | Input     | -     | SEP Key Manager SRAM memory response passthrough.                                                   | SRAM macro                 | N/A                           |
| spi_irq_i                      | logic                                          | Input     | 1     | SPI interrupt request into the SEP PIC, driven by whichever SPI controller the integration selects. | SPI controller             | Tie to '0 if unused           |
| sep_external_req_o             | sep_pkg::sep_32_64_12_axi_req_t                | Output    | -     | SEP AXI extension port request passthrough.                                                         | Adopter peripherals        | N/A                           |
| sep_external_rsp_i             | sep_pkg::sep_32_64_12_axi_rsp_t                | Input     | -     | SEP AXI extension port response passthrough.                                                        | Adopter peripherals        | Tie to DECERR if unused       |
| sep_cpu_trace_o                | sep_pkg::sep_cpu_trace_t                       | Output    | -     | SEP CPU trace output.                                                                               | Trace infrastructure       | N/A                           |
| lcc_demote_state_1_o           | logic                                          | Output    | [1:0] | SEP lifecycle demote state 1.                                                                       | External lifecycle logic   | N/A                           |
| lcc_demote_state_2_o           | logic                                          | Output    | [1:0] | SEP lifecycle demote state 2.                                                                       | External lifecycle logic   | N/A                           |
| sep_fuse_sense_done_o          | logic                                          | Output    | 1     | SEP fuse sense done output.                                                                         | System integration         | N/A                           |
| clk_sep_wdt_i                  | logic                                          | Input     | 1     | SEP watchdog timer clock passthrough.                                                               | Low-frequency clock source | N/A                           |

| Interface Name | Type                  | Direction | Width | Description                           | Connection             | Default value (if applicable) |
|----------------|-----------------------|-----------|-------|---------------------------------------|------------------------|-------------------------------|
| sep_straps_i   | sep_pkg::sep_straps_t | Input     | -     | SEP configuration straps passthrough. | GPIO padring / tie-off | N/A                           |

# Chapter 2. System Management Network (SMN) Integration

## WARNING

This section is currently under development. Content will be added to describe SMN integration procedures, network topology, and management interfaces.

# Chapter 3. Project Defines

Compile-time defines that apply across OCAH rather than to a single block.

## 3.1. SIMULATION

**SIMULATION** selects the verification view of OCAH RTL. Set it on every simulation compile and on no other kind of build:

```
+define+SIMULATION    # VCS, Xcelium
-DSIMULATION          # Verilator, Icarus
```

OCAH's own DV flows set it; a compile driven by an adopter's testbench does not.

### WARNING

Do not let **SIMULATION** reach synthesis, formal, or a gate-level flow. It enables **initial** blocks, **\$readmemh**, **\$random** and DPI exports; a tool that skips those rather than rejecting them yields a netlist that does not correspond to the RTL you verified.

Without the define, these testbench hooks are not compiled in and the plusargs are silently ignored:

| Plusarg                                                           | Effect                                                                                                                                 |
|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| +skip_fuse_sense                                                  | efuse_shadow_regs reports fuse sense complete immediately rather than sensing the bank through the adopter's SHIM.                     |
| +smc_shadow_reg_preload=<file>,<br>+sep_shadow_reg_preload=<file> | Loads the shadow registers from a hex image. Takes effect only alongside +skip_fuse_sense.                                             |
| +rom_bin64=<file>, +rom_hex=<file>                                | Preloads OCAH4CORECluster_rom_ext, the SMC CPU ROM simulation model. Applies only to integrations that use the reference memory swaps. |

The define also zeroes the RMA and security-disable token and digest bits in **efuse\_token\_processing** at time 0, which have no reset and otherwise power up X, and routes OpenTitan assertion failures in the SEP **prim\_assert** shim to **\$error** on non-UVM runs.

Three models it enables make simulation deliberately non-deterministic: **gdffsync** in the entropy source injects metastability on the second synchronizer flop, **prim\_flop\_2sync** and its Caliptra and Adams Bridge equivalents insert **prim\_cdc\_rand\_delay** ahead of the synchronizer, and **prim\_lfsr** and **prim\_double\_lfsr** randomize their default seed on simulators other than Verilator (**+prim\_lfsr\_use\_default\_seed=1** pins it). Two runs of one test are therefore not identical unless the simulator seed is fixed; record the seed with any failure you intend to reproduce.

# Chapter 4. Prim Cell Replacement and Don't-Touch Constraints

OCAH primitives (`prim_stdbuf`, `prim_and2`, `prim_clkgater`, the `prim_flop_*sync*` family, and so on) ship as behavioral models with a reference technology mapping. Adopters replace them with cells from their own library.

Two rules govern those replacements. Both are the adopter's responsibility — OCAH provides the instance names, but the names have no effect until the adopter's synthesis and place-and-route flows constrain them.

## 4.1. Naming Requirement

**Every library cell instantiated inside a replacement primitive must have `d0nt_` in its instance name.**

The marker is deliberately misspelled with a zero. `dont_touch` is a reserved keyword in several tools and a common substring in ordinary comments and signal names; `d0nt_` occurs only where the protection is intended, so a name-based match cannot pick up unrelated cells.

Clock-path cells take the extended prefix `d0nt_clk`, which selects the stricter rule below. The reference mapping uses these names:

| Instance name               | Rule        | Primitive                                         |
|-----------------------------|-------------|---------------------------------------------------|
| <code>d0nt_clkgate</code>   | Don't touch | <code>prim_clkgater</code>                        |
| <code>d0nt_clkbuf</code>    | Don't touch | <code>prim_clock_buf</code>                       |
| <code>d0nt_clkinv</code>    | Don't touch | <code>prim_clock_inv</code>                       |
| <code>d0nt_clkmux</code>    | Don't touch | <code>prim_clock_mux2</code>                      |
| <code>d0nt_clknand</code>   | Don't touch | <code>prim_clock_nand2</code>                     |
| <code>d0nt_clkxor</code>    | Don't touch | <code>prim_xor2</code>                            |
| <code>d0nt_stdbuf</code>    | Size only   | <code>prim_stdbuf</code>                          |
| <code>d0nt_stdinv</code>    | Size only   | <code>prim_flop_*sync_*</code> reset inverter     |
| <code>d0nt_an</code>        | Size only   | <code>prim_and2</code>                            |
| <code>d0nt_wrap_sync</code> | Size only   | <code>prim_flop_*sync_*</code> synchronizer flops |

Keep the same names when substituting your own cells. Adopters who rename must apply the equivalent constraints to whatever names they use.

## 4.2. Constraint Requirement

In synthesis, apply `set_size_only` to every cell matching `d0nt_`, then `set_dont_touch` to every cell matching `d0nt_clk`. Every step after synthesis needs `size_only` alone — on all `d0nt_` cells, clock cells included.

```
# Order matters: d0nt_clk* also matches d0nt_*, so the clock rule is applied
# second and overrides size-only for those cells.
set_size_only [get_cells -hierarchical -filter "full_name =~ *d0nt_*"]
set_dont_touch [get_cells -hierarchical -filter "full_name =~ *d0nt_clk*"]
```

`size_only` keeps the instance in the netlist—the tool may not delete, merge, absorb, restructure, or buffer through it—while still allowing it to be swapped for another library cell of the same logic function, typically to change drive strength or threshold-voltage flavor for timing, area, and EM closure. That is the intent for every replacement cell: these primitives exist because the structure matters, not because the logic function could not be inferred. Synthesis will otherwise remove them—an inverter pair that cancels, redundant AND gates that always agree, a buffer driving a single load are all dead weight to an optimizer.

Clock cells need the stronger `dont_touch` because that swap is exactly the problem. Under `size_only` the tool treats the logic function as the contract and every other property of the cell as free to change. On the clock path the function is the least of the reasons the cell was chosen. Clock buffers, inverters, muxes, and gaters are picked for electrical characteristics the function does not capture—matched rise and fall times above all, and with them duty-cycle preservation, balanced input-to-output delay, and low delay variation across PVT. A functionally identical cell pulled from the general-purpose library guarantees none of that. The netlist still says "inverter", functional simulation still passes, and the duty-cycle distortion shows up only in silicon or in a timing run late enough to be expensive.

`dont_touch` removes the tool's discretion: the cell the designer selected is the cell that reaches the netlist. Sizing and insertion-delay decisions on the clock path belong to clock tree synthesis, which owns them explicitly, rather than to a logic-optimization pass making a substitution it believes is free.

### 4.2.1. After Synthesis

The `dont_touch` on `d0nt_clk*` is a synthesis-only constraint. Once synthesis has handed off the netlist, release the clock cells to `size_only` and carry that—and nothing stronger—through placement, clock tree synthesis, routing, and optimization:

```
# Post-synthesis: uniform size_only, clock cells included.
set_dont_touch [get_cells -hierarchical -filter "full_name =~ *d0nt_clk*"] false
set_size_only [get_cells -hierarchical -filter "full_name =~ *d0nt_*"]
```

The stricter rule exists to stop logic optimization from choosing the cell. Downstream, clock cell sizing is CTS's job: it cannot balance skew or close transition violations with the cells it needs to resize frozen, and freezing them there buys nothing. Keeping `size_only` in place still prevents deletion, merging, and restructuring for the rest of the flow.

Restrict the physical flow to the library's clock-cell list so a resize stays inside the same clock cell family. Sizing a `CLKBUF_D4` up to a `CLKBUF_D8` keeps the matched rise and fall behavior the cell was chosen for; substituting a general-purpose buffer of equivalent drive does not.

Additional requirements:

- **Carry `size_only` through PnR.** The `dont_touch` is synthesis-only, but `size_only` is not. A cell protected in synthesis and left entirely unprotected downstream is not protected.
- **Verify the match is non-zero.** A typo in the filter, or a flow that renames instances before the constraint is read, silently protects nothing. Check the collection size and error out if it is empty.
- **Preserve the names.** Uniquification, flattening, and name mapping must not strip the `d0nt_` substring. If your flow renames instances, derive the collections before flattening and carry them forward.
- **Restore protection after ECOs.** Flows that relax preservation constraints for clock-tree or filler ECOs—clearing `eco_honor_dont_touch`, for example—must restore `size_only` before any subsequent optimization pass.
- **Check the final netlist,** not just the post-synthesis one.

Flows other than Design Compiler and Fusion Compiler must apply the equivalent `size_only` and `dont_touch` attributes to the same collections.

## 4.3. Simulation

These constraints have no functional effect and no simulation impact. All OCAH simulation flows compile with `+define+NO_LIBCELL`, which selects the behavioral model inside every primitive, so no library cell is ever elaborated by a simulator.

Confirming that the protected cells survived is therefore a gate-level netlist check. RTL simulation cannot detect their loss.

# Chapter 5. CDC Timing Constraints

OCAH ships synthesis constraints that bound every clock-domain crossing in the SMC, SEP, DTP and SMU. This section explains what those constraints do, how they are produced, and what an adopter has to change when integrating OCAH into their own flow and cell library.

The collateral ships with the harness. Each block carries its own constraints in `hw/sys/<block>/synth/`:

- `constraints.sdc` — the block's full timing intent, including the clock definitions and the asynchronous clock groups. It sources the two files below.
- `<block>_cdc_max_delay.tcl` — generated-clock periods and any hand-written overrides.
- `<block>_cdc_max_delay_generated.tcl` — the per-instance calls.

The two pieces shared by every block live in `flows/synth/constraints/`: `async_clock_groups.tcl` and `cdc_max_delay_procs.tcl`.

## 5.1. Why crossings are bounded rather than timed

A clock-domain crossing has no fixed phase relationship, so it cannot be timed against a capture edge. Its correctness still depends on delay, though: every CDC element assumes the data reaches the receiver within a bounded time.

OCAH therefore keeps the crossings visible to the tool and gives each one an explicit upper bound, sized from the clock that captures it. This asserts no phase relationship — it states only how much delay the crossing tolerates while its CDC element's guarantee still holds.

## 5.2. What is constrained

Every CDC element in each block is enumerated and given its own `set_max_delay`, sized from the clock that actually captures it as read from the netlist. Muxed domains resolve to several clocks; the tightest period wins.

The elements covered are `prim_sync2/3/4` and their reset and set variants, the pulp `sync` shim, `prim_sync_reset`, `prim_sync3_pulse`, `prim_fifo_async`, `avsbu_async_fifo`, `cdc_fifo_gray`, `axi_cdc`, `prim_reg_cdc`, `prim_sync_reqack_data`, `prim_sync_data_autohs`, `cdc_2phase`, `dmi_wrapper` and Rocket `AsyncQueue`.

The bound depends on what protects the crossing:

| Kind of path             | Bound                    | Reason                                                                                                                                     |
|--------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Synchronizer data input  | $0.5 * T_{dst}$          | Leaves half a destination period of settling margin before the capture edge the synchronizer relies on.                                    |
| Protocol-protected data  | $1.0 * T_{dst}$          | Conservative. The requirement is set by the pointer synchronizer's depth — see below.                                                      |
| Two-phase handshake nets | $\min(T_{src}, T_{dst})$ | <code>cdc_2phase</code> runs the handshake in both directions, so the tighter domain governs. This is the module's own stated requirement. |

$T_{dst}$  rather than  $\min(T_{src}, T_{dst})$  in the general case because the gating synchronizer is clocked by the destination, so its settling window is  $S * T_{dst}$  and  $T_{src}$  does not enter the guarantee. This makes slow-to-fast and fast-to-slow equally safe.

### 5.2.1. Why protocol-protected data still needs a bound

The payload of a gray-pointer FIFO or a req/ack handshake never passes through a synchronizer — only the pointer or request does. So this is not a metastability constraint, and it is reasonable to ask why such a path needs a bound at all, or why a single blanket bound covering every crossing in the design would not be enough.

Because the protocol guarantees **ordering**, not **arrival**. The read side will not look at an entry until the write pointer has crossed, which takes `SYNC_STAGES` destination cycles, where `SYNC_STAGES` is the depth of the pointer synchronizer. Nothing in the protocol ensures the data physically arrived within that window; that is a timing property, and only a `max_delay` states it. Miss it and the consumer samples the entry's previous contents. There is no violation report and no metastability event, just wrong data.

So the requirement is `data path < SYNC_STAGES * T_dst`. It is not a fixed number of nanoseconds: it scales with the clock that captures the crossing, and the destination periods within one design span an order of magnitude.

| Destination period | Guarantee at 3 stages |
|--------------------|-----------------------|
| 1 ns               | 3 ns                  |
| 5 ns               | 15 ns                 |
| 10 ns              | 30 ns                 |

That is why the bound is applied per instance rather than as one global value. A single blanket `max_delay` has to be chosen once for the whole design: loose enough to be feasible on the slowest domains, it permits several times more delay than a fast crossing's pointer scheme tolerates; tight enough for the fastest domain, it over-constrains every slow crossing and burns effort closing paths that were never at risk. Sizing each crossing from its own capturing clock is the only bound that tracks the requirement.

The blanket default that the clock-group layer applies (see [The two layers](#)) is deliberately on the loose side, and for slow destination domains it is already sufficient — there the explicit per-instance constraint is redundant. It is the fast crossings that need it.

`1.0 * T_dst` is therefore conservative by a factor of two to three. It is kept simple deliberately, but if wide payloads such as AXI data will not close, raising `cdc_data_delay` to `2.0 * T_dst` stays inside a three-stage guarantee with a cycle to spare. An adopter whose synchronizer cells use a different depth should recompute from their own `SYNC_STAGES` rather than inheriting this factor.

Enumeration and clock resolution run once, offline, against an elaborated design, and their output is an explicit per-block file of constraint calls that is reviewed and checked in. The flow itself does no discovery. This matters to an integrator because those files are written against the OCAH hierarchy and OCAH's clock names. If you instantiate a block as delivered, you do not need to regenerate or hand-edit them: set `::cdc_hier_prefix` to where the block sits in your hierarchy, and optionally `::cdc_clock_alias` to map OCAH clock names onto your own. Regeneration is needed only when the block is reconfigured in a way that changes which CDC elements exist, since no prefix can conjure constraints for elements the file never listed (see [Integration checklist](#) item 2).

## 5.3. The two layers

The per-instance bounds take effect only on top of a correctly configured clock-group layer. That layer decides whether the tool considers inter-domain paths at all, so it must be established first; the per-instance exceptions then refine it.

**Layer 1** is `set_async_clock_groups`, defined in `flows/synth/constraints/async_clock_groups.tcl` and called from each block's `constraints.sdc`. It declares the asynchronous groups with `-allow_paths` and applies a loose 20 ns / -20 ns default to every inter-group clock pair.

## IMPORTANT

`-allow_paths` is a prerequisite, not a refinement. In DC and PrimeTime exception priority, `set_false_path` outranks `set_max_delay` regardless of which is more specific, so a bare `set_clock_groups -asynchronous` silently masks **every** per-instance constraint. There is no warning; the exceptions simply have no effect.

The 20 ns default is what makes `-allow_paths` safe to enable. Once paths are allowed, every crossing without a specific constraint would otherwise be timed against a meaningless launch-capture relationship and fail. The default bounds them loosely so that only deliberately tightened paths appear as real requirements. It is emitted per ordered clock pair, never group-wide, because a 20 ns bound inside a group would mask genuine setup failures on synchronous paths.

**Layer 2** is the per-instance `set_max_delay` output of the three steps above, with two deliberate carve-outs:

- **Paths from primary inputs revert to the default.** A synchronizer input reachable from a top-level port carries that port's `set_input_delay`, which is a budget figure rather than real logic and can alone exceed the tightened bound. Only the parent block sees both ends, so there is nothing to tighten at block level.
- **Same-clock instances are skipped.** A two-clock element wired to one clock is not a crossing, and applying `set_max_delay -ignore_clock_latency` there would replace a proper setup check with a flat budget that discards skew and uncertainty. These are emitted commented out. OCAH has five, where OpenTitan AES, KMAC and OTBN instantiate `prim_sync_reqack_data` on EDN and OTP interfaces that are separate domains upstream but share a clock in SEP.

## 5.4. What is covered

| Block | CDC elements constrained | Exceptions emitted |
|-------|--------------------------|--------------------|
| SMC   | 365                      | 412                |
| SEP   | 18                       | 21                 |
| DTP   | 250                      | 270                |
| SMU   | 758                      | 838                |

SMU is a flat build containing the other three, so its instance paths carry `u_smc/`, `gen_sep.u_sep/` and `u_dtp/` prefixes. It is not the sum of the parts: DTP contributes 375 elements inside SMU against 250 standalone, because it instantiates three JTAG2AXI bridges in the SMU configuration and two on its own.

## 5.5. Integration checklist

Five things must change, three should be checked. None of them requires editing per-instance data by hand; every change is either a one-line policy edit or a regeneration.

### 5.5.1. 1. Repoint the constraints at the pin of the cell inside the prim wrapper

This is the most important change.

As shipped, the procs target the prim wrapper's own port, `prim_sync3/i_d`. That port exists only as long as the wrapper survives as a hierarchical boundary, and a prim cell is ordinary RTL: nothing stops synthesis from ungrouping it and dissolving its ports.

## WARNING

Do not rely on the wrapper port surviving. Anchor the constraint to the pin of the cell instantiated **inside** the wrapper. When you swap in your own synchronizer, that cell is a real library cell or

macro—a leaf that cannot be flattened—so its pins persist through optimisation regardless of what happens to the RTL hierarchy around it.

```
proc set_cdc_max_delay_prim_sync3 { inst dst_clk {delay {}} } {  
    cdc_emit "prim_sync3 $inst" [cdc_sync_delay $dst_clk $delay] to \  
        {} [get_pins "$inst/u_my_sync_cell/D" -quiet]  
}
```

`$inst` stays the wrapper instance path the generator recorded; you extend it down to the leaf. If your cell is instantiated in a generate loop or per bit, the path needs a glob. Note that generate scopes are separated by `.`, not `/`:

```
[get_pins "$inst/g_bits*.u_my_sync_cell/D" -quiet]
```

Anchoring this way makes hierarchy preservation a safety margin rather than a load-bearing dependency.

If you keep the prim wrappers unchanged, the shipped port names work as-is, but your flow must then preserve those wrapper boundaries. A `destination empty` warning appearing after `compile` where there was none at constraint-load time means a boundary was dissolved.

If you go further and replace a wrapper with a module of your own name, the enumeration that produced `<block>_cdc_max_delay_generated.tcl` knows nothing about it and those crossings end up with no per-instance bound at all. Nothing warns—an element that was never listed cannot report itself missing. Add the calls for it to `<block>_cdc_max_delay.tcl` by hand; the procs take an instance path and the capturing clock regardless of the module they were named after.

## 5.5.2. 2. Point the files at your hierarchy, or regenerate

The checked-in `<block>_cdc_max_delay_generated.tcl` files are written against the OCAH hierarchy, with paths relative to the block's own top and clocks named as OCAH names them. Two ways to make them apply to your design.

**If you instantiate the block as delivered**, reuse the files and set two variables. Both default to no-ops, so leaving them alone reproduces the shipped behaviour exactly.

```
# where the block sits in your hierarchy, with a trailing slash  
set ::cdc_hier_prefix "u_chiplet/u_mgmt/u_smc/"  
  
# optional: map an OCAH clock name onto the clock you actually drive it with  
set ::cdc_clock_alias(SMCCLK) MY_SYS_CLK
```

The prefix is prepended to every instance path. The alias is applied wherever a clock name reaches the tool, so OCAH's names become labels in your design. You can also skip the alias and supply the period directly with `set ::cdc_clock_period(SMCCLK) <ps>`, which is enough for the synchronizer constraints; the alias additionally matters for the data-path constraints, which name the destination clock as a timing endpoint and so need it to exist.

**If you reconfigure the block**, regenerate instead. Changing parameters that add or remove CDC elements makes the shipped file stale in the silent direction—the constraints that should exist for new elements simply will not, and nothing warns. The prefix does not help with that.

A wrong prefix fails loudly: every path resolves to nothing, so you get a warning per constraint against a zero exception count, and `SUMMARY 0 exceptions` is itself reported as a setup failure.

Regeneration is also the right answer whenever a parent build contains the block. Instance counts there can differ from the standalone build, so a parent's file is not the child's file with a prefix.

### 5.5.3. 3. Set the default bound

The 20 ns / -20 ns figures are a starting point. Both are `info exists-guarded` in `flows/synth/constraints/async_clock_groups.tcl`, so setting them before that file is sourced overrides without editing it. `set_async_clock_groups` also accepts per-call overrides, which is the cleaner way to give one block a different default:

```
set_async_clock_groups { {CLK_A} {CLK_B} } -max_delay 8000 -min_delay -8000
```

Two constraints on the choice. The default must exceed every per-instance bound you tighten, because it also governs paths from primary inputs into synchronizers; if it is tighter than  $0.5 * T_{dst}$  it becomes the binding constraint there. And tighter is not automatically better: this is a net for crossings you have not enumerated, so set it loose enough that those do not fail and rely on layer 2 for the paths that matter.

The minimum is negative deliberately. With `-allow_paths` a crossing has no meaningful hold relationship, and a zero floor would report hold violations where hold is not a real requirement.

### 5.5.4. 4. Declare your own clock groups

`set_async_clock_groups` is called from each block's `constraints.sdc` with that block's group list. Each inner list is one asynchronous group and globs are allowed. Globs resolve when the call runs, so it has to come after every `create_clock` and `create_generated_clock` it is meant to cover.

```
set_async_clock_groups {  
    {CLK_A CLK_A_DERIVED_*}  
    {CLK_B}  
    {JTAG_TCK}  
} -exclude {{PATTERN_X* PATTERN_Y*}}
```

`-exclude` takes pairs of patterns and suppresses that part of the cross product, for clock pairs that already have a `-logically_exclusive` or `-physically_exclusive` relationship that `-allow_paths` would contradict. Omit it unless you hit a `TIM-184` conflict.

### 5.5.5. 5. Declare periods for generated clocks

A generated clock exposes no readable `period` attribute in Design Compiler, so any generated clock that clocks a CDC element needs an explicit entry in `<block>_cdc_max_delay.tcl`:

```
set ::cdc_clock_period(MY_DIV_CLK) [expr {4 * $clock_periods(MY_MASTER_PERIOD)}]
```

Omitting one is loud — a `no period for clock` warning per affected element — so a clean run means you have them all.

Keep them in step with your `create_generated_clock` divide factors; nothing cross-checks them.

### 5.5.6. Also check

**Hierarchy preservation.** The reference flow applies `set_ungroup false` and `set_boundary_optimization false` to the CDC designs before `compile`. How much this matters depends on item 1: with leaf cell pins it is a safety margin, with wrapper ports it is load-bearing. Either way, your flow needs the equivalent, and it needs to name your cells rather than OCAH's.

**Delay factors.**  $0.5 * T_{dst}$  and  $1.0 * T_{dst}$  are policy, not physics. If your methodology wants more settling margin, change `cdc_sync_delay` and `cdc_data_delay` in `flows/synth/constraints/cdc_max_delay_procs.tcl`. Per-instance overrides are also available without editing procs: every proc takes an optional trailing delay in picoseconds, and re-stating a call in `<block>_cdc_max_delay.tcl` after the `source` wins.

**Tool differences.** Two Design Compiler specific choices to revisit under PrimeTime. `get_attribute <pin> clocks` is used because `get_clocks -of_objects` is PrimeTime-only, and generated clocks may expose a readable `period` in PrimeTime, making item 5 unnecessary.

## 5.6. Confirming it worked

Load the constraints and check the summary:

```
INFO: cdc_max_delay: SUMMARY <n> exceptions, 0 warnings
```

Zero warnings is necessary but not sufficient. A pin pattern that matches nothing warns, but one that matches the **wrong** thing does not. Two stronger checks:

- **Predict the count.** Multiply each proc's number of emitted exceptions by its number of calls and compare against the summary. A mismatch localises the failure, because the size of the shortfall identifies which proc did not fire.
- **Confirm the precedence.** Layer 2 is on pins and layer 1 on clocks, and the design assumes the more specific specification wins. Verify once in your tool:

```
report_timing -to [get_pins <a synchronizer>/i_d] -nworst 1
```

The `max_delay` line must show your  $0.5 * T_{dst}$  value rather than the default. If it shows the default, layer 2 is inert and nothing else in this section applies.

`SUMMARY 0 exceptions` is treated as a setup failure and warns as such. It means the constraints did not load, not that the design has no crossings.

## 5.7. Scope and limitations

The 20 ns default is loose by design. Against a 1 ns clock it permits twenty clock periods of delay into a synchronizer input. It is a safety net that keeps `-allow_paths` usable, not a real constraint; any crossing that matters belongs in layer 2.

Enumeration works from a fixed list of module names, so a hand-written two-flop synchronizer that uses none of them will not be found. Structural detection of crossings is what a static CDC tool does, and this flow does not attempt it. These constraints time crossings already known to exist; they are not a substitute for CDC signoff.

Because the instance list is generated offline, removals are loud but additions are silent. A stale instance path warns, but a synchronizer added without regenerating produces no warning: the constraint never exists, the path falls back to the default, and it passes timing. Regenerate and diff the result against what is checked in whenever the RTL changes.